NEET 2020

OFFICIAL QUESTION PAPERS WITH DETAILED SOLUTION

All questions with detailed solutions of Physics, Chemistry and Biology

CAREERS 360

Physics

Q. 1 Light with an average flux of 20 W/cm^2 falls on a non–deflection surface at normal incidence having a surface area of 20 cm^2. The energy received by the surface during a time span of 1 minute is :

Option 1:

$10 \times 10^3 J$

Option 2:

$12 \times 10^3 J$

Option 3:

$24 \times 10^3 J$

Option 4:

$48 \times 10^3 J$

Correct Answer:

$24 \times 10^3 J$

Solution:

The intensity of the light incident is given by:

$$I = \frac{E}{At}$$

Here, $I = 20 W/cm^2$, Area $A = 20 cm^2$, time $t = 1 min = 60 sec$

$$\therefore 20 = \frac{E}{20 \times 60}$$

$$\Rightarrow E = 20 \times 20 \times 60 = 24 \times 10^3 J$$

Hence the answer is the option (3).

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Q. 2 For transistor action, which of the following statement is correct?

Option 1:
Base, emitter and collector region should have same doping concentrations.

Option 2:
Base, emitter and collector region should have same size.

Option 3:
Both emitter junction as well as the collector junction are forward biased.

Option 4:
The base region must be very thin and lightly doped.

Correct Answer:
The base region must be very thin and lightly doped.

Solution:
For a transistor action, the junction must be lightly doped so that the base region is very thin. Also, the emitter junction must be reverse-biased.

Q. 3 Which of the following represents the variation of the resistivity (ρ) with temperature (T) for copper?

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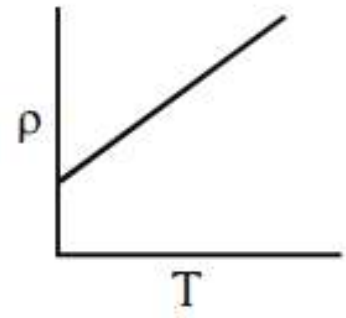
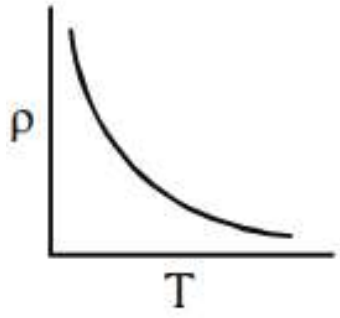
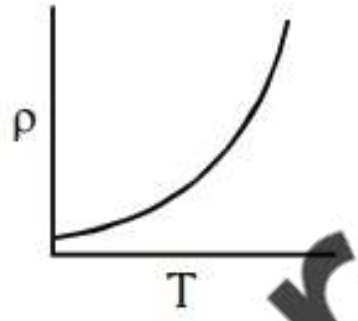
Option 1:



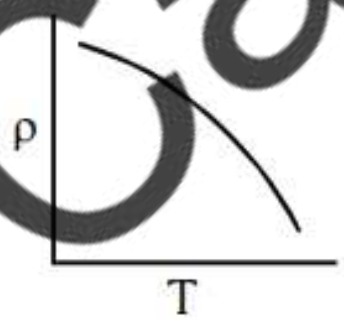
Option 2:



Option 3:



Option 4:



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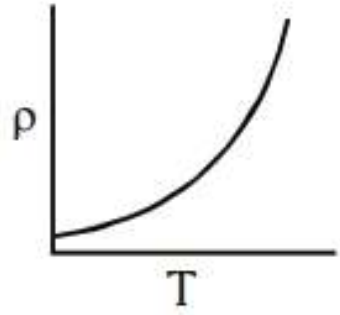
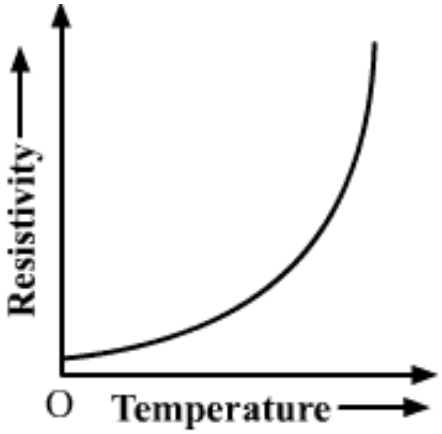
Correct Answer:



Solution:
The variation of resistivity of copper with temperature is parabolic in nature.



Q. 4 In a certain region of space with volume 0.2 m^3 the electric potential is found to be 5V throughtout. The magnitufde of electric field in this region is:

Option 1:
zero

Option 2:
0.5 N/C

Option 3:
1 N/C

Option 4:
5 N/C

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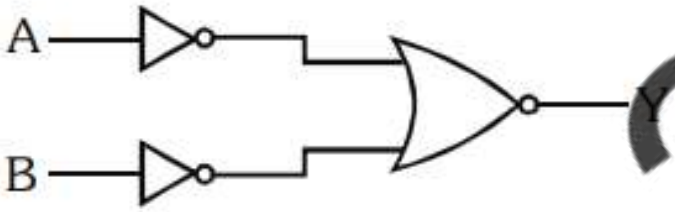
Correct Answer:
zero

Solution:

As $E = -\frac{dV}{dr}$

and V is constant throughout, therefore, $E = 0$

Q. 5 For the logic circuit shown, the truth table is:



Option 1:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

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Option 2:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Option 3:

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Option 4:

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

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Correct Answer:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Solution:

$$Y = \overline{\overline{A} + \overline{B}} = \overline{\overline{A}} \cdot \overline{\overline{B}} = A \cdot B = \text{AND gate}$$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Q. 6 A $40\mu F$ capacitor is connected to a 200V, 50Hz AC supply. The RMS value of the current in the circuit is nearly-

Option 1:
1.7 A

Option 2:
2.05 A

Option 3:
2.5 A

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Option 4:
25.1 A

Correct Answer:
2.5 A

Solution:

$$\text{Here, } C = 40 \times 10^{-6} F, E_{rms} = 200V, v = 50Hz$$
$$I_{rms} = ?$$

$$I_{rms} = \frac{E_{rms}}{X_C} = \frac{E_{rms}}{1/\omega C} = E_{rms}(\omega C) = E_{rms}(2\pi v C)$$

$$I_{rms} = 200 \times 2 \times \frac{22}{5} \times 50 \times 40 \times 10^{-6} = 2.5A$$

Hence the answer is option (3).

Q. 7 A cylinder contains hydrogen gas at a pressure of 249 kPa and temperature 27°C.

Its density is: $(\mathrm{R} = 8.3 \mathrm{Jmol^{-1}K^{-1}})$

Option 1:
0.5 kg/m^3

Option 2:
0.2 kg/m^3

Option 3:
0.1 kg/m^3

Option 4:
0.02 kg/m^3

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Correct Answer:
0.2 kg/m^3

Solution:

$$Pv = nRT$$

$$\Rightarrow PV = \frac{w}{M}RT$$

$$\Rightarrow P = \frac{w}{M} \times \frac{RT}{V} = \frac{dRT}{M} \quad [\because M = 2g/mol \ and T = 273 + 27 = 300K]$$

$$\Rightarrow d = \frac{PM}{RT} = \frac{249 \times 10^3 \times 2 \times 10^{-3}}{8.3 \times 300}$$

$$= 0.2\mathrm{Kg/m^3}$$

Hence the answer is the option (2).

Q. 8 Taking into account the significant figures, what are the values of 9.99m - 0.0099 m?

Option 1:
9.9801 m

Option 2:
9.98 m

Option 3:
9.980 m

Option 4:
9.9 m

Correct Answer:
9.98 m

Solution:
The result of an addition or subtraction in the number having different precisions should be reported to the same number of decimal places as are present in the number having the least number of decimal places.

Here 9.99 has the least number of decimal places, so

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9.99-0.0099=9.9801~9.98

Hence the answer is the option (2).

Q. 9 The mean free path for gas, with molecular diameter d and number density n, can be expressed as:

Option 1:
$\frac{1}{\sqrt{2}n\pi d}$

Option 2:
$\frac{1}{\sqrt{2}n\pi d^2}$

Option 3:
$\frac{1}{\sqrt{2}n^2\pi d^2}$

Option 4:
$\frac{1}{\sqrt{2}n^2\pi^2 d^2}$

Correct Answer:
$\frac{1}{\sqrt{2}n\pi d^2}$

Solution:
The mean free path is given by,

$$\lambda = \frac{1}{\sqrt{2}\pi n d^2}$$

Hence the answer is the option (2).

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Q. 10 An iron rod of susceptibility 599 is subject to a magnetizing field of 1200 A m^{-1}. The permeability of the material of the rod is:

$$\left(\mu_0 = 4\pi \times 10^{-7}\mathrm{TmA}^{-1}\right)$$

Option 1:
$2.4\pi \times 10^{-4}\mathrm{TmA}^{-1}$

Option 2:
$8.0 \times 10^{-5}\mathrm{TmA}^{-1}$

Option 3:
$2.4\pi \times 10^{-5}\mathrm{TmA}^{-1}$

Option 4:
$2.4\pi \times 10^{-7}\mathrm{TmA}^{-1}$

Correct Answer:
$2.4\pi \times 10^{-4}\mathrm{TmA}^{-1}$

Solution:

$\text{Relative permeability, } \mu_r = (1 + \chi)$

$\Rightarrow \mu_r = (1 + 599) = 600$

$\text{permeability of substance, } \mu = \mu_0\mu_r = 4\pi \times 10^{-7} \times 600 = 2.4\pi \times 10^{-4}$

Hence the answer is the option (1).

Q. 11 A short electric dipole has a dipole moment of $16 \times 10^{-9}\mathrm{Cm}$ The electric potential due to the dipole at a point at a distance of 0.6 m from the centre of the dipole, situated on the line making an angle of 60° with the dipole axis is:

$$\left(\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \mathrm{Nm}^2/\mathrm{C}^2\right)$$

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Option 1:
50 V

Option 2:
200 V

Option 3:
400 V

Option 4:
zero

Correct Answer:
200 V

Solution:

$$V = \frac{KP\cos\theta}{r^2} = \frac{1}{4\pi\epsilon_0} \times \frac{P\cos\theta}{r^2}$$

$$= \frac{9 \times 10^9 \times 16 \times 10^{-9} \times \cos 60^\circ}{36 \times 10^{-2}}$$

$$= \frac{9 \times 16}{36} \times 10^2 \times \frac{1}{2}$$

$$= 200\text{V}$$

Hence the answer is the Option (2).

Q. 12 A body weighs 72 N on the surface of the earth. What is the gravitational force on it, at a height equal to half of the radius of the earth?

Option 1:
48 N

Option 2:
32 N

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Option 3:
30 N

Option 4:
24 N

Correct Answer:
32 N

Solution:

Weight of body on the surface of the earth $= mg = 72N$

Acceleration due to gravity at height h is

$$g_h = \frac{gR_E^2}{(R_E+h)^2} = \frac{gR_E^2}{\left(R_E+\frac{R_E}{2}\right)^2} = \frac{4}{9}g \left(\because h = \frac{R_E}{2}\right)$$

Weight of body at height $his = mg_h = m \times \frac{4}{9}g = \frac{4}{9} \times mg = \frac{4}{9} \times 72N = 32N$

Hence the answer is the Option (2).

Q. 13 The solids which have the negative temperature coefficient of resistance are:

Option 1:
metals

Option 2:
insulators only

Option 3:
semiconductors only

Option 4:
insulators and semiconductors only.

Correct Answer:
insulators and semiconductors only.

Solution:

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A negative coefficient for a material means that its resistance decreases with an increase in temperature. The value of the temperature coefficient of resistance in semiconductors and insulators is negative.

Hence the answer is Option (4).

Q. 14 Light of frequency 1.5 times the threshold frequency is incident on a photosensitive material. What will be the photoelectric current if the frequency is halved and the intensity is doubled?

Option 1:
doubled

Option 2:
four times

Option 3:
one-fourth

Option 4:
zero

Correct Answer:
zero

Solution:
Frequency of light is 1.5 times of threshold frequency.
Now frequency is made half.
So, new frequency will be $\frac{1.5f_T}{2} = 0.75f_T$
Since, this frequency is less than threshold frequency So, there will not be ejection of electrons from the metal surface.
So, there will not be any current.

Hence the answer is option (4).

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Q. 15 A series of LCR circuits is connected to an AC voltage source. When L is removed from the circuit, the phase difference between current and voltage is $\frac{\pi}{3}$ If instead C is removed from the circuit, the phase difference is again $\frac{\pi}{3}$ between current and voltage. The power factor of the circuit is:

Option 1:
zero

Option 2:
0.5

Option 3:
1.0

Option 4:
-1.0

Correct Answer:
1.0

Solution:

$$\text{For only RC}: \phi_{RC} = \frac{\pi}{3}$$

$$\tan\left(\frac{\pi}{3}\right) = \frac{X_c}{R} \Rightarrow X_c = \sqrt{3}R$$

$$\text{For only RL}: \phi_{RL} = \frac{\pi}{3}$$

$$\tan\left(\frac{\pi}{3}\right) = \frac{X_L}{R} \Rightarrow X_L = \sqrt{3}R$$

For RL:-

$$\tan(\phi) = \frac{X_L - X_C}{R} = \frac{\sqrt{3}R - \sqrt{3}R}{R} = 0 \Rightarrow \phi = 0 \Rightarrow \cos\phi = 1$$

Hence the answer is option (3).

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Q. 16 A sphere conductor of radius 10 cm has a charge of $3.2 \times 10^{-7} C$ distributed uniformly. What is the magnitude of the electric field at a point 15 cm from the center of the sphere?

$\left(\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \mathrm{Nm}^2/\mathrm{C}^2\right)$

Option 1:
$1.28 \times 10^8 N/C$

Option 2:
$1.28 \times 10^5 N/C$

Option 3:
$1.28 \times 10^6 N/C$

Option 4:
$1.28 \times 10^7 N/C$

Correct Answer:
$1.28 \times 10^5 N/C$

Solution:
R=10 cm=0.1 m

$q = 3.2 * 10^{-7}$

r=15 cm =0.15 cm

here r>R

So $E = \frac{q}{4\pi r^2} = \frac{9 * 10^9 * 3.2 * 10^{-7}}{(0.15)^2} = 128000 = 1.28 \times 10^5 \ N/C$

Hence the answer is Option (2).

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Q. 17 Find the torque about the origin when a force of $3\hat{j}N$ acts on a particle whose position vector is $2\hat{k}m$

Option 1:
$6\hat{i}\ Nm$

Option 2:
$6\hat{j}\ Nm$

Option 3:
$-6\hat{i}\ Nm$

Option 4:
$6\hat{k}\ Nm$

Correct Answer:
$-6\hat{i}\ Nm$

Solution:

$\vec{F} = 3\hat{j}$

$\vec{r} = 2\hat{k}$

$\tau = \vec{r} \times \vec{F} = 2\hat{k} \times 3\hat{j} = -6\hat{i}$

Hence the answer is Option (3)

Q. 18 A charged particle having drift velocity of $7.5 \times 10^{-4} m\ s^{-1}$ in an electric field of $3 \times 10^{-10} V m^{-1}$ has mobility in $m^2 V^{-1} s^{-1}$ of:

Option 1:
2.25×10^{15}

Option 2:
2.5×10^{6}

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Option 3:

2.5×10^{-6}

Option 4:

2.25×10^{-15}

Correct Answer:

2.5×10^{6}

Solution:

$\mathrm{V_d} = 7.5 \times 10^{-4}$

$E = 3 \times 10^{-10}$

$$\text{Mobility of charged particle } = \mu = \frac{|\mathbf{v}_\mathrm{d}|}{\mathrm{E}} = \frac{7.5 \times 10^{-4}}{3 \times 10^{-10}} = 2.5 \times 10^{6} \mathrm{m^2 V^{-1} s^{-1}}$$

Hence the answer is Option (2).

Q. 19 A ray is incident at an angle of incidence *i* on one surface of a small angle prism (with the angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ then the angle of incidence is nearly equal to:

Option 1:

$\frac{A}{2\mu}$

Option 2:

$\frac{2A}{\mu}$

Option 3:

μA

Option 4:

$\frac{\mu A}{2}$

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Correct Answer:

μA

Solution:

The angle of incidence =i

The angle of the prism =A

The angle of emergence =e

the refractive index of the material of the prism is μ

For a small angle prism, the angle of deviation, $\delta = (\mu - 1)\mathrm{A}$

but we have $\delta = i + e - A$

Using e=0

we get $\delta = i - A$

on comparing we get

$$(\mathrm{i} - \mathrm{A}) = (\mu - 1)\mathrm{A}$$
$$\Rightarrow \mathrm{i} = \mu\mathrm{A}$$

Hence the answer is Option (3).

Q. 20 The quantities of heat required to raise the temperature of two solid copper spheres of radii r_1 and r_2 ($r_1 = 1.5r_2$) through 1K are in the ratio:

Option 1:

$\frac{27}{8}$

Option 2:

$\frac{9}{4}$

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Option 3:
$\frac{3}{2}$

Option 4:
$\frac{5}{3}$

Correct Answer:
$\frac{27}{8}$

Solution:

$$Q = mc\Delta T = \rho V s\Delta T = \rho(\frac{4}{3}\pi r^3)s\Delta T$$

Since $s, \rho, \Delta T$ is the same for both

So $Q\ \alpha\ r^3$

So $\frac{Q_1}{Q_2} = (\frac{r_1}{r_2})^3 = (\frac{1.5r_2}{r_2})^3 = (\frac{3}{2})^3 = \frac{27}{8}$

Hence, the answer is the option (1).

Q. 21 When the uranium isotope $^{235}_{92}U$ is bombarded with a neutron, it generates $^{89}_{36}Kr$, three neutrons and :

Option 1:
$^{144}_{56}Ba$

Option 2:
$^{91}_{40}Zr$

Option 3:
$^{101}_{36}Kr$

Option 4:
$^{103}_{36}Kr$

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Correct Answer:

$^{144}_{56}Ba$

Solution:

Let the unknown element is $_{A}\mathrm{X}^{Z}$

$$_{92}\mathrm{U}^{235} + {}_{0}\mathrm{n}^{1} \rightarrow {}_{36}\mathrm{Ba}^{89} + {}_{A}\mathrm{X}^{Z} + 3\ {}_{0}\mathrm{n}^{1}$$

The sum of the atomic number on L.H.S = 92

The sum of the atomic number on R.H.S= 92 = 36+A

On comparing

A=56

Similarly

The sum of the mass number on L.H.S= 235+1

The sum of the atomic number on R.H.S= 89+3+Z

On comparing

235+1=89+3+Z

So, Z=144

So the known element is $^{144}_{56}Ba$

Hence, the answer is the option (1).

Q. 22 The phase difference between displacement and acceleration of a particle in a simple harmonic motion is:

Option 1:

π rad

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Option 2:

$\frac{3\pi}{2}\ \text{rad}$

Option 3:

$\frac{\pi}{2}\ \text{rad}$

Option 4:

zero

Correct Answer:

$\pi\ \text{rad}$

Solution:

Displacement of the particle executing SHM

Acceleration of the particle — a i b

Thus phase difference between displacement and acceleration of the particle is radian.

Q. 23 A resistance wire connected in the left gap of a metre bridge balances a 10Ω resistance in the right gap at a point which divides the bridge wire in the ratio 3:2. If the length of the resistance wire is 1.5 m, then the length of 1Ω of the resistance wire is:

Option 1:

$1.0 \times 10^{-2} m$

Option 2:

$1.0 \times 10^{-1} m$

Option 3:

$1.5 \times 10^{-1} m$

Option 4:

$1.5 \times 10^{-2} m$

Correct Answer:

$1.0 \times 10^{-1} m$

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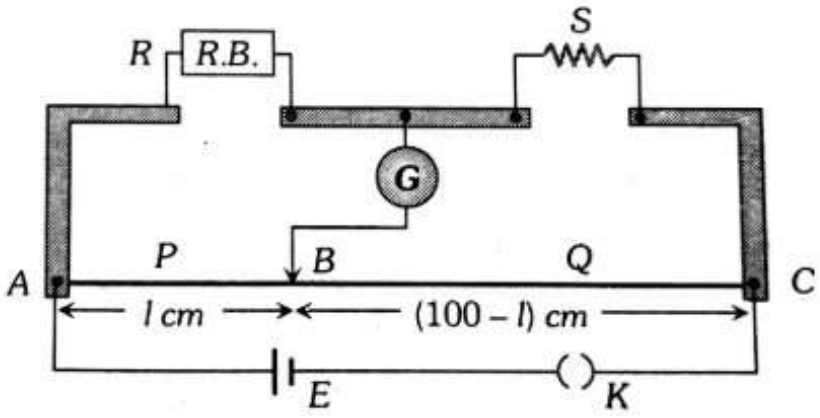
Solution:

For a meter bridge



$$\frac{P}{Q} = \frac{R}{S} = \frac{l}{100 - l}$$

From question

$$\frac{P}{Q} = \frac{3}{2}$$

$and\ S = 10\Omega$

$$So\ \frac{R}{S} = \frac{3}{2}$$

$$R = \frac{3}{2} * 10 = 15\Omega$$

I.e For 1.5 m long wire its total resistance is 15Ω

So using the unitary method

The length of 1Ω resistance wire will be = $L = \frac{1.5}{15} = 0.1 = 1 \times 10^{-1}\ m$

Q. 24 A capillary of radius r is immersed in water and water rises in it to height h. The mass of the water in the capillary is 5 g. Another capillary tube of radius 2r is immersed in water. The mass of water that will rise in this tube is:

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Option 1:
2.5 g

Option 2:
5.0 g

Option 3:
10.0 g

Option 4:
20.0 g

Correct Answer:
10.0 g

Solution:
Mass of water in the capillary tube = $m = \rho Ah$

and using

$$h = \frac{2T\cos\Theta}{\rho g r}$$

we get

$$m = \rho Ah = \rho \times (\pi r^2) \times \frac{2T\cos\Theta}{\rho g r} \Rightarrow m\ \alpha\ r$$

So ,

$$\frac{m_2}{m_1} = \frac{r_2}{r_1} = \frac{2r}{r} = 2$$

$$m_2 = 2 \times m_1 = 2 \times 5 = 10\ g$$

Hence, the answer is the option (3).

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Q. 25 The ratio of contributions made by the electric field and magnetic field components to the intensity of the electromagnetic wave is: (c = speed of electromagnetic wave)

Option 1:
C : 1

Option 2:
1 : 1

Option 3:
1 : c

Option 4:
1 : c^2

Correct Answer:
1 : 1

Solution:
The intensity of the EM wave is :

$$I = u_{av}C$$

For electric field

$$(u_{\text{av}})_E = \frac{1}{4}\varepsilon_0 E_0^2$$

and for the magnetic field,

$$(u_{\text{av}})_B = \frac{1}{4\mu_0} B_0^2$$

$$\text{Now, as we know,} \quad E_0 = cB_0 \text{ and } \mu_0\varepsilon_0 = \frac{1}{c^2} \text{ so that, } \mu_0 = \frac{1}{\varepsilon_0 c^2} \text{ and,}$$

So,

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$$\frac{1}{4\mu_0}B_0^2 = \frac{\varepsilon_0 c^2}{4}\left(\frac{E_0}{c}\right)^2 = \frac{1}{4}\varepsilon_0 E_0^2$$

So ,

$$\frac{I_E}{I_B} = 1$$

Hence, the answer is the option (2).

Q. 26 In Young's double-slit experiment, if the separation between coherent sources is halved and the distance of the screen from the coherent courses is doubled, the fringe width becomes -

Option 1:
double

Option 2:
half

Option 3:
four times

Option 4:
one-fourth

Correct Answer:
four times

Solution:

The fringe width, $\beta = \frac{\lambda D}{d}$

Where, $\lambda =$ the wavelength of light

d= distance of slit

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D= distance of the screen from the slit.
According to the question,

$$d' = \frac{d}{2}$$
$$D = 2D$$
So $\beta' = \frac{\lambda D'}{d'} = \frac{\lambda * 2D}{d/2} = 4\beta$
So, the fringe width will be quadrupled.

Hence, the answer is the option (3).

Q. 27 A long solenoid of 50 cm in length having 100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is:

$(\mu_0 = 4\pi \times 10^{-7} T m A^{-1})$

Option 1:
$6.28 \times 10^{-4} T$

Option 2:
$3.14 \times 10^{-4} T$

Option 3:
$6.28 \times 10^{-5} T$

Option 4:
$3.14 \times 10^{-5} T$

Correct Answer:
$6.28 \times 10^{-4} T$

Solution:
Given: $L = 50\text{cm} = 0.5\text{m} \quad i = 2.5\text{A} \quad N = 100 \text{ turns}$
So, number of turns per unit length $n = \frac{N}{L} = \frac{100}{0.5} = 200\text{turns/m}$
Magnetic field at the centre of the solenoid $B = \mu_o ni = 4\pi \times 10^{-7} \times 200 \times 2.5 = 6.28 \times 10^{-4}\text{T}$

Hence, the answer is the option (1).

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Q. 28 A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is: $(g = 10m/s)$

Option 1:
360 m

Option 2:
340 m

Option 3:
320 m

Option 4:
300 m

Correct Answer:
300 m

Solution:

$$v^2 - u^2 = 2as$$
$$(80)^2 - (20)^2 = 2 \times 10 \times h$$
$$h = 300m$$

Hence, the answer is the option (4).

Q. 29 For which of the following Bohr model is not valid?

Option 1:
Hydrogen atom

Option 2:
singly ionised helium atom (He^+)

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Option 3:
Deuteron atom

Option 4:
singly ionised neon atom (Ne^{+})

Correct Answer:
singly ionised neon atom (Ne^{+})

Solution:
Bohr's theory is applicable to hydrogen-like atoms (single electron system).

But Ne^{+} does not have one electron

So, the Bohr model is not valid for Ne^{+}

Hence, the answer is the option (4).

Q. 30 The average thermal energy for a mono-atomic gas is: (k_B is Boltzmann constant and T, absolute temperature)

Option 1:
$\frac{1}{2}k_BT$

Option 2:
$\frac{3}{2}k_BT$

Option 3:
$\frac{5}{2}k_BT$

Option 4:
$\frac{7}{2}k_BT$

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Correct Answer:

$\frac{3}{2}k_BT$

Solution:

The thermal energy of a system = translational Kinetic energy + rotational kinetic energy + vibrational energy

$$E_{th} = \frac{3}{2}Nk_BT = \frac{3}{2}nRT$$

Hence, the answer is the option (2).

Q. 31 Two particles of mass 5kg and 10kg respectively are attached to the two ends of a rigid rod of length 1m with negligible mass.

The centre of mass of the system from the 5kg particle is nearly at a distance of:

Option 1:

33 cm

Option 2:

50 cm

Option 3:

67 cm

Option 4:

80 cm

Correct Answer:

67 cm

Solution:

The position of COM from m_1 mass is $\frac{m_2 r}{(m_1 + m_2)}$

Here, m_1 = 5 kg

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r = 1 m

m_2 = 10 kg

By putting this value we will get 0.666667 meter

or we can write = 67 cm

Hence, the answer is the option (3).

Q. 32 In a guitar, two strings, A and B made of some material are slightly out of tune and produce beats of frequency 6Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be:

Option 1:
523 Hz

Option 2:
524 Hz

Option 3:
536 Hz

Option 4:
537 Hz

Correct Answer:
524 Hz

Solution:

$Beat\ frequency\ = |f_A - f_B| = 6$

since tension in B is decreased so it's frequency will also decrease

New beat frequency =7 (increased by 1) So B must have smaller frequency

Hence frequency of B $= f_A - 6 = 530 - 6 = 524Hz$

Hence, the answer is the option (2).

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Q. 33 Two cylinders A and B of equal capacity are connected via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is:

Option 1:
isothermal

Option 2:
adiabatic

Option 3:
isochoric

Option 4:
isobaric

Correct Answer:
adiabatic

Solution:
Since the process is very fast and instantaneous or sudden there is no time to do evacuation is almost negligible. So there is no time to heat transfer, so no heat transfer occurs. So, this process is an Adiabatic process.

Hence, the answer is the option (2).

Q. 34 The capacitance of a parallel plate capacitor with air as a medium is $6\mu F$ With the introduction of a dielectric medium, the capacitance becomes $30\mu F$. The permittivity of the medium is:

$$\left(\varepsilon_0 = 8.875 \times 10^{-12} C^2 N^{-1} m^{-2}\right)$$

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Option 1:

$0.44 \times 10^{-12} C^2 N^{-1} m^{-2}$

Option 2:

$1.77 \times 10^{-12} C^2 N^{-1} m^{-2}$

Option 3:

$0.44 \times 10^{-10} C^2 N^{-1} m^{-2}$

Option 4:

$500\ C^2 N^{-1} m^{-2}$

Correct Answer:

$0.44 \times 10^{-10} C^2 N^{-1} m^{-2}$

Solution:

Capacitance of air capacitor $C_0 = \frac{\varepsilon_0 A}{d} = 6\mu F$

When a dielectric of permittivity ε_r and dielectric constant K is introduced between the plates, then capacitance,

$C = \frac{K\varepsilon_0 A}{d} = 30\mu F$

Dividing eq. (ii) by (i),

we get $\frac{C}{C_0} = \frac{\frac{K\varepsilon_0 A}{d}}{\frac{\varepsilon_0 A}{d}} = \frac{30}{6}$

$\Rightarrow K = 5$

$\therefore$ permittivity of the medium

$\varepsilon_0 = \varepsilon_0 K = 8.85 \times 10^{-12} \times 5 = 0.44 \times 10^{-10}$

Hence, the answer is the option (3).

Q. 35 An electron is accelerated from the rest through a potential difference of V volt. If the de Broglie wavelength of the electron is $1.227 \times 10^{-2} nm$ the potential difference is:

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Option 1:
10V

Option 2:
10^2V

Option 3:
10^3V

Option 4:
10^4V

Correct Answer:
10^4V

Solution:

$$\text{By using } \lambda = \frac{12.24}{\sqrt{V}}\ Angstorm$$

By putting the value of wavelength = 1.227 x 10^{-3} Angstorm

We will get the answer = 10000 Volt = 10^4 Volt

Hence, the answer is the option (4).

Q. 36 A wire of length L. area of cross-section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's Modulus is:

Option 1:
$\frac{MgL_1}{AL}$

Option 2:
$\frac{Mg(L_1 - L)}{AL}$

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Option 3:
$\frac{MgL}{AL_1}$

Option 4:
$\frac{MgL}{A(L_1-L)}$

Correct Answer:
$\frac{MgL}{A(L_1-L)}$

Solution:

$$Y=\frac{stress}{strain}=\frac{\frac{Mg}{A}}{\frac{\Delta L}{L}}=\frac{\frac{Mg}{A}}{\frac{L_1-L}{L}}$$

$$Y=\frac{MgL}{A(L_1-L)}$$

Hence, the answer is the option (4).

Q. 37 The Brewsters angle i_b for an interface should be:

Option 1:
$0^\circ < i_b < 30^\circ$

Option 2:
$30^\circ < i_b < 45^\circ$

Option 3:
$45^\circ < i_b < 90^\circ$

Option 4:
$i_b = 90^\circ$

Correct Answer:
$45^\circ < i_b < 90^\circ$

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Solution:

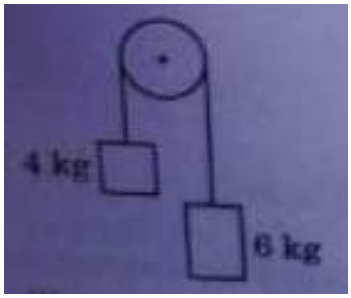
$tan\ i_b = \mu$

Since μ is always greater than 1 and less than infinite

So, $45^o < i_b < 90^o$

Hence, the answer is the option (3).

Q. 38 Two bodies of mass 4kg and 6kg are tied to the ends of a massless string. The string passes over a pulley which is frictionless (see figure). The acceleration of the system in terms of acceleration due to gravity (g) is:



Option 1:

g

Option 2:

$\frac{g}{2}$

Option 3:

$\frac{g}{5}$

Option 4:

$\frac{g}{10}$

Correct Answer:

$\frac{g}{5}$

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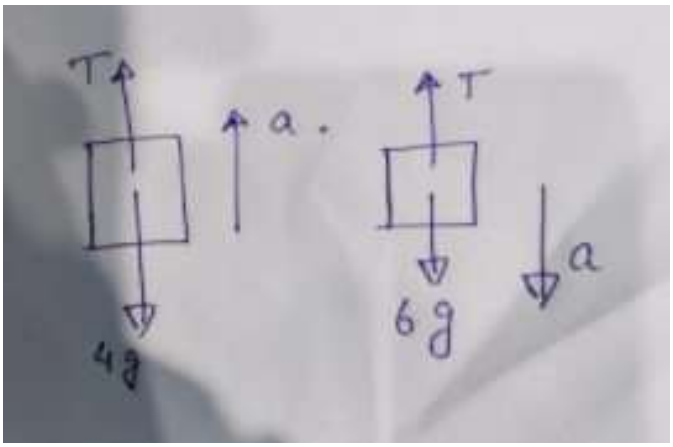
Solution:
For block of 4 kg and 6 kg -



So,

For 4 kg block -

T - 4g = 4a . .....(1)

And for 6 kg block -

6g - T = 6a ..... (2)

By adding (1) and (2)

$a = \frac{g}{5}$

Q. 39 The dimensions of Stress are:

Option 1:
$[MLT^{-2}]$

Option 2:
$[ML^{2}T^{-2}]$

Option 3:
$[ML^{0}T^{-2}]$

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Option 4:
$[ML^{-1}T^{-2}]$

Correct Answer:
$[ML^{-1}T^{-2}]$

Solution:

$$Stress = \frac{Force}{Area} = \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$$

Hence, the answer is the option (4).

Q. 40 A screw gauge has the least count of 0.01 mm and there are 50m divisions in its circular scale. The pitch of the screw gauge is:

Option 1:
0.01mm

Option 2:
0.25 mm

Option 3:
0.5 mm

Option 4:
1.0 mm

Correct Answer:
0.5 mm

Solution:

As, Least count of screw gauge = $\frac{Pitch}{Number\ of\ circular\ division}$

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So, $$0.01 = \frac{Pitch}{50}$$

$$Pitch = 0.5\ mm$$

Hence, the answer is the option (3).

Q. 41 The energy required to break one bond in DNA is 10^{-20}J. This value in eV is nearly:

Option 1:
6

Option 2:
0.6

Option 3:
0.06

Option 4:
0.006

Correct Answer:
0.06

Solution:
As,

$$1eV = 1.6 \times 10^{-19} J$$

$$1J = \frac{1}{1.6 \times 10^{-19}} eV$$

$$So, 10^{-20} J = \frac{10^{-20}}{1.6 \times 10^{-19}} eV = 0.06eV$$

Hence, the answer is the option (3).

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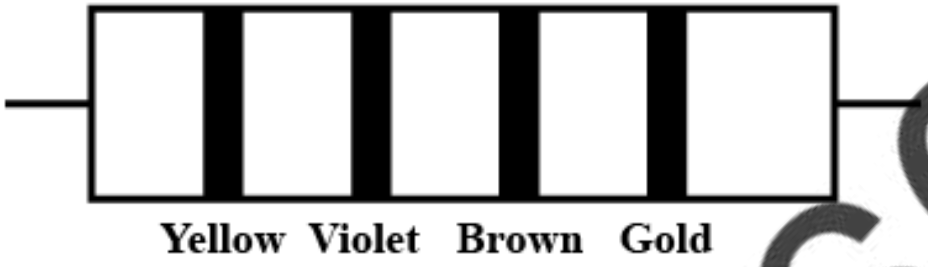
Q. 42 The colour code of resistance is given below:



The value of resistance and tolerance, respectively are:

Option 1:
470k Ω , 5%

Option 2:
47k Ω ,10%

Option 3:
4.7k Ω , 5%

Option 4:
470Ω , 5%

Correct Answer:
470Ω , 5%

Solution:
From the color coding, we know that for -

Yellow = 4

Violet = 7

Brown = Multiplier factor = 10

Gold = Tolerance = 5 %

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So, Resistance = 47 x 10 = 470Ω

And Tolerance = 5 %

Q. 43 Assume that light of wavelength 600nm is coming from a star. The limit of resolution of telescope whose objective has a diameter of 2m is:

Option 1:
$3.66 \times 10^{-7} rad$

Option 2:
$1.83 \times 10^{-7} rad$

Option 3:
$7.32 \times 10^{-7} rad$

Option 4:
$6.00 \times 10^{-7} rad$

Correct Answer:
$3.66 \times 10^{-7} rad$

Solution:

$$\text{Limit of resolution} = \frac{1.22\lambda}{D} = \frac{1.22 \times 600 \times 10^{-9}}{2}$$
$$= 3.66 \times 10^{-7} \text{rad.}$$

Q. 44 The increase in the width of the depletion region in a p-n junction diode is due to:

Option 1:
forward bias only

Option 2:
reverse bias only

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Option 3:
both forward bias reverse bias

Option 4:
increase in forward current

Correct Answer:
reverse bias only

Solution:
In reverse biasing, the positive terminal of the battery is connected to the n-type whereas the negative terminal is connected to the p-type junction. So the positive terminal tends to pull the electrons (near to the depletion layer) in n-type towards itself whereas the negative terminal pulls the holes towards itself which results in increase in the width of depletion layer.

Hence, the answer is the option (2).

Q. 45 The energy equivalent of 0.5g of substance is:

Option 1:
$4.5 \times 10^{16} J$

Option 2:
$4.5 \times 10^{13} J$

Option 3:
$1.5 \times 10^{13} J$

Option 4:
$0.5 \times 10^{13} J$

Correct Answer:
$4.5 \times 10^{13} J$

Solution:

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$$E = mc^2 = \frac{0.5}{1000} \times \left(3 \times 10^8\right)^2$$
$$= 4.5 \times 10^{13} J$$

Hence, the answer is the option (2).

Chemistry

Q. 1 The number of protons, neutrons, and electrons $^{175}_{71}\mathrm{Lu}$ respectively are:

Option 1:
175, 104, and 71

Option 2:
71, 104 and 71

Option 3:
104, 71 and 71

Option 4:
71, 71 and 104

Correct Answer:
71, 104 and 71

Solution:
The atomic number is equal to the number of protons or electrons. Thus, the number of protons and electrons is 71. Now, the atomic mass = number of protons + number of neutrons.
Thus, number of neutrons = 175 - 71 = 104

Hence, the answer is the option (2).

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Q. 2 The following metal ion activates many enzymes, participates in the oxidation of glucose to produce ATP and with Na, is responsible for the transmission of nerve signals.

Option 1:
Potassium

Option 2:
Iron

Option 3:
Copper

Option 4:
Calcium

Correct Answer:
Potassium

Solution:
The transportation of potassium ions inside and outside the cell is very crucial for cardiovascular and nerve functions. When potassium ions enter the cell, the sodium-potassium pump activates which creates electrical potential and allows the nerve signals to transmit.

Therefore, **Option(1) is correct.**

Q. 3 Which of the following is not correct about carbon monoxide?

Option 1:
It is produced due to incomplete combustion.

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Option 2:
it forms carboxyhemoglobin

Option 3:
It reduces the oxygen-carrying ability of blood

Option 4:
The carboxyhemoglobin (hemoglobin bound to CO is less stable than oxyhemoglobin).

Correct Answer:
The carboxyhemoglobin (hemoglobin bound to CO is less stable than oxyhemoglobin).

Solution:
Carbon monoxide is formed by incomplete combustion of carbon.

$$2C + O_2 \rightarrow 2CO$$

carbon monoxide forms a complex with hemoglobin which is much stronger than oxygen-hemoglobin, also carbon monoxide can substitute oxygen and reduce oxygen carrying blood.

Carbon monoxide binds with Hb to form a 300 times more stable compound than the oxyhemoglobin complex.

Therefore, **Option(4) is correct.**

Q. 4 Which one of the following has a maximum number of atoms?

Option 1:
1g of Li(s) [Atomic mass of Li = 7]

Option 2:
1g of Ag(s) [Atomic mass of Ag = 108]

Option 3:
1g of Mg(s) [Atomic mass of Mg = 24]

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Option 4:
1g of O_2(g) [Atomic mass of O = 16]

Correct Answer:
1g of Li(s) [Atomic mass of Li = 7]

Solution:
For Li:

Number of atoms in 1 g $\mathrm{Li} = \frac{\mathrm{N_A}}{7}$

For Ag:

Number of atoms in 1 g $\mathrm{Ag} = \frac{\mathrm{N_A}}{108}$

For Mg:

Number of atoms in 1 g $\mathrm{Mg} = \frac{\mathrm{N_A}}{24}$

For O_2:

Number of atoms in 1 g $\mathrm{O_2} = \frac{(2 \times \mathrm{N_A})}{32} = \frac{\mathrm{N_A}}{16}$

Thus, the number of atoms in 1 g of Li is the highest among these.

Hence, the answer is the option (1).

Q. 5 Paper chromatography is an example of:

Option 1:
Column chromatography

Option 2:
Adsorption chromatography

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Option 3:
Partition chromatography

Option 4:
Thin layer Chromatography

Correct Answer:
Partition chromatography

Solution:
Paper chromatography is an example of partition chromatography because in paper chromatography the components are separated based on the difference in solubility of sample components in the stationary or mobile phase while in adsorption chromatography separation of components is done based on the rate of adsorption of components on the stationary phase. Thin-layer and column chromatographies are adsorption chromatography.

Hence, the answer is the option (3).

Q. 6 Which of the following in a natural polymer?

Option 1:
Poly (butadiene-acrylonitrile)

Option 2:
cis-1,4-polyisoprene

Option 3:
poly (Butadiene-styrene)

Option 4:
Polybutadiene

Correct Answer:
cis-1,4-polyisoprene

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Solution:
cis-1,4-polyisoprene is natural rubber polymer.

Therefore, **Option(2) is correct.**

Q. 7 The mixture which shows positive deviation from Raoult's Law is:

Option 1:
Chloroethene + Bromoethene

Option 2:
Ethanol + Acetone

Option 3:
benzene + Toluene

Option 4:
Acetone + Cloroform

Correct Answer:
Ethanol + Acetone

Solution:
A mixture of ethanol and acetone shows a positive deviation from Raoult's Law. Pure ethanol possesses hydrogen bonding. The introduction of acetone between the molecules of ethanol results in the breaking of some of these hydrogen bonds. Due to the weakening of interactions, the solution shows a positive deviation from Raoult's law.

Hence, the answer is the option (2).

Q. 8 The calculated spin-only magnetic moment of Cr^{2+} the ion is :

Option 1:
2.84 BM

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Option 2:
3.87 BM

Option 3:
4.90 BM

Option 4:
5.92 BM

Correct Answer:
4.90 BM

Solution:
The calculated spin-only magnetic moment is given as:

$$\mu = \sqrt{n(n+2)}$$

$$Cr^{2+} \rightarrow 3d^4 4s^0$$

So, for Cr^{2+}, there are 4 unpaired electrons.

Thus, the spin-only magnetic moment is:

$$\mu = \sqrt{4(4+2)}$$

$$\mu = 4.90$$

Hence, the answer is the option (3).

Q. 9 A tertiary butyl carbocation is more stable than a secondary butyl carbocation because of which of the following?

Option 1:
Hyperconjugation

Option 2:
-I effect of -CH_3 groups

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Option 3:
+R effect of CH_3 groups

Option 4:
-R effect of -CH_3 groups

Correct Answer:
Hyperconjugation

Solution:

is Tert butyl Carbocation

It has 9-α hydrogens and thus it is stabilized through hyperconjugation.

Therefore, **Option(1) is correct.**

Q. 10 The correct option for free expansion of an ideal gas under adiabatic conditions is:

Option 1:
$q > 0, \Delta T > 0$ and $w > 0$

Option 2:
$q = 0, \Delta T = 0$ and $w = 0$

Option 3:
$q = 0, \Delta T < 0$ and $w > 0$

Option 4:
$q < 0, \Delta T = 0$ and $w = 0$

Correct Answer:
$q = 0, \Delta T = 0$ and $w = 0$

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Solution:
For an adiabatic process, $q = 0$ and free expansion, $w = 0$, Therefore $\Delta T = 0$.

The temperature during adiabatic expansion does not change.

$$\Delta U - q + w = 0 + 0 = 0$$
$$\Delta U = 0 = C_v dT$$
$$Now, C_v \neq 0 \quad so\ \Delta T = 0$$

Hence, the answer is the option (2).

Q. 11 Elimination reaction of 2-bromo-pentane to form pent-2-ene is:

(a) β -elemination reaction

(b) Follows Zaitsev's rule

(c) Dehydrohalogenation reaction

(d) Dehydration reaction

Option 1:
(a) (b) (d)

Option 2:
(a) (b) (c)

Option 3:
(a) (c) (d)

Option 4:
(b) (c) (d)

Correct Answer:
(a) (b) (c)

Solution:

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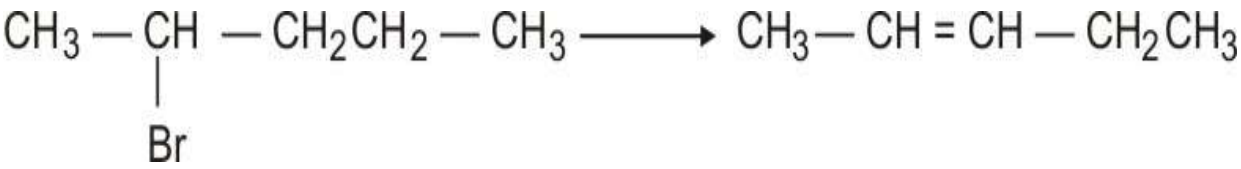
The reaction occurs as follows:



It is an example of an elimination reaction which follows Saytzeff rule.

Q. 12 Identify the correct statements from the following:

(a) $CO_2(g)$ is used as refrigerent for ice-cream and frozen food.

(b) The structure of C_{60} contains twelve six carbon rings and 25 carbon rings.

(c) ZSM- 5, a type of zeolite, is used to convert alcohols into gasoline.

(d) CO is colourless and odourless gas.

Option 1:
(c) and (d) only

Option 2:
(a), (b) and (c) only

Option 3:
(a) and (c) only

Option 4:
(b) and (c) only

Correct Answer:
(c) and (d) only

Solution:
CO_2 is not used as a refrigerant for ice creams and frozen food because high working pressure is required for its cycles but dry ice is used to keep frozen food and ice-creams.

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C_{60} has twelve 5-member rings and twenty 6-member rings.

$ZSM-5$ is a type of zeolite which is used to convert alcohols into gasoline.

CO is Colorless and odorless gas.

Therefore, **Option(1) is correct.**

Q. 13 Which of the following is the correct order of increasing the field strength of ligands to form coordination compounds?

Option 1:
$CN^- < C_2O_4^{2-} < SCN < F^-$

Option 2:
$SCN < F^- < C_2O_4^{2-} < CN^-$

Option 3:
$SCN < F^- < CN^- < C_2O_4^{2-}$

Option 4:
$F^- < SCN < C_2O_4^{2-} < CN^-$

Correct Answer:
$SCN < F^- < C_2O_4^{2-} < CN^-$

Solution:
The correct order of increasing field strength of ligands to form coordination Compounds -

$$SCN^- < F^- < C_2O_4^{-2} < CN^-$$

Hence, the answer is the option (2).

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Q. 14 Hydrolysis of sucrose is given by the following reaction:

$$Sucrose + H_2O \rightleftharpoons Glucose + Fructose$$

If the equilibrium constant (K_c) is 2×10^{13} at 300K, the value $\Delta_r G^{\Theta}$ at the same temperature will be:

Option 1:
$-8.314 Jmol^{-1}K^{-1} \times 300K \times ln(4 \times 10^{13})$

Option 2:
$-8.314 Jmol^{-1}K^{-1} \times 300K \times ln(2 \times 10^{13})$

Option 3:
$8.314 Jmol^{-1}K^{-1} \times 300K \times ln(2 \times 10^{13})$

Option 4:
$8.314 Jmol^{-1}K^{-1} \times 300K \times ln(3 \times 10^{13})$

Correct Answer:
$-8.314 Jmol^{-1}K^{-1} \times 300K \times ln(2 \times 10^{13})$

Solution:
The chemical reaction occurs as follows:

$$\text{Sucrose} + \text{H}_2\text{O} \rightleftharpoons \text{Glucose} + \text{fructose}$$

$$\text{k}_\text{c} = 2 \times 10^{13} \text{ at } 300\text{K}$$

Now, $\Delta_r G^{\circ}$ is given as :

$$\Delta_\text{r}\text{G}^0 = -\text{RT} \ln \text{K}_\text{c}$$

$$\Delta_r G^0 = -8.31 \times 300 \ln \left(2 \times 10^{13}\right)$$

Hence, the answer is the option (2).

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Q. 15 Identify the incorrect match:

Name	IUPAC official name
(a) Unnilunium	(i) Mendelevium
(b) Unniltrium	(ii) Lawrencium
(c) Unnilhexium	(iii) Seaborgium
(d) Ununnnium	(iv) Darmstaditum

Option 1:
(d), (iv)

Option 2:
(a), (i)

Option 3:
(b), (ii)

Option 4:
(c), (iv)

Correct Answer:
(d), (iv)

Solution:
Ununnunium is Roentgenium.

Hence, the answer is the option (1).

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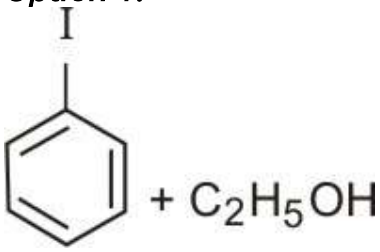
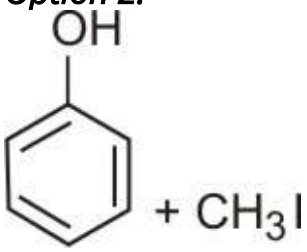
Q. 16 Anisole on cleavage with HI gives:

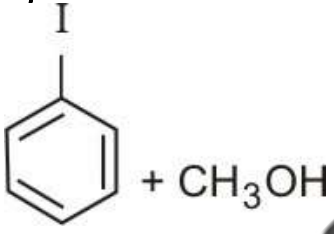
Option 1:



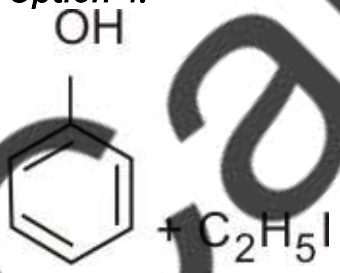
Option 2:



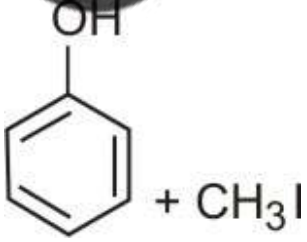
Option 3:



Option 4:



Correct Answer:



Solution:

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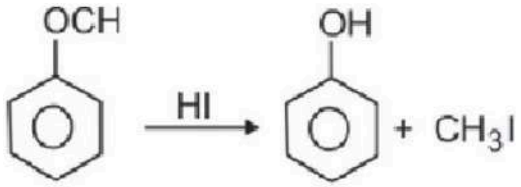

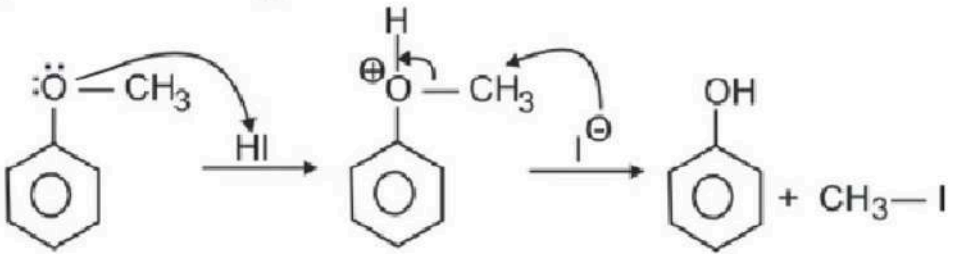

Therefore, **Option(2) is correct.**

Q. 17 Identify the correct statement from the following:

Option 1:
Pig iron can be modulated into a variety of shapes.

Option 2:
Wrought iron is impure iron with 4% carbon.

Option 3:
Bilster copper has bilstered apperance due to evolution of CO_2

Option 4:
vapour phase refining is carried out for Nickel by Van Arkel method.

Correct Answer:
Pig iron can be modulated into a variety of shapes.

Solution:
Pig iron can be molded in a variety of shapes.

Wrought iron is the purest form of iron among pig, cast, and wrought, usually having carbon less than 0.1%.

The blistered appearance of copper is due to the evolution of SO_2 gas.

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Vapor phase refining of Ni is done by Mond's process.

Therefore, **Option(1) is correct.**

Q. 18 For reaction, $2Cl(g) \rightarrow Cl_2(g)$ the correct option is:

Option 1:
$\Delta_r H < 0$ and $\Delta_r S < 0$

Option 2:
$\Delta_r H > 0$ and $\Delta_r S > 0$

Option 3:
$\Delta_r H > 0$ and $\Delta_r S < 0$

Option 4:
$\Delta_r H < 0$ and $\Delta_r S > 0$

Correct Answer:
$\Delta_r H < 0$ and $\Delta_r S < 0$

Solution:
The reaction occurs as follows: $2\mathrm{Cl(g)} \longrightarrow \mathrm{Cl_2(g)}$

Since the bond formation is taking place between two Chlorine atoms, an exothermic process change in enthalpy will be negative.

On the other hand change in entropy will also be negative as the number of particles decreases.

$\mathrm{Thus,\ \Delta_rH < 0\ and\ \Delta_rS < 0}$

Hence, the answer is the option (1).

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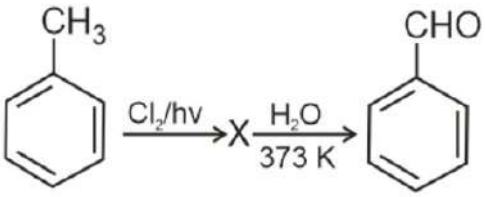
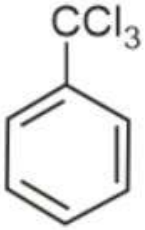
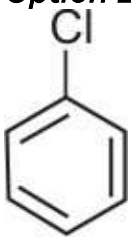
Q. 19 Identify compound X in the following sequence of reactions:



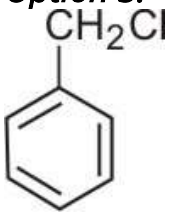
Option 1:



Option 2:

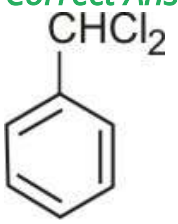


Option 3:



Option 4:

Correct Answer:



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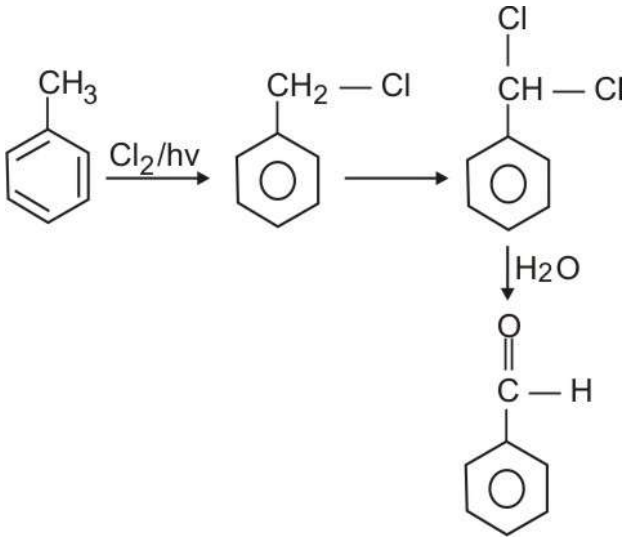
Solution:



Therefore, **Option (4) is correct.**

Q. 20 Match the following :

Oxide	Nature
(a) CO	(i) Basic
(b) BaO	(ii) Neutral
(c) Al_2O_3	(iii) Acidic
(d) Cl_2O_7	(iv) Amphoteric

Which of the following is correct option ?

Option 1:

(a)	**(b)**	**(c)**	**(d)**
(iv)	(iii)	(ii)	(i)

Option 2:

(a)	**(b)**	**(c)**	**(d)**
(i)	(ii)	(iii)	(iv)

Option 3:

(a)	**(b)**	**(c)**	**(d)**
(ii)	(i)	(iv)	(iii)

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Option 4:

(a)	(b)	(c)	(d)
(iii)	(iv)	(i)	(ii)

Correct Answer:

(a)	(b)	(c)	(d)
(ii)	(i)	(iv)	(iii)

Solution:

CO is neutral in nature.

BaO is basic in nature.

Al_2O_3 is amphoteric in nature.

Cl_2O_7 is acidic in nature.

Thus, (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

Therefore, **Option(3) is correct.**

Q. 21 Urea reacts with water to form A which will decompose to form B. B when passed through $Cu^{2+}(aq)$ deep blue color solution C is formed. What is the formula of C from the following?

Option 1:

$CuCO_3.Cu(OH)_2$

Option 2:

$CuSO_4$

Option 3:

$[Cu(NH_3)_4]^{2+}$

Option 4:

$Cu(OH)_2$

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Correct Answer:
$[Cu(NH_3)_4]^{2+}$

Solution:
Urea with water given ammonium hydroxide which dissociates to give ammonia.

$$NH_2CONH_2 + 3H_2O \rightarrow CO_2 + 2NH_4OH$$

$$NH_4OH \rightarrow NH_3 + H_2O$$

Ammonia with $Cu^{2+}(aq)$ gives a deep blue color solution.

$$Cu^{2+}(aq.) + 4NH_3 \rightarrow [Cu(NH_3)_4]^{2+}(aq.)$$

Then C will be $[Cu(NH_3)_4]^{2+}$

Therefore, **the correct option is (3).**

Q. 22 On electrolysis of dilute sulphuric acid using Platinum (Pt) electrode, the product obtained at the anode will be :

Option 1:
$SO_2\ gas$

Option 2:
Hydrogen gas

Option 3:
Oxygen gas

Option 4:
$H_2S\ gas$

Correct Answer:
Oxygen gas

Solution:
On electrolysis of of dilute sulphuric acid,

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At anode - $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

At Cathode - $4H^+ + 4e^- \rightarrow 2H_2$

On electrolysis of dilute sulphuric acid using the Platinum (Pt) electrode, the product obtained at the anode will be oxygen gas while at the cathode it is hydrogen gas.

In the case of Concentrated sulfuric acid, an anode $H_2S_2O_8$ is formed.

Therefore, **the correct option is (3).**

Q. 23 An element has a body-centred cubic (bcc) structure with a cell edge of 288pm. The atomic radius is:

Option 1:

$\frac{4}{\sqrt{2}} \times 288 \text{ pm}$

Option 2:

$\frac{\sqrt{3}}{4} \times 288 \text{ pm}$

Option 3:

$\frac{\sqrt{2}}{4} \times 288 \text{ pm}$

Option 4:

$\frac{4}{\sqrt{3}} \times 288 \text{ pm}$

Correct Answer:

$\frac{\sqrt{3}}{4} \times 288 \text{ pm}$

Solution:

We know the relation between r and a for the BCC unit cell

$4\text{r} = \sqrt{3}\text{a}$

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So,

$$\mathrm{r} = \frac{\sqrt{3}}{4}\mathrm{a}$$

$$\mathrm{r} = \frac{\sqrt{3}}{4} \times 288 \text{ pm}$$

Therefore, the correct option is (2).

Q. 24 Sucrose on hydrolysis gives :

Option 1:
$\alpha - D - Fructose + \beta - D - Fructose$

Option 2:
$\beta - D - Glucose + \alpha - D - Fructose$

Option 3:
$\alpha - D - Glucose + \beta - D - Glucose$

Option 4:
$\alpha - D - Glucose + \beta - D - Fructose$

Correct Answer:
$\alpha - D - Glucose + \beta - D - Fructose$

Solution:
The reaction occurs as follows:

$$\text{Sucrose} \xrightarrow{H_2O} + \beta - \text{D Fructose}$$

Hence, the answer is the option (4).

Q. 25 Which of the following is a basic amino acid?

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Option 1:
Lysine

Option 2:
Serine

Option 3:
Alanine

Option 4:
Tyrosine

Correct Answer:
Lysine

Solution:
Lysine has an extra NH_2 group to make amino acid as a basic amino acid.

Hence, the answer is the option (1).

Q. 26 Which of the following sets of molecules will have zero dipole moment?

Option 1:
Boron trifluriode, beryllium difluoride, carbon dioxide, 1,4- dichlorobenzene

Option 2:
Ammonia, breyllium difluriode ,water, 1,4- dichlorobenzene

Option 3:
Boron trifluoride, hydrogen fluoride, carbon dioxide, 1,3- dichlorobenzene

Option 4:
Nitrogen trifluoride

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Correct Answer:
Boron trifluriode, beryllium difluoride, carbon dioxide, 1,4- dichlorobenzene

Solution:
BF_3, BeF_2, CO_2, 1,4-dichlorobenzene each compound's total dipole moment is zero.

Hence, the answer is the option (1).

Q. 27 The freezing point depression constant $(\mathrm{K_f})$ of benzene is $5.12\ \mathrm{K\ kg\ mol^{-1}}$. The freezing point of depression for the solution of molality 0.078 m containing a non-electrolyte solute in benzene is (rounded off up to two decimal places):

Option 1:
0.60K

Option 2:
0.20K

Option 3:
0.80K

Option 4:
0.40K

Correct Answer:
0.40K

Solution:
The formula for depression in freezing point is (for non-electrolyte solute) -

$$\Delta \mathrm{T_f} = \mathrm{K_f} \times \mathrm{m}$$

$$\Delta \mathrm{T_f} = 5.12 \times 0.078 = 0.399\ \mathrm{K}$$

$$\Delta \mathrm{T_f} \approx 0.40\ \mathrm{K}$$

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Hence, the answer is the option (4).

Q. 28 Which of the following oxoacid of sulphur has $-O-O-$ linkage?

Option 1:
$H_2S_2O_7$ pyrosulphuric acid

Option 2:
H_2SO_3 sulphurous acid

Option 3:
H_2SO_4 sulphuric acid

Option 4:
$H_2S_2O_8$ peroxodisulphuric acid

Correct Answer:
$H_2S_2O_8$ peroxodisulphuric acid

Solution:
To find an oxyacid that has peroxy linkage - If the calculated oxidation state is more than that of the maximum oxidation state, then the oxyacid contains peroxy linkage,

The calculated oxidation state of S in,

$H_2S_2O_7 \quad O.S. = +6$

$H_2SO_3 \quad O.S. = +4$

$H_2SO_4 \quad O.S. = +6$

$H_2S_2O_8 \quad O.S. = +7$

$H_2S_2O_8$, peroxodisulphuric acid has $-O-O-$ a linkage.

Hence, the answer is the option (4).

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Q. 29 HCl was passed through a solution of $CaCl_2, MgCl_2, NaCl$. Which of the following compound(s) crystallize (s)?

Option 1:
$NaCl, MgCl_2 and CaCl_2$

Option 2:
Both $MgCl_2 and CaCl_2$

Option 3:
Only $NaCl$

Option 4:
Only $MgCl_2$

Correct Answer:
Only $NaCl$

Solution:
Solubility of $NaCl$ is less than that of the $CaCl_2 \ and \ MgCl_2$. This is because the hydration enthalpy of $Ca^{2+} \ and \ Mg^{2+}$ is greater than Na^+ due to the smaller size and higher charge. The energy released on the dissolution $CaCl_2 \ and \ MgCl_2$ is higher because the hydration enthalpy compensates for the lattice enthalpy required to break the compound. While in the case of $NaCl$, hydration energy is the least and least soluble. On passing $HCl, NaCl$ is precipitated out first because of the common ion effect.

Hence, the answer is the option (3).

Q. 30 Measuring Zeta potential is useful in determining which property of colloidal solution?

Option 1:
Size of the collodial particles

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Option 2:
Viscosity

Option 3:
solubility

Option 4:
Stability of the collodial particles

Correct Answer:
Stability of the collodial particles

Solution:
Zeta potential is useful in measuring the stability of colloidal particles

Therefore, **Option(4) is correct.**

Q. 31 The rate constant for a first order reaction is $4.606 \times 10^{-3} s^{-1}$ The time required to reduce $2g$ of the reactant to $0.2g$ is:

Option 1:
1000s

Option 2:
100s

Option 3:
200 s

Option 4:
500s

Correct Answer:
500s

Solution:
We are given:

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$K = 4.606 \times 10^{-3} s^{-1}$

Thus, the reaction is of the first order.

Thus, we have:

$$K\,t\; =\; ln\frac{A_o}{A_t}$$

$$4.606\,\mathrm{x}\,10^{-3}\,\mathrm{t}\; =\; 2.303\log\left(\frac{2}{0.2}\right)$$

Thus, $t = 0.5 \times 10^3$ sec

$\mathrm{t} = 500$ sec

Therefore, **Option(4) is correct.**

Q. 32 Which of the following alkanes cannot be made in good yield by the Wurtz reaction?

Option 1:
n- butane

Option 2:
n- Hexane

Option 3:
2,3- Dimethylbutane

Option 4:
n- Hepatne

Correct Answer:
n- Hepatne

Solution:

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Wurtz reaction is a good method to prepare even alkanes.

Hence, the answer is the option (4).

Q. 33 Match the following and identify the correct option.

(a) $CO(g) + H_2(g)$	(i) $Mg(HCO_3)_2 + Ca(HCO_3)_2$
(b) Temporary hardness	(ii) An electron-deficient hydride
(c) B_2H_6	(iii) Synthesis of gas
(d) H_2O_2	(iv) Non - planar structure

Option 1:

(a)	(b)	(c)	(d)
i	iii	ii	iv

Option 2:

(a)	(b)	(c)	(d)
iii	i	ii	iv

Option 3:

(a)	(b)	(c)	(d)
iii	ii	i	iv

Option 4:

(a)	(b)	(c)	(d)
iii	iv	ii	i

Correct Answer:

(a)	(b)	(c)	(d)
iii	i	ii	iv

Solution:

$CO + H_2$ is Synthesis gas ; (a)- (iii)

Temporary hardness is caused due to carbonates and bicarbonates; (b)- (i)

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B_2H_6 is electron deficient molecule; (c)- (ii)

H_2O_2 has a non Planar structure; (d)- (iv)

Therefore, **Option(2) is correct.**

Q. 34 Find out the solubility of $Ni(OH)_2$ in $0.1MNaOH$. Given that the ionic product of $Ni(OH)_2$ is 2×10^{-15}.

Option 1:
$1 \times 10^{8} M$

Option 2:
$2 \times 10^{-13} M$

Option 3:
$2 \times 10^{-8} M$

Option 4:
$1 \times 10^{-13} M$

Correct Answer:
$2 \times 10^{-13} M$

Solution:
The reaction occurs as follows:

$$\underset{(s)}{\mathrm{Ni(OH)_2}} \rightleftharpoons \underset{(s)}{\mathrm{Ni^{2+}}} + \mathrm{2OH^-} \qquad 0.1 + 2s \sim 0.1$$

Thus, $s(0.1)^2 = 2 \times 10^{-15}$

$s = 2 \times 10^{3}$

Hence, the answer is the option (2).

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Q. 35 Identify the molecule which does not exist.

Option 1:
O_2

Option 2:
He_2

Option 3:
Li_2

Option 4:
C_2

Correct Answer:
He_2

Solution:
The electronic configuration of the molecules is:

$O_2 - \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2 \pi^* 2p_x^1 = \pi^* 2p_y^1 \quad BO = \frac{10-6}{2} = 2$

$He_2 - \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \quad BO = \frac{4-4}{2} = 0$

$Li_2 - \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \quad BO = \frac{4-2}{2} = 1$

$C_2 - \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^2 \quad BO = \frac{8-4}{2} = 2$

The bond order of the He_2 molecule is zero and hence it does not exist.

Hence, the answer is the option (2).

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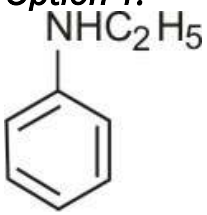
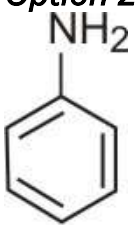
Q. 36 Which of the following amine will give the carylamine test ?

Option 1:

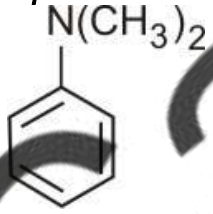


Option 2:

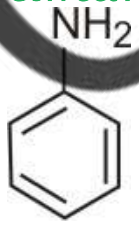


Option 3:

Option 4:



Correct Answer:



Solution:

Carbyamine reaction is used to detect primary amine and phenol.

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Therefore, **Option(2) is correct.**

Q. 37 Identify the incorrect statment.

Option 1:
The oxidation states of chromium in CrO_4^{2-} and $Cr_2O_7^{2}$ are not the same.

Option 2:
$Cr^{2+}(d^4)$ is a stronger reducing agent than $Fe^{2+}(d^6)$ in water.

Option 3:
The transition metals and their compounds are known for the their catalytic activity due to their aility to adopt multiple oxidation states and to form complexes.

Option 4:
Interstitial compounds those that are formed when sma;l atoms like H, C or N are trapped inside the crystal lattices of metals.

Correct Answer:
The oxidation states of chromium in CrO_4^{2-} and $Cr_2O_7^{2}$ are not the same.

Solution:
The oxidation states of Chromium are the same in ${CrO_4}^{2-}$ and ${Cr_2O_7}^{2-}$ and equal to +6. Hence the first statement is incorrect.

Hence, the answer is the option (1).

Q. 38 An increase in the concentration of the reactants of a reaction leads to a change in:

Option 1:
Collision frequency

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Option 2:
Activation energy

Option 3:
heat of reaction

Option 4:
threshold energy

Correct Answer:
Collision frequency

Solution:
An increase in the concentration of the reactants leads to an increase in the collision frequency. Threshold energy, activation energy, and heat of reaction are not affected by the increase in concentration. This is because due to the increase in concentration, the number of particles increases and hence the number of collisions. Whereas the activation energy is affected by the temperature, threshold energy can be changed by using a catalyst or a poison while the enthalpy of the reaction can not be changed by either temperature catalyst, or concentration.

Hence, the answer is the option (1).

Q. 39 A mixture of N_2 and Ar gases in a cylinder contains 7 g of N_2 and 8 g of Ar. If the total pressure of the mixture of the gases in the cylinder is 27 bar, the partial pressure of N_2 is : [Use atomic masses (in g mol^{-1}) $N = 14, Ar = 40$]

Option 1:
18 bar

Option 2:
9 bar

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Option 3:
12 bar

Option 4:
15 bar

Correct Answer:
15 bar

Solution:

Moles of $N_2 = \frac{7g}{28g/mol} = 0.25mol$

Moles of $Ar = \frac{8g}{40g/mol} = 0.2mol$

Mole fraction of N_2 is $\frac{0.25}{0.45} = \frac{5}{9}$

From Dalton's law of partial pressure,

Therefore, Pressure exerted by is N_2 is $\frac{5}{9} \times 27bar = 15bar$

Therefore, **Option(4) is correct.**

Q. 40 The number of Faradays (F) required to produce 20 g of calcium from molten $CaCl_2$ (Atomic mass of $Ca = 40g\ mol^{-1}$) is :

Option 1:
4

Option 2:
1

Option 3:
2

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Option 4:
3

Correct Answer:
1

Solution:
Moles of Ca present in 20 g is 0.5.

$$\mathrm{Ca}^{2+} \xrightarrow{+2e^-} \mathrm{Ca}$$

For 1 mol of Ca need 2 moles of electrons.

So, for 0.5 mole of Ca need 1 mole of electrons.

Hence, 1 mole of electrons means 1 Faraday of charge will be required for the process.

Hence, the answer is the option (2).

Q. 41 Reaction betweeen benzaldehyde and acetophenone in presence of dilute $NaOH$ is known as :

Option 1:
Cross Aldol condensation

Option 2:
Aldol condensation

Option 3:
Cannizzaro's reaction

Option 4:
Cross Cannizzaro's reaction

Correct Answer:
Cross Aldol condensation

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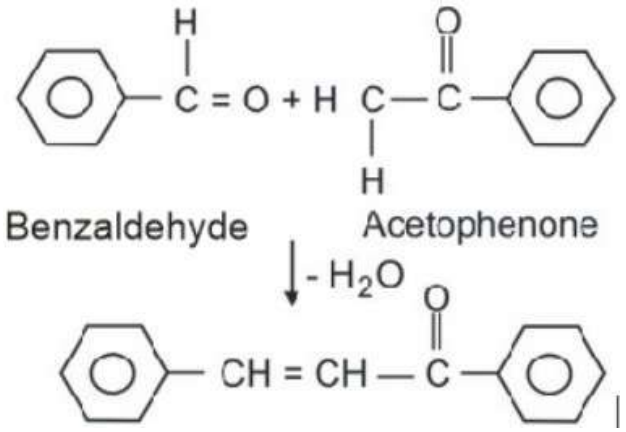
Solution:



Therefore, **Option (1) is correct.**

Q. 42 What is the change in oxidation number of carbon in the following reaction ?

$$CH_4(g) + 4Cl_2(g) \rightarrow CCl_4(l) + 4HCl(g)$$

Option 1:
0 to -4

Option 2:
+4 to +4

Option 3:
0 to +4

Option 4:
-4 to +4

Correct Answer:
-4 to +4

Solution:
In CH_4 the oxidation number of carbon is -4 while in CCl_4 the oxidation number of Carbon is +4.

Hence, the oxidation state changes from -4 to +4.

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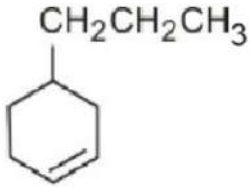
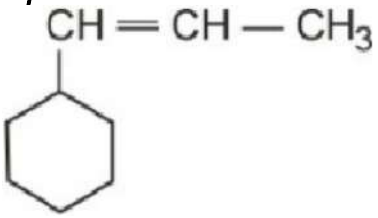
Therefore, **Option(4) is correct.**

Q. 43 An alkene on ozonolysis gives methanal as one of the product. Its structure is:

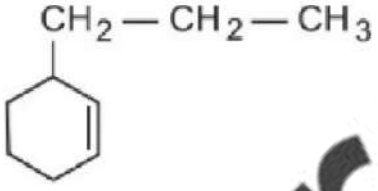
Option 1:



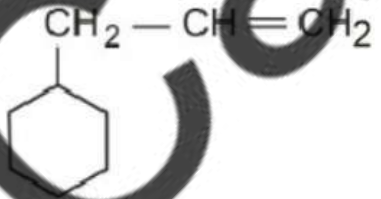
Option 2:



Option 3:



Option 4:



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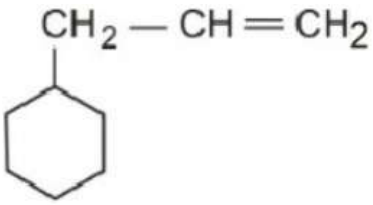
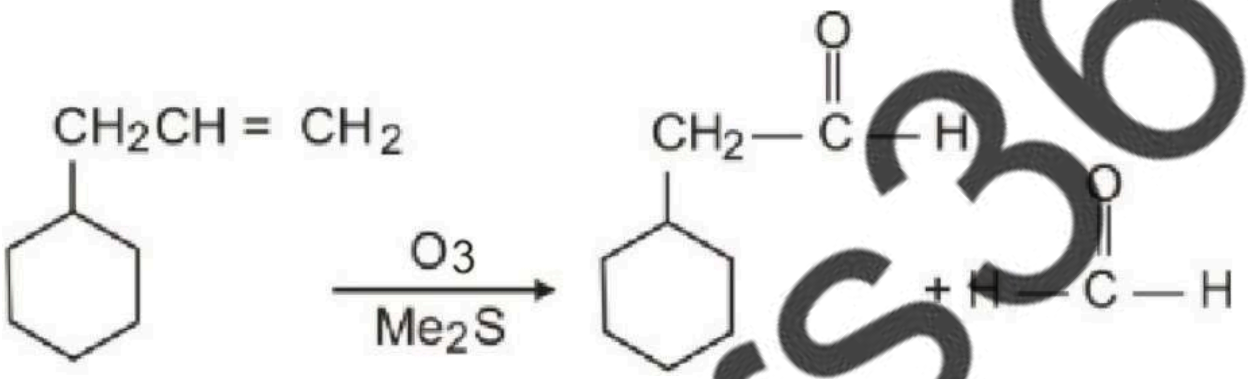
Correct Answer:



Solution:



Therefore, **Option(4) is correct.**

Q. 44 Reaction between acetone and methylmagnesium chloride followed by hydrolysis will give:

Option 1:
isobutyul alcohol

Option 2:
Isopropyl alcohol

Option 3:
Sec. butyl alcohol

Option 4:
tert. butyl alcohol

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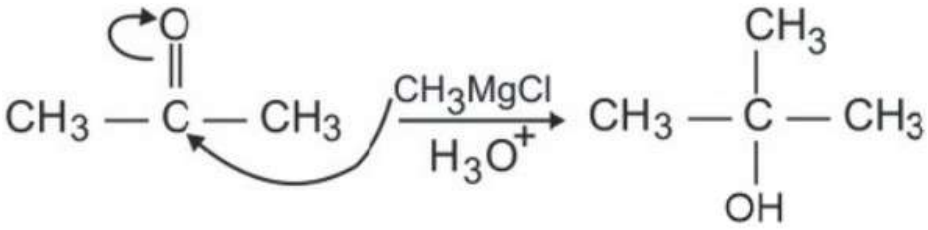
Correct Answer:
tert. butyl alcohol

Solution:



Therefore,

Tertiary Butyl alcohol

Option (4) is correct.

Q. 45 Which of the following is a cationic detergent?

Option 1:
Sodium dodecylbenzene sulphonate

Option 2:
sodium lauryl sulphate

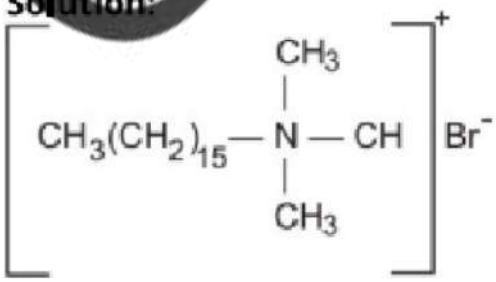
Option 3:
Sodium sterate

Option 4:
Cetyltrimethyl ammonium bromide

Correct Answer:
Cetyltrimethyl ammonium bromide

Solution:

Solution:



This cationic group is involved in cleansing action.

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Therefore, **Option(4) is correct.**

Biology

Q. 1 Which of the following refer to correct example(s) of organisms which have evolved due to changes in the environment brought about by anthropogenic action?

A Darwin's Finches of Galapagos islands.
B Herbicide-resistant weeds
C Drug-resistant eukaryotes
D Man created breeds of domesticated animals like dogs

Option 1:
only (a)

Option 2:
(a) and (c)

Option 3:
(b),(c) and (d)

Option 4:
only (d)

Correct Answer:
(b),(c) and (d)

Solution:

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Anthropogenic actions are due to the direct or indirect effects of human actions on the environment. Resistance due to antibiotics, use of herbicide and artificial breeding of domesticated animals are examples of anthropogenic actions. Hence, the correct option is (c).

Q. 2 Match the following column and select the correct option

Column-1	Column-2
(A) Organ of Corti	(i) Connect middle ear and pharynx
(B) Cochlea	(ii)Coiled part of the labyrinth
(C) Eustachian tube	(iii)Attached to the oval window
(D) Stapes	(iv)Located on the basilar membrane

Option 1:

	(a)	(b)	(c)	(d)
.	(ii)	(iii)	(i)	(iv)

Option 2:

	(a)	(b)	(c)	(d)
2	(iii)	(i)	(iv)	(ii)

Option 3:

	(a)	(b)	(c)	(d)
3	(iv)	(ii)	(i)	(iii)

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Option 4:

	(a)	(b)	(c)	(d)
4	(i)	(ii)	(iv)	(iii)

Correct Answer:

	(a)	(b)	(c)	(d)
3	(iv)	(ii)	(i)	(iii)

Solution:
The correct match is:

The organ of Corti - Located on the basilar membrane

Cochlea - Coiled part of the labyrinth

Eustachian tube - Connects middle ear and pharynx

Stapes - Attached to the oval window

Hence, the correct option is (c).

Q. 3 Identify the wrong statement which reference to immunity

Option 1:
When exposed to antigen (living or dead) antibodies are produced in the host's body.It is called " Active immunity"

Option 2:
When ready made antibodies are directly given it is called "Passive immunity"

Option 3:
Active immunity is quick and gives full response

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Option 4:
Foetus receives some antibodies from mother, it is an example for passive immunity

Correct Answer:
Active immunity is quick and gives full response

Solution:
Active immunity is slow and takes time to give its full effective response. However, the antibody may not be quick but are long-lasting.

Hence, the correct option is (c).

Q. 4 Select the correct events that occur during inspiration

A Contraction of the diaphragm
B Contraction of external intercostal muscles
C Pulmonary volume decreases
D Intrapulmonary pressure increases

Option 1:
(a) and (b)

Option 2:
(c) and (d)

Option 3:
(a) , (b) and (d)

Option 4:
only (d)

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Correct Answer:
(a) and (b)

Solution:
During inspiration, the diaphragm and external intercostal muscles contract, pulmonary volume increases and the intrapulmonary pressure decreases.

Q. 5 The oxygenation activity of RuBisCo enzyme in photorespiration leads to the formation of :

Option 1:
2 molecules of 3- C compound

Option 2:
1 molecule of a 3-C compound

Option 3:
1 molecule of a 6-C compound

Option 4:
1 molecule of a 4-C compound and 1 molecule of a 2-C compound

Correct Answer:
1 molecule of a 3-C compound

Solution:
The RuBisCo by oxygenation activity forms 1 molecule of 3PGA which is a 3-carbon compound.

Q. 6 The infectious stage of Plasmodium that enter the human body is :

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Option 1:
Trophozoites

Option 2:
Sporozoites

Option 3:
Female gametocytes

Option 4:
Male gametocytes

Correct Answer:
Sporozoites

Solution:
Sporozoite is the infection stage of Plasmodium that enters the human body.

Q. 7 Which of the following statements about inclusion bodies is incorrect?

Option 1:
They are not bound by any membrane

Option 2:
These are involved in the ingestion of food particles

Option 3:
They lie free in the cytoplasm

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Option 4:
These represent reserve material in the cytoplasm

Correct Answer:
These are involved in the ingestion of food particles

Solution:
Inclusion bodies are not involved in the ingestion of food particles.

Q. 8 Dissolution of the synaptonemal complex occurs during

Option 1:
Pachytene

Option 2:
Zygotene

Option 3:
Diplotene

Option 4:
Leptotene

Correct Answer:
Diplotene

Solution:
The dissolution of the synaptonemal complex occurs during the diplotene stage.

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Q. 9 Ray florets have

Option 1:
Inferior ovary

Option 2:
Superior ovary

Option 3:
Hypogynous ovary

Option 4:
Half inferior ovary

Correct Answer:
Inferior ovary

Solution:
Ray florets of sunflowers have inferior ovaries.

Q. 10 In ..... separated DNA fragments can be visualized with the help of:

Option 1:
AAcetocarmine in bright blue light

Option 2:
Ethidium bromide in UV radiation

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Option 3:
Acetocarmine in UV radiation

Option 4:
Ethidium bromide in infrared radiation

Correct Answer:
Ethidium bromide in UV radiation

Solution:
In AGE or Agarose Gel Electrophoresis, DNA fragments can be separated and visualized with the help of EtBr in UV radiation. Hence, the correct option is (b).

Q. 11 In which of the following techniques the embryos are transferred to assist those females who cannot conceive?

Option 1:
ZIFT and IUT

Option 2:
GIFT and ZIFT

Option 3:
ICSI and ZIFT

Option 4:
GIFT and ICSI

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Correct Answer:
ZIFT and IUT

Solution:
ZIFT - Zygote Intra-Fallopian tube Transfer is a form of assisted reproductive technology (ART). In this technique, we are going to put the zygote which has started developing into the embryo in the fallopian tube rather than in the uterus and we will wait for it to travel the whole length of the fallopian tube and reach the uterus on its own. On the other hand, IUT (Intrauterine transfer) is a technique in which embryos are placed into the uterus of a female with the intent to establish a pregnancy.

Hence, the correct option is (a).

Q. 12 Select the option including all sexually transmitted diseases

Option 1:
Gonorrhea, Syphilis, Genital herpes

Option 2:
Gonorrhea, Malaria, Genital herpes

Option 3:
AIDS, Malaria, Filaria

Option 4:
Cancer, AIDS, Syphilis

Correct Answer:
Gonorrhea, Syphilis, Genital herpes

Solution:

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Gonorrhoea and Syphilis are sexually transmitted diseases caused by bacteria. Gonorrhea is caused by *Neisseria gonorrhoeae* and Syphilis is caused by the *Treponema pallidum*. On the other hand, Genital Herpes is a sexually transmitted disease caused by the virus called Herpes simplex virus.

Q. 13 Identify the wrong statement with reference to the transport of oxygen

Option 1:
Binding of oxygen with hemoglobin is mainly related to the partial pressure of O_2

Option 2:
The partial pressure of CO_2 can interfere with O_2 binding with hemoglobin

Option 3:
Higher H^+ conc. In alveoli favors the formation of oxyhemoglobin

Option 4:
Low pCO_2 in alveoli favors the formation of oxyhemoglobin

Correct Answer:
Higher H^+ conc. In alveoli favors the formation of oxyhemoglobin

Solution:
In alveoli, all the factors favorable for the formation of oxyhemoglobin are high pO_2, low pCO_2, less H^+ concentration and low temperature.

Hence, the correct option is (c).

Q. 14 Identify the incorrect statement

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Option 1:
Heartwood does not conduct water but gives mechanical support

Option 2:
Sapwood is involved in the conduction of water and minerals from the root to the leaf

Option 3:
Sapwood is the innermost secondary xylem and is lighter in color

Option 4:
Due to the deposition of tannine resins, oil, etc heartwood is dark in color.

Correct Answer:
Sapwood is the innermost secondary xylem and is lighter in color

Solution:
Sapwood is the outermost secondary Xylem. The sapwood may become dark and discolored such as when decaying. Hence, the correct option is (c).

Q. 15 Identify the wrong statement with regards to Restriction Enzymes

Option 1:
Each restriction enzyme functions by inspecting the length of a DNA sequence

Option 2:
They cut the strand of DNA at palindromic sites

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Option 3:
They are useful in genetic engineering

Option 4:
Sticky ends can be joined by using DNA ligases.

Correct Answer:
Sticky ends can be joined by using DNA ligases.

Solution:
They recognize and bind to specific sequences of DNA, called restriction sites. Each restriction enzyme recognizes just one or a few restriction sites. When it finds its target sequence, a restriction enzyme will make a double-stranded cut in the DNA molecule. Hence, the correct option is (d).

Q. 16 Floridian starch has a structure similar to

Option 1:
Starch and cellulose

Option 2:
Amylopectin and glycogen

Option 3:
Mannitol and algin

Option 4:
Laminarin and cellulose

Correct Answer:
Amylopectin and glycogen

Solution:

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In Rhodophyta, the reserve food is Floridean Starch which is similar to amylopectin and glycogen.

Q. 17 Choose the correct option from the following

Option 1:
Ligases - Join the two DNA molecules

Option 2:
Polymerases -Break the DNA into a fragment

Option 3:
Nucleases -Separate the two strands of DNA

Option 4:
Exonucleases -Make cut at a specific position within DNA

Correct Answer:
Ligases - Join the two DNA molecules

Solution:
Ligases are used to join the DNA molecules by establishing a new chemical bond.

Q. 18 Embryological support for evolution was disapproved by

Option 1:
Karl Ernst von Baer

Option 2:
Alfred Wallace

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Option 3:
Charles Darwin

Option 4:
Oparin

Correct Answer:
Karl Ernst von Baer

Solution:
Karl Ernst von Baer has proposed four laws of animal development, which later referred to as von Baer's laws of embryology.

Hence, the correct option is (a).

Q. 19 The first phase of translation is

Option 1:
Binding of mRNA to the ribosome

Option 2:
Recognition of DNA molecule

Option 3:
Aminoacylation of tRNA

Option 4:
Recognition of an anti-codon

Correct Answer:
Binding of mRNA to the ribosome

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Solution:
The first phase of translation is an initiation in which there are the two ribosomal subunits (small and large), The mRNA to be translated, the first (formyl) aminoacyl tRNA (the tRNA charged with the first amino acid),

Hence, the correct option is (a).

Q. 20 The plant parts which consist of two generations on within the other

A Pollen grains inside the anther
B Germinated pollen grains with two male gametes
C Seed inside the fruit
D Embryo sac inside the ovule

Option 1:
(a) only

Option 2:
(a), (b) and (c)

Option 3:
(c) and (d)

Option 4:
(a) and (d)

Correct Answer:
(a) and (d)

Solution:
The plants which consist of two generations on within the other are: pollen grains inside the anther and embryo sac inside the ovule.

Hence, the correct option is (d).

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Q. 21 The number of substrate-level phosphorylations in one turn of the citric acid cycle is

Option 1:
Zero

Option 2:
One

Option 3:
Two

Option 4:
Three

Correct Answer:
Two

Solution:
The number of substrate-level phosphorylations in one turn of the citric acid cycle is 2.

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Q. 22 **Match the following column and select the correct option**

	Column-1		Column -2
(a)	Floating Ribe	(i)	Located between Second and seventh Ribs
(b)	Acromion	(ii)	Head of the humerus
(c)	Scapuls	(iii)	Clavicle
(d)	Glenoid cavity	(iv)	Do not connect with the sternum

Option 1:

(a)	(b)	(c)	(d)
(ii)	(iv)	(i)	(iii)

Option 2:

(a)	(b)	(c)	(d)
(i)	(iii)	(ii)	(iv)

Option 3:

(a)	(b)	(c)	(d)
(iii)	(ii)	(iv)	(i)

Option 4:

(a)	(b)	(c)	(d)
(iv)	(iii)	(i)	(ii)

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Correct Answer:

(a)	(b)	(c)	(d)
(iv)	(iii)	(i)	(ii)

Solution:
Option number 4 contains the correct match.

Q. 23 Match the following diseases with the causative organism and select the correct option.

	Column-1		Column -2
(a)	Typhoid	(i)	Wuchereria
(b)	Pneumonia	(ii)	Plasmodium
(c)	Filariasis	(iii)	Salmonella
(d)	Malaria	(iv)	Haemophilus

Option 1:

(a)	(b)	(c)	(d)
(i)	(iii)	(i1)	(iv)

Option 2:

(a)	(b)	(c)	(d)
(iii)	(iv)	(i)	(ii)

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Option 3:

(a)	(b)	(c)	(d)
(ii)	(i)	(iii)	(iv)

Option 4:

(a)	(b)	(c)	(d)
(iv)	(i)	(ii)	(iii)

Correct Answer:

(a)	(b)	(c)	(d)
(iii)	(iv)	(i)	(ii)

Solution:
Option **number 2** contains the diseases with their causative organism.

Q. 24 Montreal Protocol was signed in 1987 for control of

1 Transport of Genetically modified organisms from one country to another
2 Emission of ozone-depleting substances
3 Release of Green House Gases
4 Disposal of e-waste

Option 1:
Transport of Genetically modified organisms from one country to another

Option 2:
Emission of ozone-depleting substances

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Option 3:
Release of Green House Gases

Option 4:
Disposal of e-waste

Correct Answer:
Emission of ozone-depleting substances

Solution:
Montreal Protocol was signed in 1987 for control of emission of ozone-depleting substances. Hence, the correct option is (b).

Q. 25 The QRS complex in a standard ECG represents

Option 1:
Repolarisation of auricles

Option 2:
Depolarisation of auricles

Option 3:
Depolarisation of ventricles

Option 4:
Repolarisation of ventricles

Correct Answer:
Depolarisation of ventricles

Solution:

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The QRS complex in a standard ECG represents Depolarisation of ventricles.

Q. 26 Name the plant growth regulator which upon spraying on sugarcane crop, increases the length of the stem, thus increasing the yield of sugarcane crop

Option 1:
Cytokinin

Option 2:
Gibberellin

Option 3:
Ethylene

Option 4:
Abscisic acid

Correct Answer:
Gibberellin

Solution:
Gibberellin is that plant growth regulator which upon spraying on sugarcane crop, increases the length of the stem, thus increasing the yield of sugarcane crop.

Hence, the correct option is (b).

Q. 27 How many true-breeding pea plant varieties did Mendel select as pairs, which were similar except in one character with contrasting traits

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Option 1:
4

Option 2:
2

Option 3:
14

Option 4:
8

Correct Answer:
14

Solution:
Total 14 true-breeding pea plant varieties did Mendel select as pairs, which were similar except in one character with contrasting traits.Hence, the correct option is (c).

Q. 28 Bilaterally symmetrical and acoelomate animals are exemplified by

Option 1:
Ctenophora

Option 2:
Platyhelminthes

Option 3:
Aschelminthes

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Option 4:
Annelida

Correct Answer:
Platyhelminthes

Solution:
Bilaterally symmetrical and acoelomate animals are exemplified by platyhelminthes.

Q. 29 Cuboidal epithelium with brush border of microvilli is found in

Option 1:
Lining of intestine

Option 2:
ducts of salivary glands

Option 3:
proximal convoluted tubule of the nephron

Option 4:
eustachian tube

Correct Answer:
proximal convoluted tubule of the nephron

Solution:
Cuboidal epithelium with brush border of microvilli is found in the proximal convoluted tubule of the nephron.

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Q. 30 Which is the important site of information of glycoproteins and glycolipids in eukaryotic cells?

Option 1:
Endoplasmic reticulum

Option 2:
Peroxisomes

Option 3:
Golgi bodies

Option 4:
Polysomes

Correct Answer:
Golgi bodies

Solution:
Golgi bodies are the important sites of information of glycoproteins and glycolipids in eukaryotic cells.

Q. 31 In light reaction plastoquinone facilitates the transfer of electrons from:

Option 1:
d to π DP i

Option 2:
dd to N complex

Option 3:
complex to d

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Option 4:

d to D synthase

Correct Answer:

dd to N complex

Solution:

Correct option is D. dd to complex

In light reaction, PQ facilitates the transfer of electrons from PS-II to complex, which then reduces plastocyanin.

So, the correct answer is ' PS-II to complex'.

Q. 32 Match the following concerning essential elements and their function in plants:

a	Iron	(i)	Photolysis of water
b	Zinc	(ii)	Pollen germination
c	Boron	(iii)	Required for chlorophyll biosynthesis
d	Manganese	(iv)	IAA biosynthesis

Option 1:

(a)	(b)	(c)	(d)
(ii)	(i)	(iv)	(iii)

Option 2:

(a)	(b)	(c)	(d)
(iv)	(iii)	(ii)	(i)

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Option 3:

(a)	(b)	(c)	(d)
(iii)	(iv)	(ii)	(i)

Option 4:

(a)	(b)	(c)	(d)
(iv)	(i)	(ii)	(iii)

Correct Answer:

(a)	(b)	(c)	(d)
(iv)	(iii)	(ii)	(i)

Solution:
In option (b) the essential elements are matched with their correct function in plants.

Q. 33 The root that originated from the base of the stem are :

Option 1:
Fibrous root

Option 2:
Primary root

Option 3:
Prop root

Option 4:
Lateral root

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Correct Answer:
Fibrous root

Solution:
Fibrous root originates from the base of the stem.

Q. 34 From the experiments, S.I. Miller produced amino acids by mixing the following in a closed flask:

Option 1:
CH_4, H_2, NH_3 and water vapor at 800 C^o

Option 2:
CH_3, H_2, NH_4 and water vapor at 800 C^o

Option 3:
CH_4, H_2, NH_3 and water vapor at 600 C^o

Option 4:
CH_3, H_2, NH_3 and water vapor at 600 C^o

Correct Answer:
CH_4, H_2, NH_3 and water vapor at 800 C^o

Solution:
S.I. Miller mixed CH_4, H_2, NH_3 and water vapour at 800 C^o.

Q. 35 dentify the basic amino acid from the following.

Option 1:
Tyrosine

Option 2:
Glutamic Acid

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Option 3:
Lysine

Option 4:
Valine

Correct Answer:
Lysine

Solution:
Lysine is the basic amino acid from the following.

Q. 36 The process of growth is maximum during

Option 1:
Log phase

Option 2:
Lag Phase

Option 3:
Senescence

Option 4:
Dormancy

Correct Answer:
Log phase

Solution:

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The growth rate becomes more rapid in the log phase. Hence, the correct option is (a).

Q. 37 Presence of which of the following conditions in urine are indicative of Diabetes Mellitus?

Option 1:
Uremia and ketonuria

Option 2:
Uremia and Renal Calculi

Option 3:
Ketonuria and Glycosuria

Option 4:
Renal calculi and Hyperglycemia

Correct Answer:
Ketonuria and Glycosuria

Solution:
The presence of Ketonuria and Glycosuria conditions in urine is indicative of Diabetes Mellitus. Hence, the correct option is (c).

Q. 38 Select the correct match

Option 1:
Haemophilia -Ylinked

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Option 2:
Phenylketonuria - Autosomal dominant trait

Option 3:
Sickle cell anaemia -Autosomal recessive trait chromosome 11

Option 4:
Thalassemia - X linked

Correct Answer:
Sickle cell anaemia -Autosomal recessive trait chromosome 11

Solution:
Sickle cell anaemia - Autosomal recessive trait chromosome 11 is the correct one.

Hence, the correct option is (c).

Q. 39 Strobili or cones are found in :

Option 1:
Salvinia

Option 2:
Pteris

Option 3:
Marchantia

Option 4:
Equisetum

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Correct Answer:
Equisetum

Solution:
Strobili or cones are found in Equisetum. hence, the correct option is (d).

Q. 40 Identify the wrong statement with reference to the gene 'I' that controls ABO blood groups

Option 1:
The gene (I) has three alleles.

Option 2:
A person will have only two of the three alleles.

Option 3:
When I^A and I^B are present together, they express same type of sugar.

Option 4:
Allele "i" does not produce any sugar.

Correct Answer:
When I^A and I^B are present together, they express same type of sugar.

Solution:
When $I^A$and I^B are present together, they both express a different types of sugars.

Q. 41 Identify the correct statement with reference to the human digestive system.

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Option 1:
The ileum opens into the small intestine.

Option 2:
Serosa is the innermost layer of the alimentary canal.

Option 3:
The ileum is a highly coiled part.

Option 4:
The vermiform appendix arises from the duodenum.

Correct Answer:
The ileum is a highly coiled part.

Solution:
The ileum is a highly coiled part, so this is the correct statement.

Q. 42 Which of the following would help in the prevention of diuresis?

Option 1:
More water reabsorption due to underreaction of ADH

Option 2:
Reabsorption of Na and water from renal tubules to aldosterone

Option 3:
Atrial natriuretic factor causes vasoconstriction

Option 4:
Decreases in secretion renin by JG cells

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Correct Answer:

More water reabsorption due to underreaction of ADH

Solution:

More water reabsorption due to underreaction of ADH would help in the prevention of diuresis.

Q. 43 Match the following with respect to meiosis:

(a) Zygotene (i) Terminalization

(b) Pachytene (ii) Chiasmata

(c) Diplotene (iii) Crossing over

(d) Diakinesis (iv) Synapsis

Option 1:

(a)	(b)	(c)	(d)
(iii)	(iv)	(i)	(ii)

Option 2:

(a)	(b)	(c)	(d)
(iv)	(iii)	(ii)	(i)

Option 3:

(a)	(b)	(c)	(d)
(i)	(ii)	(iv)	(iii)

Option 4:

(a)	(b)	(c)	(d)
(ii)	(iv)	(iii)	(i)

Correct Answer:

(a)	(b)	(c)	(d)
(iv)	(iii)	(ii)	(i)

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Solution:
(a) (iv) (b) (iii) (c) (ii) (d) (i)

(a) Zygotene - Terminalization

(b) Pachytene - Chiasmata

(c) Diplotene - Synapsis

(d) Diakinesis - Crossing Over

Q. 44 Which of the following is not an inhibitory substance governing seed dormancy?

Option 1:
Gibberellic acid

Option 2:
Abscisic acid

Option 3:
Phenolic acid

Option 4:
Para-ascorbic acid

Correct Answer:
Gibberellic acid

Solution:
Gibberellic acid is not an inhibitory substance governing seed dormancy.

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Q. 45 Match the following columns and select the correct option.

Column -I	**Column -II**
(a) Bt-cotton	(i) Gene therapy
(b) Adenosine deaminase deficiency	(ii) Cellular defence
(c) RNAi	(iii) Detection of HIV infection
(d) PCR	(iv) Bacillus thuringiensis

Option 1:

(a)	**(b)**	**(c)**	**(d)**
iv)	(i)	(ii)	(iii)

Option 2:

(a)	**(b)**	**(c)**	**(d)**
(iii)	(ii)	(i)	(iv)

Option 3:

(a)	**(b)**	**(c)**	**(d)**
(ii)	(iii)	(iv)	(i)

Option 4:

(a)	**(b)**	**(c)**	**(d)**
(i)	(ii)	(iii)	(iv)

Correct Answer:

(a)	**(b)**	**(c)**	**(d)**
iv)	(i)	(ii)	(iii)

Solution:

(a)	**(b)**	**(c)**	**(d)**
iv)	(i)	(ii)	(iii)

(a) Bt-cotton - Bacillus thuringiensis

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(b) Adenosine deaminase deficiency - Gene therapy

(c) RNAi - Cellular defence

(d) PCR - Detection of HIV

Q. 46 **Match the following :**

(a) Inhibitor of catalytic activity	(i) Rieia
(b) Posses peptide bonds	(ii) Malorinta
(c) Cell wall material in fungi	(iii) Chitin
(d) Secondary metabolite	(iv) Collagen

Choose the correct option from the following:

Option 1:

(a)	**(b)**	**(c)**	**(d)**
(ii)	(iv)	(iii)	(i)

Option 2:

(a)	**(b)**	**(c)**	**(d)**
(iii)	(i)	(iv)	(ii)

Option 3:

(a)	**(b)**	**(c)**	**(d)**
(iii)	(iv)	(i)	(ii)

Option 4:

(a)	**(b)**	**(c)**	**(d)**
(ii)	(iii)	(i)	(iv)

Correct Answer:

(a)	**(b)**	**(c)**	**(d)**
(ii)	(iv)	(iii)	(i)

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Solution:
(a) (ii) (b) (iv) (c) (iii) (d) (i)

Q. 47 The sequence that controls the copy number of the linked DNA in the vector, is termed:

Option 1:
Selectable marker

Option 2:
Ori-site

Option 3:
Palindromic sequence

Option 4:
Reognitions site

Correct Answer:
Ori-site

Solution:
Ori-site is the site where DNA replication begins and controls the copy number of the linked DNA.

Q. 48 Snow- blindnese in Antarctic region

Option 1:
Freezing of fluids in the temperature

Option 2:
Inflammation of cornea due UV-B radiation

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Option 3:
High reflection of light from

Option 4:
Damage to retina caused by infra

Correct Answer:
Inflammation of cornea due UV-B radiation

Solution:
Snow- blindnese is caused because of direct exposure of UV-B radiations in the Antarctic region as it causes inflammation in the cornea.

Q. 49 According to Robert May, the global species diversity is about :

Option 1:
1.5 million

Option 2:
20 million

Option 3:
50 million

Option 4:
7 million

Correct Answer:
7 million

Solution:
The global species diversity is about 7 million according to Robert May. Hence, the correct option is (d).

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Q. 50 By which method was a new breed 'Hisardale' of sheep formed by using Bikaneri ewes and Marino rams?

Option 1:
Out crossing

Option 2:
Mutational breeding

Option 3:
Cross breeding

Option 4:
Inbreeding

Correct Answer:
Cross breeding

Solution:
Crossbreeding is the technique used in developing 'Hisardale' breed of sheep.

Q. 51 Which of the following regions of the globe exhibits highest species diversity?

Option 1:
Western Ghats of India

Option 2:
Madagascar

Option 3:
Himalayas

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Option 4:
Amazon Forests

Correct Answer:
Amazon Forests

Solution:
Amazon forests have the highest diversity on the earth. Hence, the correct option is (d).

Q. 52 Match the following columns and select the correct option.

Column I	**Column II**
(a) 6- 15 pairs of gill slits	(i) Trygon
(b) Heterccercal caudal fin	(ii) Cyclostones
(c) Air Bladder	(iii) Chondrichthyes
(d) Poison sting	(iv) Osteichthyes

Option 1:

(a)	**(b)**	**(c)**	**(d)**
(ii)	(iii)	(iv)	(iv)

Option 2:

(a)	**(b)**	**(c)**	**(d)**
(iii)	(iv)	(i)	(ii)

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Option 3:

(a)	(b)	(c)	(d)
(iv)	(ii)	(iii)	(i)

Option 4:

(a)	(b)	(c)	(d)
(i)	(iv)	(iii)	(ii)

Correct Answer:

(a)	(b)	(c)	(d)
(ii)	(iii)	(iv)	(iv)

Solution:
6-15 gill slits are found in Cyclostomata which is given only in option 1.

Q. 53 Which of the following statements is not correct ?

Option 1:
In man insulin is synthesised as a proinsulin

Option 2:
The proinsulin has an extra peptide called C-peptide

Option 3:
The functional insulin has A and B chains linked together by hydrogen bonds

Option 4:
Genetically engineered insulin is produced in E-coli.

Correct Answer:
The functional insulin has A and B chains linked together by hydrogen bonds

Solution:
The functional insulin has A and B chain bonded by the disulfide bridge. Hence, the correct option is (c).

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Q. 54 Match the organisms with its use in biotechnology

(a) Bacillus thuringiensis	(i) Cloning vector
(b) Thermus aquaticus	(ii) Construction of first r DNA
(c) Agrobacterium tumefaciens	(iii) DNA polymerase
(d) Salmonella typhimurium	(iv) Cry proteins

Select the correct option from the following :

Option 1:

(a)	**(b)**	**(c)**	**(d)**
(ii)	(iv)	(iii)	(i)

Option 2:

(a)	**(b)**	**(c)**	**(d)**
(iv)	(iii)	(i)	(ii)

Option 3:

(a)	**(b)**	**(c)**	**(d)**
(iii)	(ii)	(iv)	(i)

Option 4:

(a)	**(b)**	**(c)**	**(d)**
(iii)	(iv)	(i)	(ii)

Correct Answer:

(a)	**(b)**	**(c)**	**(d)**
(iv)	(iii)	(i)	(ii)

Solution:

Bacillus thuringiensis produces cry protein and that is given only in option 2.

Hence, the correct option is (b).

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Q. 55 Which of the following pairs is of unicellular algae ?

Option 1:
Laminaria and Sargassum

Option 2:
Gelidium and Gracilaria

Option 3:
Anabaena and Volvox

Option 4:
Chlorella and spirulina

Correct Answer:
Chlorella and spirulina

Solution:
Chlorella and spirulina are examples of unicellular algae.

Q. 56 Meiotic division of the secondary oocyte is completed :

Option 1:
Prior to ovulation

Option 2:
At the time of copulation

Option 3:
After zygote formation

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Option 4:
At the time of fusion of a sperm with an ovum

Correct Answer:
At the time of fusion of a sperm with an ovum

Solution:
Meiosis is completed when the secondary oocyte fuses with sperm. Hence, the correct option is (d).

Q. 57 Secondary metabolites such as nocotine, strychnine and caffein are produced by plants for their :

Option 1:
Nutritive value

Option 2:
Growth response

Option 3:
Defence action

Option 4:
Effect on reproduction

Correct Answer:
Defence action

Solution:

Secondary metabolites in plants helps in defence action

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Q. 58 Which of the following statements are true for the phylum -Chordata?

(a) In a Urochrdata notochord extends from head to tail and it is present throughout their life.

(b) In Vertebrata notochord is present during the embryonic period only

(c) Central nervous system is dorsal and hallow

(d) Chordata is divided into 3 subphyla: Hermichordata, Tunicata and Cephalochordata

Option 1:
(d) and (c)

Option 2:
(c) and (a)

Option 3:
(a) and (b)

Option 4:
(b) and (c)

Correct Answer:
(b) and (c)

Solution:
In the phylum Chordata:

In Vertebrata, the notochord is present during the embryonic stage only

The nerve cord is dorsal and hollow.

Hence, the correct option is (d).

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Q. 59 Bt cotton variety that was developed by the introduction of toxin gene of Bacillus thuringiensis (Bt) in resistant to :

Option 1:
Insect pests

Option 2:
Fungal diseases

Option 3:
Plant nematodes

Option 4:
Insect predators

Correct Answer:
Insect pests

Solution:
toxins from Bacillus thuringiensis kill Bollworm that eats cotton plant

Q. 60 The product (s) of reaction catalyzed by nitrogenase in root nodules of leguminous plants is /are :

Option 1:
Ammonia alone

Option 2:
Nitrate alone

Option 3:
Ammonia and oxygen

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Option 4:
Ammonia and hydrogen

Correct Answer:
Ammonia alone

Solution:
Nitrogenase converts nitrogen into ammonia. Hence, the correct option is (a).

Q. 61 Match the following columns and select the correct option,

Column -I	**Column -II**
(a) Pituitary gland	(i) Grave's disense
(b) Thyroid gland	(ii) Diabetes mellitus
(c) Adrenal gland	(iii) Diabetes insipidus
(d) Pancreases	(iv) Addison's disesase

Option 1:
(a) (b) (c) (d)
(iv) (iii) (i) (ii)

Option 2:
(a) (b) (c) (d)
(iii) (ii) (i) (iv)

Option 3:
(a) (b) (c) (d)
(iii) (i) (iv) (ii)

Option 4:
(a) (b) (c) (d)
(ii) (i) (iv) (iii)

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Correct Answer:

(a) (b) (c) (d)
(iii) (i) (iv) (ii)

Solution:
The correct match is:

Pituitary gland - Diabetes insipidus

Thyroid gland - Graves disease

Adrenal gland - Addison's disease

Pancreas - Diabetes mellitus

(a) (b) (c) (d)
(iii) (i) (iv) (ii)

Q. 62 Which one of the following is the most abundant protein in the animals?

Option 1:
Haemoglobin

Option 2:
Collagen

Option 3:
Lactin

Option 4:
Insulin

Correct Answer:
Collagen

Solution:

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Collagen is the most abundant proteins in animals

Q. 63 Identify the correct statement with regard to G_1 phase (Gap 1) of interphase.

Option 1:
DNA synthesis or replication takes place

Option 2:
Reorganisation of all cell components takes place

Option 3:
The cell is metabolically active, and grows but does not replicate its DNA.

Option 4:
Nuclear Division takes place.

Correct Answer:
The cell is metabolically active, and grows but does not replicate its DNA.

Solution:
In the G1 phase, the cell is metabolically active, and grows but does not replicate its DNA.

Q. 64 Match the trophic levels with their correct species examples in grassland ecosystem.

(a) Fourth trophic level (i) Crow

(b) Second trophic level (ii) Vulture

(c) First trophic level (iii) Rabbit

(d) Third trophic level (iv) Grass

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Option 1:
(a) (b) (c) (d)
(ii) (iii) (iv) (i)

Option 2:
(a) (b) (c) (d)
(iii) (ii) (i) (iv)

Option 3:
(a) (b) (c) (d)
(iv) (iii) (ii) (i)

Option 4:
(a) (b) (c) (d)
(i) (ii) (iii) (iv)

Correct Answer:
(a) (b) (c) (d)
(ii) (iii) (iv) (i)

Solution:
First trophic level is grass and only option 1 shows it.

Q. 65 The ovary is half inferiors in :

Option 1:
Brinjal

Option 2:
Mustard

Option 3:
Sunflower

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Option 4:
Plum

Correct Answer:
Plum

Solution:
The ovary of plum is half inferior. Hence, the correct option is (d).

Q. 66 The body of the ovule is fused within the funicle at :

Option 1:
Hilum

Option 2:
Micropyle

Option 3:
Nucellus

Option 4:
Chalaza

Correct Answer:
Hilum

Solution:

Ovule is fused with the futile at the hilum.

Q. 67 The specific palindromic sequence which is recognized by EcoRI is :

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Option 1:
$5' - GAATTC - 3'$
$3' - CTTAAG - 5'$

Option 2:
$5' - GGAACC - 3'$
$3' - CCTTGG - 5'$

Option 3:
$5' - CTTAAG - 3'$
$3' - GAATTC - 5'$

Option 4:
$5' - GGATCC - 3'$
$3' - CCTAGG - 5'$

Correct Answer:
$5' - GAATTC - 3'$
$3' - CTTAAG - 5'$

Solution:
The palindrome identified by EcoRI is 5'-GAATTC-3' / 3'-CTTAAG-5'. Hence, the correct option is (a).

Q. 68 Which of the following is correct about viroids ?

Option 1:
They have RNA with protein coat

Option 2:
They have free RNA without protein coat

Option 3:
They have DNA with protein coat

Option 4:
They have free DNA without protein coat

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Correct Answer:
They have free RNA without protein coat

Solution:
-Viroids have single-stranded circular RNA with no protein coat.

- Viroids only effect plants and not humans so they do not cause any disease in humans.

Q. 69 In water hyacinth and waterlily, pollination takes place by :

Option 1:
insects or wind

Option 2:
water currents only

Option 3:
wind and water

Option 4:
Insects and water

Correct Answer:
insects or wind

Solution:
Due to attractive conspicuous flowers, water hyacinth and water lily are pollinated by insects and some by the wind.

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Q. 70 The transverse action of a plant shows following anatomical features:

(a) Large number of scattered vascular bundles surrounded by bundle sheath

(b) Large conspicuous parenchym tissue

(c) Vascular bundles conjoint and

(d) Phloem parenchyma absent

Identify the category of plant and its part:

Option 1:
Monocotyledonous stem

Option 2:
Monocotyledonous root

Option 3:
Dicotyledonous stem

Option 4:
Dicotyledonous root

Correct Answer:
Monocotyledonous stem

Solution:
Monocotyledonous stems have all the characters mentioned in the options.

Q. 71 Which of the following statements is correct?

Option 1:
Adenine pairs with thymine through two H-bonds

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Option 2:
Adenine pairs with thymine through one H-bond

Option 3:
Adenine pairs with thymine through three H-bonds.

Option 4:
Adenine does not pair with thymine

Correct Answer:
Adenine pairs with thymine through two H-bonds

Solution:
In double-stranded DNA, Adenine pairs with Thymine with two Hydrogen bonds that is (A=T) and Guanine pairs with Cytosine with three hydrogen bonds.

Q. 72 Select the correct statement.

Option 1:
Glucocorticoids stimulate gluconecgenesis

Option 2:
Glucogon is associted with hypoglycemia

Option 3:
Insulin acts on pancreatic cells and adipocytes

Option 4:
Insulin is associated with hyperglycemia

Correct Answer:
Glucocorticoids stimulate gluconecgenesis

Solution:
Glucocorticoids are steroid hormones that promote gluconeogenesis in the liver.

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Q. 73 Match the following columns and select the correct option.

Column I	Column II
(a) gregarnious polyphagus pest	(i) Asterias
(b) Adult with the radial symmetry and larva with bilateral symmetry	(ii) Scorpion
(c) Book lungs	(iii) Ctenoplanes
(d) Boioluminesces	(iv) Locust

Option 1:

(a)	(b)	(c)	(d)
i	iii	ii	iv

Option 2:

(a)	(b)	(c)	(d)
iv	i	ii	iii

Option 3:

(a)	(b)	(c)	(d)
iii	ii	i	iv

Option 4:

(a)	(b)	(c)	(d)
ii	i	iii	iv

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Correct Answer:

(a)	(b)	(c)	(d)
iv	i	ii	iii

Solution:
a. gregarious polyphagous pest - Locust

b. Adult with the radial symmetry and larva with bilateral symmetry - Asterias

c. Book lungs - Scorpion

d. Boioluminesces - Ctenoplanes

Q. 74 match the following columns and select the correct option

Column -I	Column -II
(a) Eosinophils	(i) Immune response
(b) basophils	(ii) Phagocytes
(c) Neutrophils	(iii) Release histamine, destructive enzymes
(d) Lymphocytes	(iv) Release granules containing histamine

Option 1:

(a)	(b)	(c)	(d)
iii	iv	ii	i

Option 2:

(a)	(b)	(c)	(d)
iv	i	ii	iii

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Option 3:

(a)	(b)	(c)	(d)
i	ii	iv	iii

Option 4:

(a)	(b)	(c)	(d)
ii	i	iii	iv

Correct Answer:

(a)	(b)	(c)	(d)
iii	iv	ii	i

Solution:
a. Eosinophils - release histamine, destructive enzymes

b. Basophils - release granules containing histamine

c. Neutrophils - Phagocytes

d. Lymphocytes - Immune response

Hence, the correct option is (a).

Q. 75 If the head of a cockroach is removed, it may live for few days because:

Option 1:
The supra-oesophageal ganglia of the cockroach are situated in ventral part of the abdomen.

Option 2:
The cockroach does not have a nervous system

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Option 3:
The head holds a small proportion of a nervous system while the rest is situated along the ventral part of its body.

Option 4:
The head holds 1/3rd of a nervous system while the rest is situated along the dorsal part of the body

Correct Answer:
The head holds a small proportion of a nervous system while the rest is situated along the ventral part of its body.

Solution:
The head holds a bit of a nervous system while the rest is situated along the ventral part of its body .The brain is represented by supra-oesophageal ganglion which supplies nerves to antennae and compound eyes.

Q. 76 Name the enzymes that facilitate the opening of DNA helix during transcription.

Option 1:
DNA ligase

Option 2:
DNA helicase

Option 3:
DNA polymerase

Option 4:
RNA polymerase

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Correct Answer:
RNA polymerase

Solution:
During initiation in the process of transcription, the RNA polymerase holoenzyme binds to the promoter region of the transcription unit. The RNA polymerase-sigma complex binds to the promoter and initiates transcription of a specific strand of DNA. The DNA strand unwinds by the action of this complex.

Q. 77 Flippers of Penguins and Dolphins are examples of:

Option 1:
Adaptive radiation

Option 2:
Convergent evolution

Option 3:
Industrial melain

Option 4:
Natural selection

Correct Answer:
Convergent evolution

Solution:
Flippers of penguin and dolphins have different structures but similar functions and thus are categorized under analogous organs. Also, Penguin and dolphins are not closely related to each other but have evolved the similar traits (as per this question - flippers) therefore, show convergent type of evolution.

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Q. 78 Which of the following hormone levels will cause the release of the ovum (ovulation) from the graffian follicle?

Option 1:
The high concentration of estrogen

Option 2:
The high concentration of progesterone

Option 3:
low concentration of LH

Option 4:
low concentration of FSH

Correct Answer:
The high concentration of estrogen

Solution:
- The high concentration of Estrogen will suppress the release of FSH through which the ovary won't be able to grow to become a fully mature Graafian follicle
- High level of estrogen also causes LH surge which is important for ovulation

Q. 79 If the distance between two consecutive base pairs is 0.34nm and the total number of base pairs of a DNA helix in a typical mammalian cell is 6.6×10^9 bp, then the length of the DNA is approximate:

Option 1:
2.0 meters

Option 2:
2.5 meters

Option 3:
2.2 meters

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Option 4:
2.7 meters

Correct Answer:
2.2 meters

Solution:
Length of the DNA = Distance between two base pairs x Total no. of base pairs.

Length of the DNA = $0.34 \times 6.6 \times 10^5 = 2.244$

Q. 80 Match the following columns and select the correct option.

Column - I	Column -II
(a) Placenta	(i) Androgens
(b) Zona pellucida	(ii) Human Chorionic Gonadotropin (hCG)
(c) Bulbo-urethral gland	(iii) Layer of the ovum
(d) Leydig cells	(iv) Lubrication of the Penis

Option 1:

(a)	(b)	(c)	(d)
iv	iii	i	ii

Option 2:

(a)	(b)	(c)	(d)
i	iv	ii	iii

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Option 3:

(a)	(b)	(c)	(d)
iii	ii	iv	i

Option 4:

(a)	(b)	(c)	(d)
ii	iii	iv	i

Correct Answer:

(a)	(b)	(c)	(d)
ii	iii	iv	i

Solution:
a. Placenta - Human Chorionic Gonadotropin (hCG)

b. Zona pellucida - Layer of the ovum

c. Bulbourethral glands - Lubrication of the Penis

d. Leydig cells - Androgens

Hence, the correct option is (d).

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Q. 81 Match of the following columns and select the correct otion.

Column I	Column II
(a) Clostridium butylicum	(i) Cyclosporin - A
(b) Trichoderma polysprom	(ii) Butyric acid
(c) Monasucus purpureus	(iii) Citric acid
(d) Aspergillus niger	(iv) Blood cholestrol lowering agent

Option 1:

(a)	(b)	(c)	(d)
iii	iv	ii	i

Option 2:

(a)	(b)	(c)	(d)
ii	i	iv	iii

Option 3:

(a)	(b)	(c)	(d)
i	ii	iv	iii

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Option 4:

(a)	(b)	(c)	(d)
iv	iii	ii	i

Correct Answer:

(a)	(b)	(c)	(d)
ii	i	iv	iii

Solution:
a. Clostridium butylicum - Butyric acid

b. Trichoderma polysprom - Cyclosporin - A

c. Monasucus purpureus - Blood cholestrol lowering agent

d. Aspergillus niger - Citric acid

Hence, the correct option is (b).

Q. 82 Goblet cells of the alimentary canal are modified from:

Option 1:
Squamous epithelial cells

Option 2:
Columnar epithelial cells

Option 3:
Chondrocytes

Option 4:
Compound epithelial cells

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Correct Answer:
Columnar epithelial cells

Solution:
Goblet cells of the alimentary canal are modified from columnar epithelial cells.

Q. 83 Experimental verification of the chromosomal theroy of the inheritance was done by:

Option 1:
Mendel

Option 2:
Sutton

Option 3:
Boveri

Option 4:
Morgan

Correct Answer:
Morgan

Solution:
The experimental verification of the chromosomal theory of inheritance was done by Thomas Hunt Morgan and his colleagues which led to discovering the basis for the variation that sexual reproduction produced.

Q. 84 The process responsible for facililating loss of water in liquid form from the tip of grass blades at night and early morning is:

Option 1:
Transpiration

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Option 2:
Root pressure

Option 3:
Inhibition

Option 4:
Plasmolysis

Correct Answer:
Root pressure

Solution:
Effects of root pressure is observable at night and early morning when evaporation is low, and excess water collects in the form of droplets around special openings of veins near the tip of grass blades, and leaves of many herbaceous parts. Such water loss in its liquid phase is known as guttation.

Q. 85 Identify the substance having a glycosidic bond and peptide bond, respectively in their structure:

Option 1:
Chittin, cholestrol

Option 2:
Glycerol, trypsin

Option 3:
Cellulose, lecithin

Option 4:
Inulin, insulin

Correct Answer:
Inulin, insulin

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Solution:
Inulin and Insulin are the substances having a glycosidic bond and peptide bond, respectively in their structure.

Q. 86 Which of the following is not an attribute of a population?

Option 1:
Sex ratio

Option 2:
natality

Option 3:
mortality

Option 4:
species interaction

Correct Answer:
species interaction

Solution:
The factors considered under the population attributes are Birth Rate or Natality, Death Rate or Mortality and Sex Ratio. Hence, the correct option is (d).

Q. 87 The enzyme enterkinase helps in conversion of:

Option 1:
protein into polypeptide

Option 2:
trypsinogen into trypsin

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Option 3:
caseinogen into casein

Option 4:
pepsinogen into pepsin

Correct Answer:
trypsinogen into trypsin

Solution:
The enzyme enterokinase helps in the conversion of trypsinogen into trypsin. The enzyme trypsin further helps in the breakdown of proteins for the process of digestion.

Q. 88 Some dividing cells exit the cell cycle and enter the vegetative inactive stage. This is called the quiescent stage (G_0). This process occurs at the end of:

Option 1:
M phase

Option 2:
G_1 phase

Option 3:
S phase

Option 4:
G_2 phase

Correct Answer:
M phase

Solution:
G1 is commonly called Pre DNA synthesis phase or first gap phase. The cell recovers from the previous division.

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Q. 89 In relation to gross primary productivity and Net primary productivity of an ecosystem which one of the following statements is correct?

Option 1:
Gross primary productivity is always less than net primary productivity

Option 2:
Gross primary productivity is always more than net primary productivity

Option 3:
Gross primary productivity and net primary productivity are one and the same

Option 4:
There is no relationship between Gross primary productivity and net primary productivity.

Correct Answer:
Gross primary productivity is always more than net primary productivity

Solution:
Gross primary productivity is always more than net primary productivity. It is because GPP is the total solar energy trapped in the food material by photosynthesis. On the other hand, Gross primary productivity minus respiration losses (R), is the net primary productivity (NPP). Hence, the correct option is (b).

Q. 90 Which of the following is put into Anaerobic digester for further sewage treatment?

Option 1:
Primary sludge

Option 2:
Floating debris

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Option 3:
Effluents of primary treatment

Option 4:
Activated sludge

Correct Answer:
Activated sludge

Solution:
Activated sludge is a type of wastewater system as aerated sewage containing aerobic microorganisms which help to break it down.

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Physics

Q. 1 Three stars A, B, C have surface temperature T_A, T_B, T_C respectively. Star A appears bluish, star B appears reddish and star C yellowish. Hence

Option 1:
$T_B > T_C > T_A$

Option 2:
$T_C > T_B > T_A$

Option 3:
$T_A > T_C > T_B$

Option 4:
$T_A > T_B > T_C$

Correct Answer:
$T_A > T_C > T_B$

Solution:
The appearance of a star depends on the wavelength (λm) at which it radiates maximum energy.

By Wien's law,

$$\lambda_m \alpha \frac{1}{T}$$

As $\lambda_{blue} < \lambda_{yellow} < \lambda_{red}$.

Star A (bluish) radiates the shortest λ and must be at the highest temperature, while star B (reddish) must be at the lowest temperature.

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Q. 2 Which of the following gate is called universal gate.

Option 1:
AND gate

Option 2:
NAND gate

Option 3:
NOT gate

Option 4:
OR gate

Correct Answer:
NAND gate

Solution:
NAND Gate and NOR Gate are universal gates, as we can make all the gates with the help of NAND and NOR

Hence, the answer is the option (2).

Q. 3 The mean free path l for a gas molecule depends upon the diameter, d of the molecule as:

Option 1:
$l \propto d$

Option 2:
$l \propto d^2$

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Option 3:

$l \propto \frac{1}{d}$

Option 4:

$l \propto \frac{1}{d^2}$

Correct Answer:

$l \propto \frac{1}{d^2}$

Solution:

$$\text{Mean free path} = l = \frac{1}{\sqrt{2}\pi n d^2}$$

Hence, the answer is the option (4).

Q. 4 The magnetic flux linked with a coil (in Wb) is given by the equation $\phi = 5t^2 + 3t + 16$. The magnitude of induced emf in the coil at the fourth second will be :

Option 1:

43 V

Option 2:

108 V

Option 3:

10 V

Option 4:

33 V

Correct Answer:

43 V

Solution:

The magnitude of induced emf is given by

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$$|e| = \frac{d\phi}{dt} = \frac{d(5t^2 + 3t + 16)}{dt}$$

$$|e| = 10t + 3 + 0$$

at t = 4 s

$$|e| = 40 + 3 = 43$$

Hence, the answer is the option (1).

Q. 5 The angular speed of the wheel of a vehicle is increased from 360 rpm to 1200 rpm in 14 seconds. Its angular acceleration is

Option 1:
$28\pi\ rad/s^2$

Option 2:
$120\pi\ rad/s^2$

Option 3:
$1\ rad/s^2$

Option 4:
$2\pi\ rad/s^2$

Correct Answer:
$2\pi\ rad/s^2$

Solution:

$$\omega_{initial} = 360rpm = 12\pi rad/s$$

$$\omega_{final} = 1200rpm = 40\pi rad/s$$

$$\because 1rpm = 2\pi rad/60$$

$$\text{Angular acceleration} = \frac{\Delta\omega}{\Delta t} = \frac{40\pi - 12\pi}{14} = 2\pi rad/s^2$$

Hence, the answer is the option (4).

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Q. 6 Two coherent sources of light interfere and produce a fringe pattern on a screen. For central maximum, the phase difference between the two waves will be

Option 1:
π

Option 2:
$\frac{3\pi}{2}$

Option 3:
$\frac{\pi}{2}$

Option 4:
zero

Correct Answer:
zero

Solution:
For central maximum, path difference is zero$(\Delta x = 0)$
hence phase difference for coherent sources of light will be zero.
therefore,
$\phi = (2\pi/\lambda)\Delta x$
where$\Delta x$is path difference and$\phi$is phase difference

Hence, the answer is the option (4).

Q. 7 What is the depth at which the value of acceleration due to gravity becomes 1/n times the value at the surface of the earth? (radius of earth =R)

Option 1:
$R(n-1)/n$

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Option 2:
$Rn/(n-1)$

Option 3:
R/n

Option 4:
R/n^2

Correct Answer:
$R(n-1)/n$

Solution:

Acceleration due to gravity at depth d $= g_d = g(1-\frac{d}{R})$

$g_d = g/n = g(1-\frac{d}{R})$

$1/n = 1(1-\frac{d}{R})$

$\frac{d}{R} = 1 - 1/n$

$d = R(n-1)/n$

Hence, the answer is the option (1).

Q. 8 An object is placed on the principal axis of a concave mirror at a distance of $1.5f$(f is the focal length). The image will be at

Option 1:
$1.5f$

Option 2:
$-1.5f$

Option 3:
$3f$

Option 4:
$-3f$

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Correct Answer:
$-3f$

Solution:
u=-1.5f

focal length = f

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}$$

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{-1.5f}$$

$$\frac{1}{v}=\frac{1}{3f}$$

$$v=-3f$$

Hence, the answer is the option (4).

Q. 9 The acceleration of an electron due to the mutual attraction between the electron and a proton when they are $1.6\mathring{A}$ apart is,

$(me \simeq 9 \times 10^{31} kg, e = 1.6 \times 10^{-19} C)$
$\left(Take \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 Nm^2C^{-2}\right)$

Option 1:
$10^{23} m/s^2$

Option 2:
$10^{22} m/s^2$

Option 3:
$10^{25} m/s^2$

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Option 4:

$10^{24} m/s^2$

Correct Answer:

$10^{22} m/s^2$

Solution:

$$F = \frac{1}{4\pi\varepsilon_0} \times \frac{q_1 q_2}{r^2}$$

$$F = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{(1.6 \times 10^{-19})^2}$$

$$F = 9 \times 10^{-9} N$$

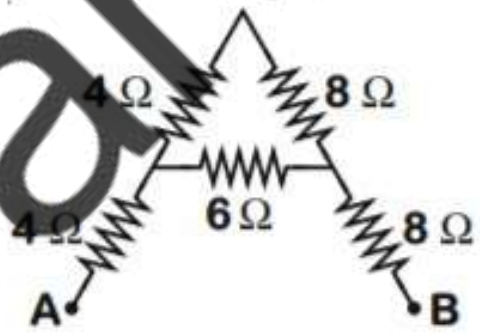
Acceleration of electron

$$a_e = \frac{F}{m_e} = \frac{9 \times 10^{-9}}{9 \times 10^{31}} kg$$

$$a_e = 10^{22} m/s^2$$

Hence, the answer is the option (2).

Q. 10 The equivalent resistance between A and B for the mesh shown in the figure is



Option 1:

$16\ \Omega$

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Option 2:

$30\ \Omega$

Option 3:

$4.8\ \Omega$

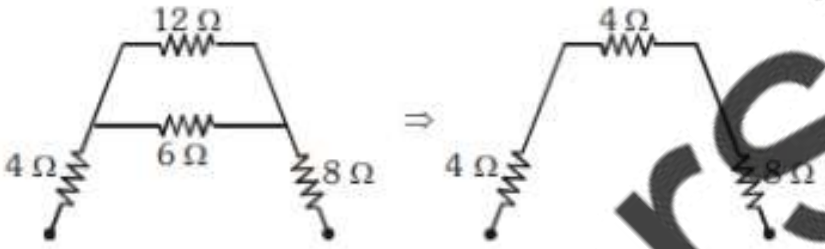
Option 4:

$7.2\ \Omega$

Correct Answer:

$16\ \Omega$

Solution:



$R_{AB} = 16\Omega$

Q. 11 Identify the function which represents a periodic motion -

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Option 1:
$\log_e(\omega t)$

Option 2:
$\sin \omega t + \cos \omega t$

Option 3:
$e^{-\omega t}$

Option 4:
$e^{\omega t}$

Correct Answer:
$\sin \omega t + \cos \omega t$

Solution:
For periodic function

$$f(t) = f(t+T)$$

T - Time period

(1) For $f(t) = \log_e(\omega t)$

$$f(t+T) = \log_e(\omega(t+T))$$

$$f(t) \neq f(t+T)$$

(2) For $f(t) = \sin \omega t + \cos \omega t$

$$f(t+T) = \sin(\omega(t+T)) + \cos(\omega(t+T))$$

$(\because \omega T = 2\pi)$

$$f(t) = f(t+T)$$

(3) $f(t) = e^{-\omega t}$

$$f(t+T) = e^{-\omega(t+T)}$$

$$f(t) \neq f(t+T)$$

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(4) $f(t) = e^{\omega t}$

$f(t+T) = e^{\omega(t+T)}$

$f(t) \neq f(t+T)$

So $f(t) = \sin \omega t + \cos \omega t$ is the periodic function

Hence, the answer is the option (2).

Q. 12 A liquid does not wet the solid surface if the angle of contact is :

Option 1:
equal to 60^o

Option 2:
greater than 90^o

Option 3:
zero

Option 4:
equal to 45^o

Correct Answer:
greater than 90^o

Solution:
We know that the liquids make the solid substances wet when they get into physical contact with each other. This depends on the contact angle between the liquid and solid surfaces. When the liquid does not wet the sides of a solid object, it means that the angle of contact is more than 90° but less than 180°.

A liquid does not wet the solid surface if the angle of contact is obtuse (i.e. θ>90°).

Hence, the answer is the option (2).

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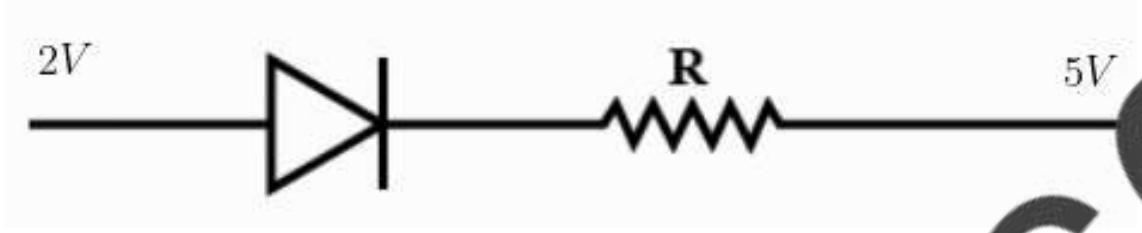
Q. 13 Out of the following which one is a forward biased diode?

Option 1:

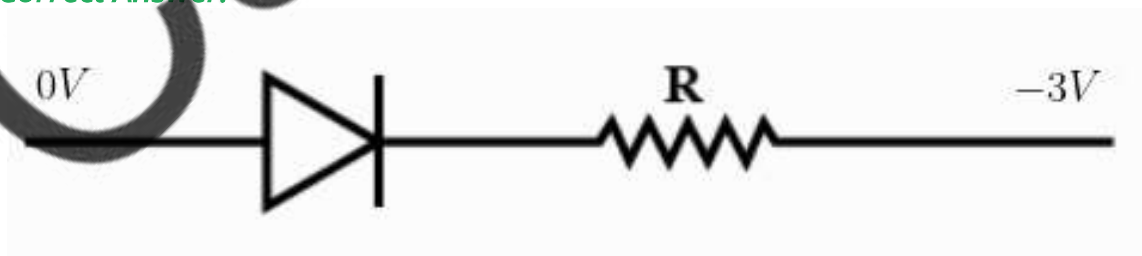


Option 2:

Option 3:

Option 4:

Correct Answer:



Solution:

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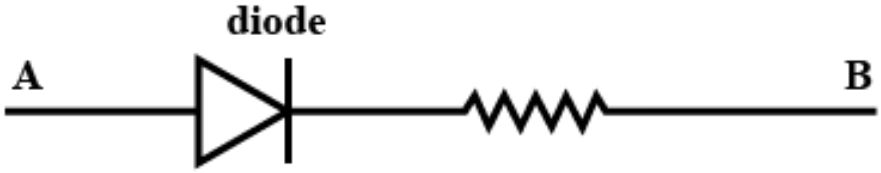
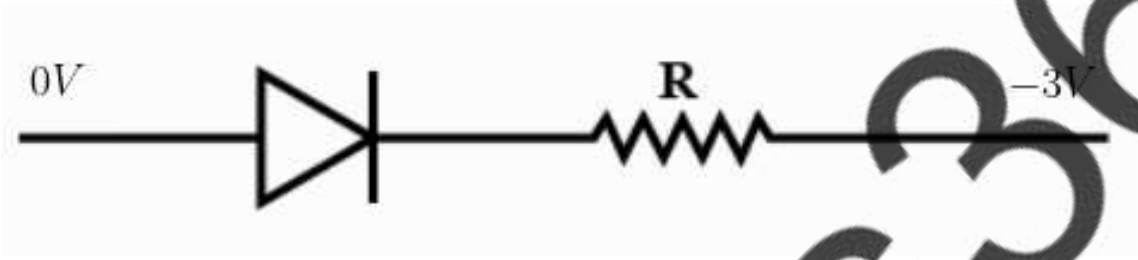


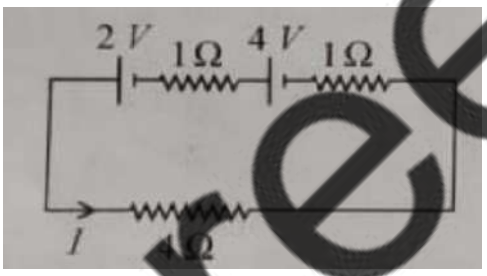
A diode is said to be forward biased when potential at A is higher (more positive) than at B

ie. $V_A > V_B$



For the above case, 0V is greater than (-3V) so it is a case of forward biased.

Q. 14 For the circuit shownin the figure, the current I will be



Option 1:
1A

Option 2:
1.5 A

Option 3:
0.5 A

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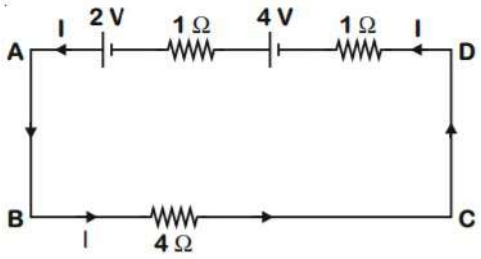
Option 4:
0.75 A

Correct Answer:
1A

Solution:



By KVL in a closed loop ABCDA,

$V_A - I \times 4 - I \times 1 + 4 - I \times 1 + 2 = V_A$

$-6I + 6 = 0$

$I = 1$ A

Q. 15 Time intervals measured by a clock give the following readings:

1.21 s,1.25 s, 1.24 s, 1.27 s, and 1.28 s. What is the percentage relative error of the observation?

Option 1:
4 %

Option 2:
16%

Option 3:
1.6%

Option 4:
2%

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Correct Answer:
1.6%

Solution:
Given,

t_1 = 1.25s

t_2 = 1.24s

t_3 = 1.27s

t_4 = 1.28s

$t_5 = 1.21s$

$$t_{mean} = \frac{t_1 + t_2 + t_3 + t_4 + t_5}{5}$$

$$t_{mean} = \frac{1.25 + 1.24 + 1.27 + 1.28 + 1.21}{5} = 1.25$$

$$E_1 = 1.25 - 1.25 = 0$$
$$E_2 = 1.25 - 1.24 = 0.01$$
$$E_3 = 1.25 - 1.27 = -0.02$$
$$E_4 = 1.25 - 1.28 = -0.03$$
$$E_1 = 1.25 - 1.21 = 0.04$$

$$\text{Relative Error} = \frac{\text{Sum of Absolute error}}{\text{Mean value} \times n} \times 100\%$$

$$= \frac{|E_1| + |E_2| + |E_3| + |E_4| + |E_5|}{1.25 \times 5} \times 100\%$$

$$= \frac{0 + 0.01 + 0.02 + 0.03 + 0.04}{1.25 \times 5} \times 100\%$$

$$= 1.6\%$$

Hence, the answer is the option (3).

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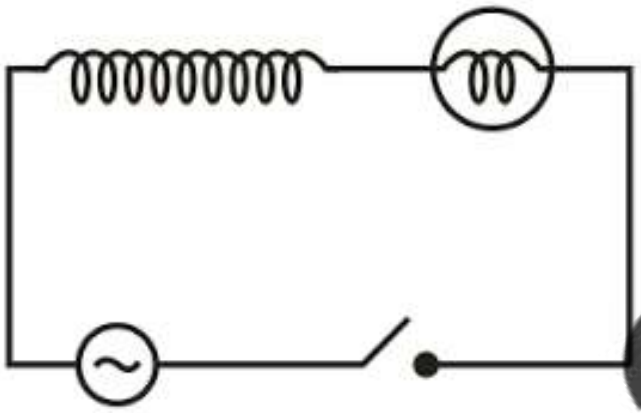
Q. 16 A light bulb and an inductor coil are connected to an ac source through a key as shown in the figure below. The key is closed and after sometime an iron rod is inserted into the interior of the inductor. The glow of the light bulb



Option 1:
remain unchanged

Option 2:
will fluctuate

Option 3:
increases

Option 4:
decreases

Correct Answer:
decreases

Solution:

$$L = \frac{\mu_0 N^2 (\pi r^2)}{l}$$

After inserting the iron rod

$$L_{final} = \frac{\mu N^2 (\pi r^2)}{l} \qquad \mu > \mu_0$$

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$$e = -L\left(\frac{di}{dt}\right)$$

As L increases, emf induced across the inductor will increase leaving less voltage across the bulb. Therefore, the glow of the light bulb decreases.

Q. 17 Two solid conductors are made up of the same material, have the same length, and same resistance. One of them has a circular cross-section of area A_1 and the other one has a square cross-section of area A_2. The ratio A_1/A_2 is

Option 1:
1

Option 2:
0.8

Option 3:
2

Option 4:
1.5

Correct Answer:
1

Solution:

$R_1 = R_2$

Also

$s_1 = s_2$ (same material)

$l_1 = l_2$

$\frac{s_1 l_1}{A_1} = \frac{s_2 l_2}{A_2}; \frac{A_1}{A_2} = \frac{1}{2}$

Hence, the answer is the option (1).

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Q. 18 The magnetic field in a plane electromagnetic wave is given by,

$B_y = 2 \times 10^{-7} sin\left(\pi 10^3 x + 3\pi \times 10^{11} t\right)$ calculate the wavelength.

Option 1:

$2 \times 10^{-3} m$

Option 2:

$2 \times 10^{3} m$

Option 3:

$\pi \times 10^{-3} m$

Option 4:

$\pi \times 10^{3} m$

Correct Answer:

$2 \times 10^{-3} m$

Solution:

$B_y = B_o \sin(kx + \omega t)$

$B_y = 2 \times 10^{-7} \sin(\pi \times 10^3 (x) + 3\pi \times 10^{11} t)$

$K = \pi \times 10^3 = \frac{2\pi}{\lambda}$

$\lambda = 2 \times 10^{-3} m$

Hence, the answer is the option (1).

Q. 19 The varition of electrostatic potential with radial distance r from the centre of a positively charged metallic thin shell of radius R is given by the graph

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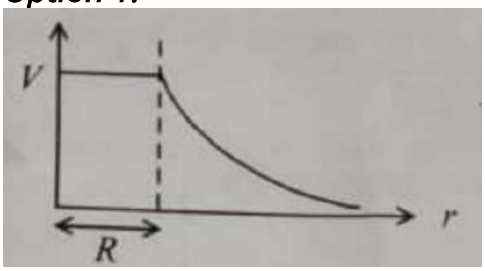
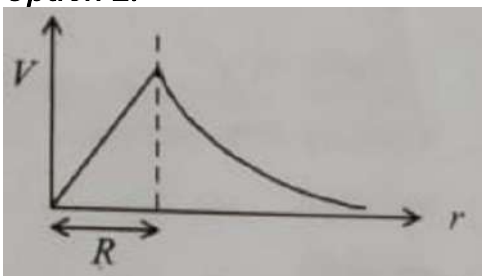
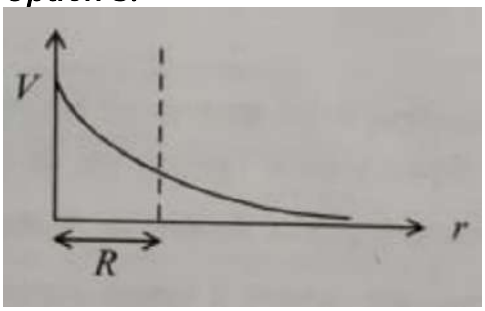
Option 1:



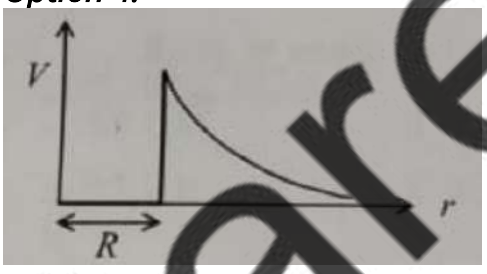
Option 2:



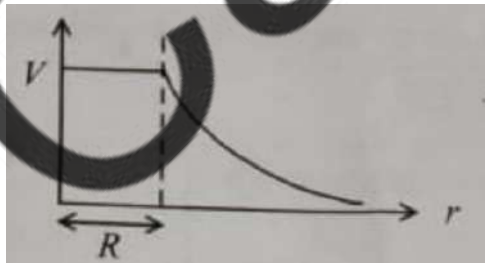
Option 3:



Option 4:



Correct Answer:



Solution:

$$V = \frac{1}{4\pi r_0}\frac{Q}{R} \qquad (0 < r \leq R)$$

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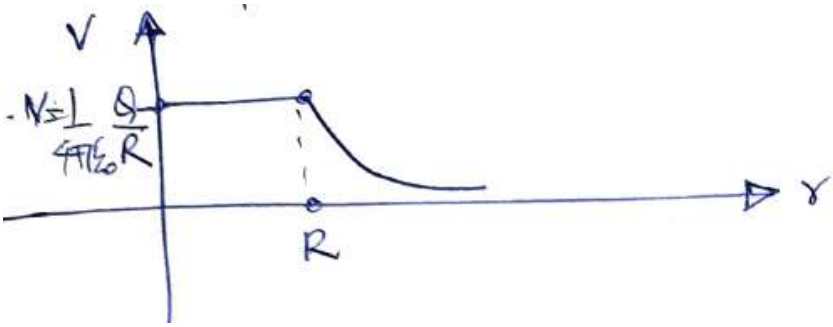
$$V = \frac{1}{4\pi r_0}\frac{Q}{r} \qquad (r > R)$$



Q. 20 The efficiency of a Carnot engine depends upon

Option 1:
the temperatures of the source and sink

Option 2:
the volume of the cylinder of the engine.

Option 3:
the temperature of the source only

Option 4:
the temperature of the sink only

Correct Answer:
the temperatures of the source and sink

Solution:

$$\eta = I - \frac{T_2}{T_1}$$

T_2 - Temperature of sink

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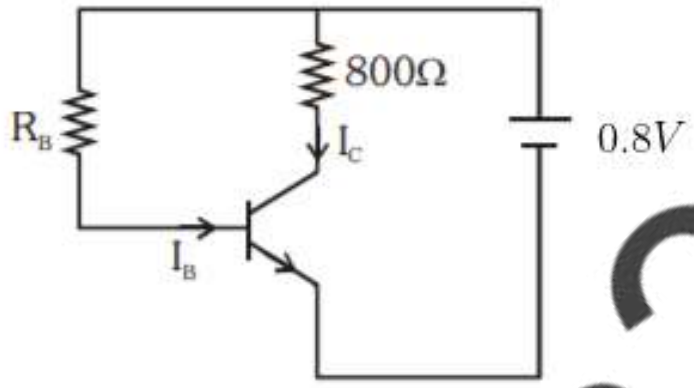
T_1 - Temprature of source

Q. 21 A n-p-n transistor is connected in common emitter configuration (see figure) in which collector voltage drop across load resistance $(800\ \Omega)$ connected to the collector circuit is 0.8 V. The collector current is,



Option 1:
$0.1\ mA$

Option 2:
$1\ mA$

Option 3:
$0.2\ mA$

Option 4:
$2\ mA$

Correct Answer:
$1\ mA$

Solution:
$v_{load} = I_C \times R_C$

$0.8 = I_C \times 800$

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$I_C = 10^{-3}A$

Q. 22 The total energy of an electron in the nth stationary orbit of the hydrogen atom can be obtained by.

Option 1:

$E_n = \frac{13.6}{n^2}eV$

Option 2:

$E_n = -\frac{1.36}{n^2}eV$

Option 3:

$E_n = -13.6 \times n^2 eV$

Option 4:

$E_n = \frac{13.6}{n^2}eV$

Correct Answer:

$E_n = -\frac{1.36}{n^2}eV$

Solution:

$E_n = \frac{-13.6z^2}{n^2}eV$

For hydrogen atom, z=1

$\therefore E_n = \frac{-13.6}{n^2}eV$

Hence, the answer is the option (2).

Q. 23 The de Broglie wavelength of an electron moving with kinetic energy of 144 eV is nearly :

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Option 1:

$102 \times 10^{-4} nm$

Option 2:

$102 \times 10^{-5} nm$

Option 3:

$102 \times 10^{-2} nm$

Option 4:

$102 \times 10^{-3} nm$

Correct Answer:

$102 \times 10^{-2} nm$

Solution:

$E = 144eV$

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 144 \times 1.6 \times 10^{-19}}}$$

$\lambda = 1.02 \times 10^{-9} m$

$\lambda = 102 \times 10^{-2} nm$

Hence, the answer is the option (3).

Q. 24 If the critical angle for total internal reflection from a medium to vacuum is 45^0, then the velocity of light in the medium is :

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Option 1:

$\frac{3}{\sqrt{2}} \times 10^8 m/s$

Option 2:

$\sqrt{2} \times 10^8 m/s$

Option 3:

$3 \times 10^8 m/s$

Option 4:

$1.5 \times 10^8 m/s$

Correct Answer:

$\frac{3}{\sqrt{2}} \times 10^8 m/s$

Solution:

$i_c = 45^\circ$

$$\mu_d^R = \frac{1}{\sin i_c} = \frac{V_{rarer}}{V_{denser}}$$

$$\frac{1}{\sin 45^\circ} = \frac{3 \times 10^8}{V_{denser}}$$

$$V_{denser} = \frac{3}{\sqrt{2}} = 2.121 \times 10^8 \frac{m}{s}$$

Hence, the answer is the option (1).

Q. 25 An ideal gas equation can be written as $P = \frac{\rho RT}{M_0}$ Where $\rho \; and \; M_0$ are respectively -

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Option 1:
Numberdensity, molar mass

Option 2:
mass density, molar mass

Option 3:
number density, mass of the gas

Option 4:
mass density, mass of the gas

Correct Answer:
mass density, molar mass

Solution:

$$PV = nRT$$

$$\Rightarrow P = \frac{1}{V} \cdot \frac{m}{M_0} \cdot RT$$

$$= \left(\frac{m}{V}\right)\left(\frac{RT}{M_0}\right)$$

$$\Rightarrow P = \frac{\rho RT}{M_0}$$

$$\rho = \frac{m}{v} = \text{ mass density}$$

$$M_0 = \text{ molar mass}$$

Hence, the answer is the option (2).

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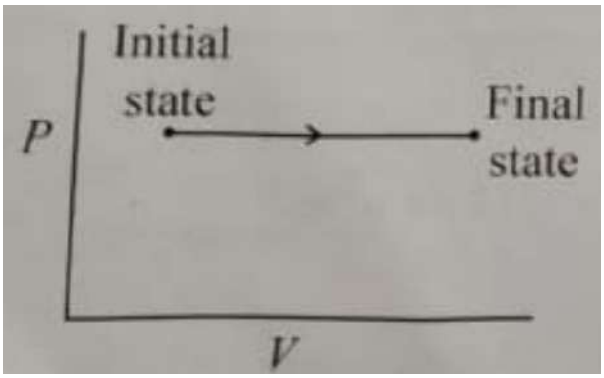
Q. 26 The $P-V$ diagram for an ideal gas in a position cylinder assembly undergoing a thermodynamic process is shown in the figure. The process is



Option 1:
isochoric

Option 2:
isobaric

Option 3:
isothermal

Option 4:
adiabatic

Correct Answer:
isobaric

Solution:
For the process pressure is same so it is an isobaric process.

Q. 27 The half life of a radioactive sample undergoing $\alpha-decay\ is\ 1.4\times10^{17}s$. If the number of nuclei in the sample is 2.0×10^{21}, the activity of the sample is nearly

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Option 1:
$10^5\ Bq$

Option 2:
$10^6\ Bq$

Option 3:
$10^3\ Bq$

Option 4:
$10^4\ Bq$

Correct Answer:
$10^4\ Bq$

Solution:
$T_{\frac{1}{2}} = 1.4 \times 10^7 s$

Activity = λN

$$= \left(\frac{ln(2)}{t_{\frac{1}{2}}}\right) \times N$$

$$= \frac{0.693}{1.41 \times 10^{17}} \times 2 \times 10^{21}$$

$$= 10^4 dps = 10^4 Bq$$

[1 disinteration per second = 1Bq]

Q. 28 A barometer is constructed using a liquid $\left(density = 760\ kg/m^3\right)$ .What would be the height of the liquid column, when a mercury barometer reads 76 cm ?

$\left(density\ of\ mercury = 13600 kg/m^3\right)$

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Option 1:
13.6 m

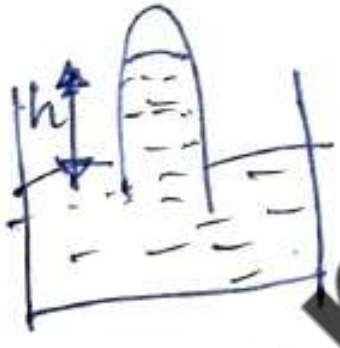
Option 2:
136 m

Option 3:
0.76 m

Option 4:
1.36 m

Correct Answer:
13.6 m

Solution:
d = 760 kg/m^3



$P_0 = sgh$ - Ready of barometer

$s_1 g_1 h_1 = s_2 g_2 h_2$

$760 \times h_1 = 13600 \times 76cm$

$h_1 = 1360cm$

$h_1 = 13.6m$

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Q. 29 What happens to the mass number and atomic number of an element when it emits γ -radiation?

Option 1:
Mass number and atomic number remain unchanged

Option 2:
Mass number remain unchanged while atomic number decreases by one

Option 3:
Mass number increases by four and atomic number increases by two.

Option 4:
Mass number decreases by four and atomic number decreases by two.

Correct Answer:
Mass number and atomic number remain unchanged

Solution:
γ - radiation are chargeless and massless

A gamma-ray is an electromagnetic wave. It is at the small wavelength end of the electromagnetic spectrum. A gamma-ray has no mass and no charge.

Emitting a gamma-ray makes no difference to the mass number or the atomic number but will make the nucleus more stable.

Q. 30 The E.M. wave with the shortest wavelength among the following is.

Option 1:
X -rays

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Option 2:
Gamma - rays

Option 3:
Microwaves

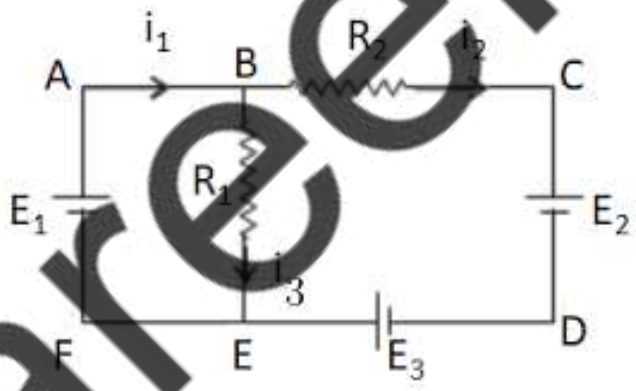
Option 4:
Ultraviolet rays

Correct Answer:
Gamma - rays

Solution:
Gamma rays have the shortest wavelength.

Hence, the answer is the option (2).

Q. 31 For the circut given below, the Kirchoft's loop rule for the loop BCDEB is given by the equation



Option 1:
$i_2R_2 + E_2 - E_3 - i_3R_1 = 0$

Option 2:
$i_2R_2 + E_2 + E_3 + i_3R_1 = 0$

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Option 3:

$-i_2R_2 + E_2 + E_3 + i_3R_1 = 0$

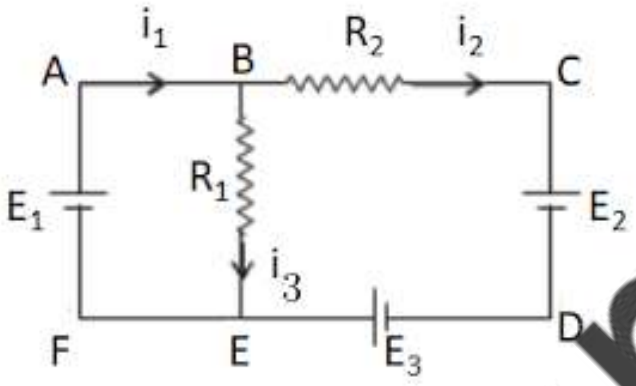
Option 4:

$-i_2R_2 + E_2 - E_3 + i_3R_1 = 0$

Correct Answer:

$i_2R_2 + E_2 - E_3 - i_3R_1 = 0$

Solution:



In loop BCDEB

$-i_2R_2 - E_2 + E_3 + i_3R_1 = 0$

$i_2R_2 + E_2 - E_3 - i_3R_1 = 0$

Q. 32 A plano-convex lens of unknown material and unknown focal length is given. With the help of a spherometer, we can measure the,

Option 1:

radius of curvature of the curved surface

Option 2:

aperature of the lens

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Option 3:
refractive index of the material

Option 4:
focal length of the lens

Correct Answer:
radius of curvature of the curved surface

Solution:
Spherometer is a device that measures the radius of curvature of lens.

Hence, the answer is the option (1).

Q. 33 The power of the biconvex lens is 10 dioptre and the radius of curvature of each surface is 10 cm. Then the refractive index of the material of the lens is,

Option 1:
$\frac{9}{8}$

Option 2:
$\frac{5}{3}$

Option 3:
$\frac{3}{2}$

Option 4:
$\frac{4}{3}$

Correct Answer:
$\frac{3}{2}$

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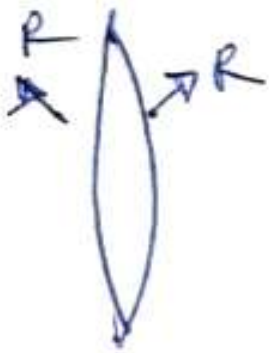
Solution:
P = 10D

R=10cm =0.1 m



$$\frac{1}{f} = (u-1)\left(\frac{1}{+R} - \frac{1}{(-R)}\right)$$

$$\frac{1}{f} = \frac{2(u-1)}{R}$$

$$10 = \frac{2(u-1)}{0.1}; u = \frac{3}{2}$$

Q. 34 A point mass 'm' is moved in a vertical circle of radius 'r' with the help of a string. The velocity of the mass is $\sqrt{7gr}$ at the lowest point. The tension in the string at the lowest point is

Option 1:
7 mg

Option 2:
8 mg

Option 3:
1 mg

Option 4:
6 mg

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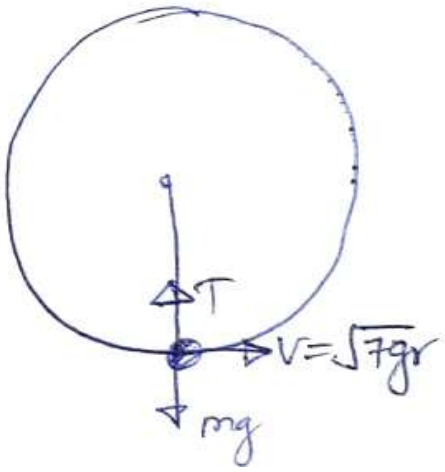
Correct Answer:
8 mg

Solution:

$v = \sqrt{7gr}$

At lowest option

$$T - mg = \frac{mv^2}{r}$$



$T = mg + 7mg$

$T = 8mg$

Q. 35 A wire of length L metre carrying a current of I ampere is bent in the form of a circle. Its magnetic moment is

Option 1:
$I\pi L^2/4 \ A\,m^2$

Option 2:
$2IL^2/\pi \ A\,m^2$

Option 3:
$IL^2/4\pi \ A\,m^2$

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Option 4:
$I L^2/4\ A\ m^2$

Correct Answer:
$I L^2/4\pi\ A\ m^2$

Solution:
$l = 2\pi r$

$M = IA = I \times \pi r^2$

$M = \frac{Il^2}{4\pi}$

Hence, the answer is the option (3).

Q. 36 An intrinsic semiconductor is converted into an n-type extrinsic semiconductor by doping it with

Option 1:
Aluminium

Option 2:
Silver

Option 3:
Germanium

Option 4:
Phosphorous

Correct Answer:
Phosphorous

Solution:
→ n-type extrinsic semiconductor is doped with pentavalent impurity

→ Phosphorous is pentavalent

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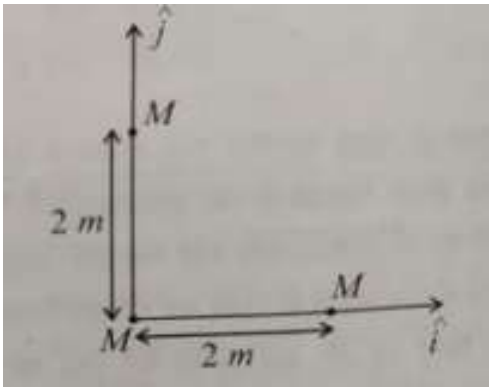
Hence, the answer is the option (4).

Q. 37 Three identical spheres, each of mass M, are placed at the corner of a right angle triangle with mutually perpendicular side equal to 2 m (see figure). Taking the point of intersection of the two mutually perpendicular sides as the origin. find the position vector of centre of mass



Option 1:
$\left(\hat{i}+\hat{j}\right)$

Option 2:
$\frac{2}{3}\left(\hat{i}+\hat{j}\right)$

Option 3:
$\frac{4}{3}\left(\hat{i}+\hat{j}\right)$

Option 4:
$2\left(\hat{i}+\hat{j}\right)$

Correct Answer:
$\frac{2}{3}\left(\hat{i}+\hat{j}\right)$

Solution:

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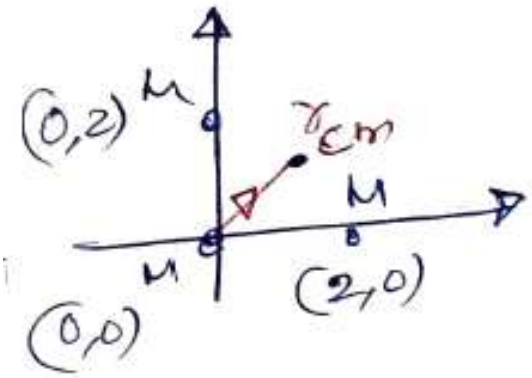

$$y_{cm} = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3}$$

$$y_{cm} = \frac{0 + 0 + m(2)}{3m} = \frac{-2}{3}$$

$$x_{cm} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$

$$x_{cm} = \frac{m(2) + 0 + 0}{3m}$$

$$x_{cm} = \frac{2}{3}$$

$$(y_{cm}, x_{cm}) = \left(\frac{2}{3}, \frac{2}{3}\right)$$

$$\vec{r}_{cm} = x_{cm}\hat{i} + y_{cm}\hat{j}$$

$$\vec{r}_{cm} = \frac{2}{3}\hat{i} + \frac{2}{3}\hat{j}$$

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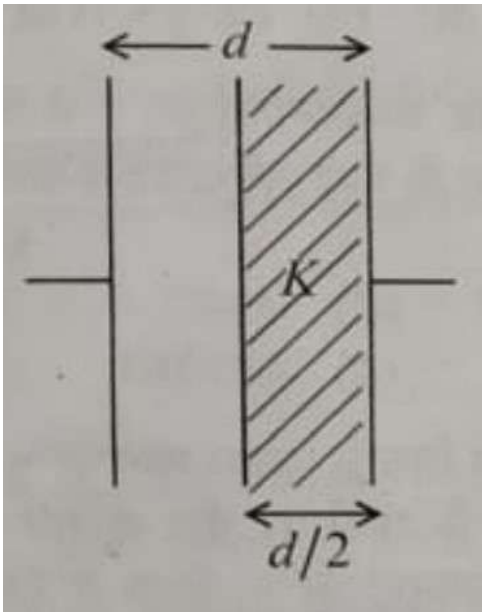
Q. 38 A parallel plate capacitor having cross sectional area *A* and separation *d* has air in between the plates . Now an insulating slab of same area but thickness *d/2* is inserted between the plates s shown in figure having dielectric constant $K(=4)$. The ratio of new capacitance to its original capacitance will be.



Option 1:
$8:5$

Option 2:
$6:5$

Option 3:
$4:1$

Option 4:
$2:1$

Correct Answer:
$8:5$

Solution:

$$C=\frac{A\varepsilon_0}{\frac{t_1}{K_1}+\frac{t_2}{K_2}}$$

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$$C_{new} = \frac{A\varepsilon_0}{\frac{\frac{d}{2}}{1} + \frac{\frac{d}{2}}{4}} = \frac{8A\varepsilon_0}{5d}$$

$$C_{original} = \frac{A\varepsilon_0}{d}$$

$$\frac{C_{new}}{C_{original}} = \frac{8}{5}$$

Q. 39 A person sitting in the ground floor of a building notices through the window, of height 1.5 m, a ball dropped from the roof of the building crosses the window in 0.1 s. What is the velocity of the ball when it is at the topmost point of the window? (g=10 m/s^2)

Option 1:
$14.5\ m/s$

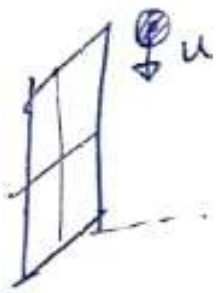
Option 2:
$4.5\ m/s$

Option 3:
$20\ m/s$

Option 4:
$15.5\ m/s$

Correct Answer:
$14.5\ m/s$

Solution:



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$$s = ut + \frac{1}{2}at^2$$

$$-1.5 = u(0.1) + \frac{1}{2}(-10)(0.1)^2$$

$$-1.45 = u(0.1)$$

$$u = -14.5m/s$$

At topmost point of window speed of the ball is 14.5m/s in downward direction.

Q. 40 The angle of 1' (minute of arc) in radian is nearly equal to :

Option 1:
$4.85 \times 10^{-4} rad$

Option 2:
$4.80 \times 10^{-6} rad$

Option 3:
$1.75 \times 10^{-2} rad$

Option 4:
$2.91 \times 10^{-4} rad$

Correct Answer:
$2.91 \times 10^{-4} rad$

Solution:

Angle in radian $= \frac{1^\circ}{60} \times \frac{\pi}{180}$

$= \frac{\pi}{10800}$

Hence, the answer is the option (4).

Q. 41 The wave nature of electrons was experimentally verified by,

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Option 1:
Hertz

Option 2:
Einstein

Option 3:
Davisson and Germer

Option 4:
de Broglie

Correct Answer:
Davisson and Germer

Solution:
Davission and Germer experiment was the experiment proof of De-brogile hypnothesis of matter wave.

Q. 42 The length of the string of a musical instrument is 90 cm and has a fundamental frequency of 120 Hz. Where should it be pressed to produce a fundamental frequency of 180 Hz?

Option 1:
60 cm

Option 2:
45 cm

Option 3:
80 cm

Option 4:
75 cm

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Correct Answer:
60 cm

Solution:

$L_1 = 0.9m$

$f_1 = 120Hz$

$f_1 = \frac{V}{2L_1} = 120$

$f_2 = \frac{V}{2L_2}$

$f_1 l_1 = f_2 l_2$

$120 \times 0.9 = 180 \times l_2$

$l_2 = 0.6$

Hence, the answer is the option (1).

Q. 43 A wheel with 20 metallic spokes each 1 m long is rotated with a speed of 120 rpm in a plane perpendicular to a magnetic field of 0.4 G. The induced emf between the axle and rim of the wheel will be, (1G = 10^{-4} T)

Option 1:
$2.51 \times 10^{-4} V$

Option 2:
$4.0 \times 10^{-5} V$

Option 3:
$2.51 V$

Option 4:
$4.0 \times 10^{-4} V$

Correct Answer:
$2.51 \times 10^{-4} V$

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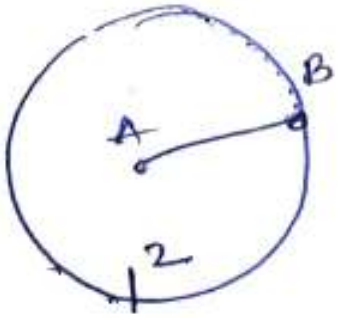
Solution:

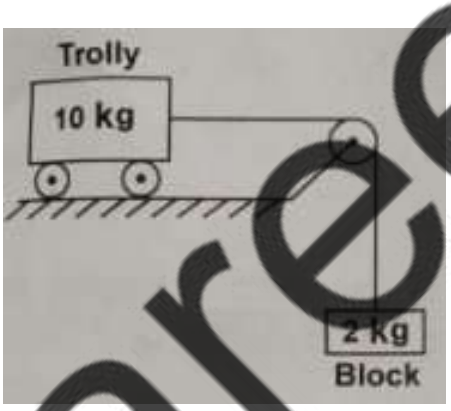
$$e_{AB} = \frac{1}{2} Bwl^2$$

$$e_{AB} = \frac{1}{2} \times 0.4 \times 10^{-4} \times 120 \times \frac{2\pi}{60}$$

$$e_{AB} = 0.4 \times 10^{-4}$$



Q. 44 Calculate the acceleration of the block and trolly system shown in the figure.The coefficient of kinetic friction between the trolly and the surface is 0.05.(g = 10 m/s^2. mass of the string is negligible and no other friction exists}



Option 1:
1.50 m/s^2

Option 2:
1.66 m/s^2

Option 3:
1.00 m/s^2

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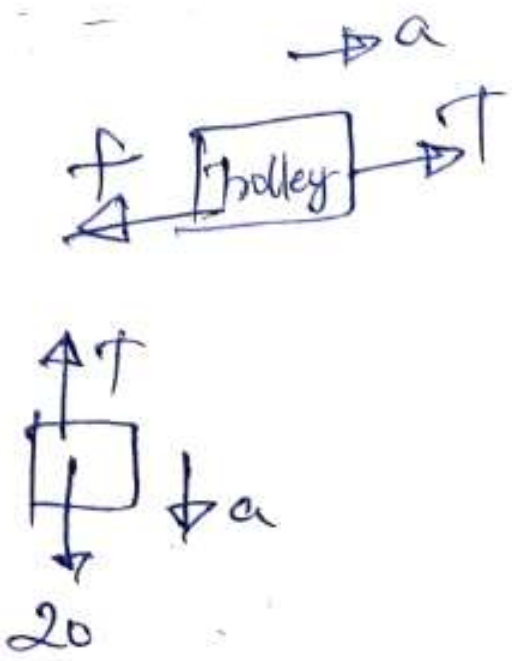
Option 4:
1.25 m/s^2

Correct Answer:
1.25 m/s^2

Solution:



$$T - f = ma$$

$$T - 5 = (10)a \qquad \text{.....(1)}$$

$$a = \frac{15}{12} = \frac{5}{4}$$

Q. 45 The electric field at a point on the equatorial plane at a distance r from the centre of a dipole having dipole moment $\vec{p}$ is given by ,($r >>$ separation of two charge forming the dipole, ϵ_0 - permittivity of free space)

Option 1:

$$\vec{E} = \frac{2\vec{P}}{4\pi\epsilon_0 r^3}$$

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Option 2:

$$\vec{E} = \frac{\vec{P}}{4\pi\epsilon_0 r^2}$$

Option 3:

$$\vec{E} = \frac{\vec{P}}{4\pi\epsilon_0 r^3}$$

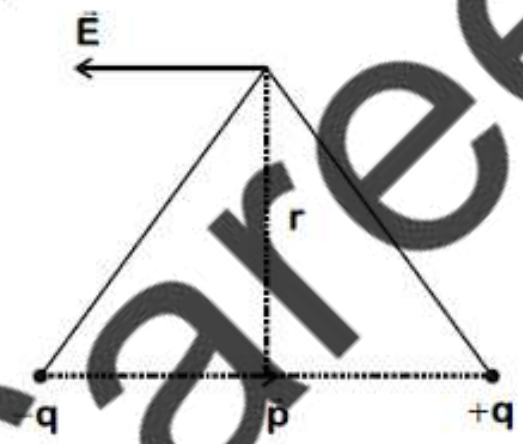
Option 4:

$$\overrightarrow{E} = \frac{-\overrightarrow{p}}{4\pi\varepsilon_0 r^3}$$

Correct Answer:

$$\overrightarrow{E} = \frac{-\overrightarrow{p}}{4\pi\varepsilon_0 r^3}$$

Solution:



$$\overrightarrow{E} = \frac{-K\overrightarrow{p}}{r^3}$$

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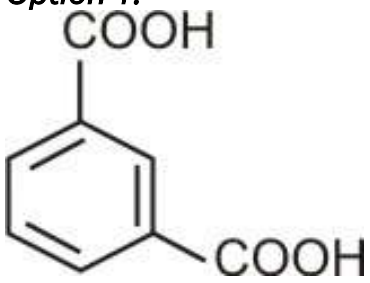
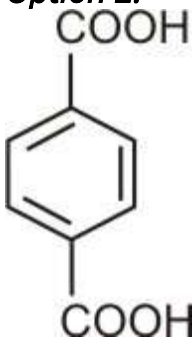
Chemistry

Q. 1 Which of the following acid will form an (a) Anhydride on heating and (b) Acid imide on strong heating with ammonia ?

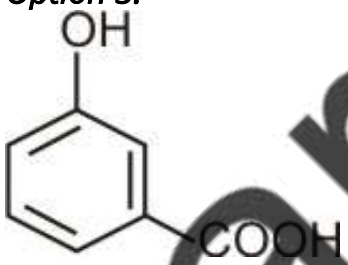
Option 1:



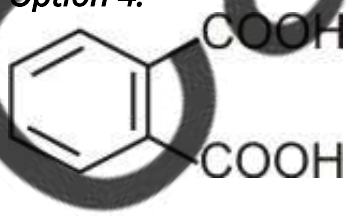
Option 2:



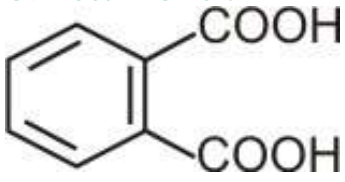
Option 3:



Option 4:



Correct Answer:



Solution:

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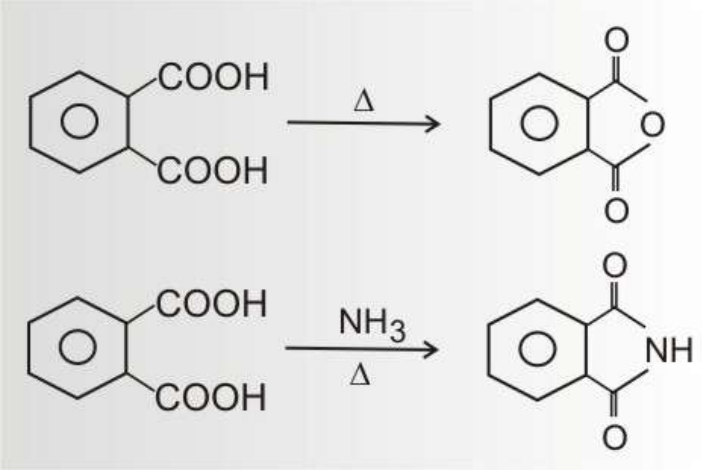
Phthalic acid forms phthalic anhydride and phthalimide



Hence, the correct answer is **option 4.**

Q. 2 Deficiency of which vitamin causes osteomalacia ?

Option 1:
Vitamin D

Option 2:
Vitamin K

Option 3:
Vitamin E

Option 4:
Vitamin A

Correct Answer:
Vitamin D

Solution:
Osteomalacia is caused by a deficiency of Vitamin D.

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Hence, the answer is the option (1).

Q. 3 In which of the sols, the colloidal particles are with negative charge ?

Option 1:
Haemoglobin

Option 2:
Starch

Option 3:
Hydrated Al_2O_3

Option 4:
TiO_2

Correct Answer:
Starch

Solution:
Starch forms a colloid with negatively charged particles.

Hence, the correct answer is **option 2.**

Q. 4 Identify the reaction from following having top position in EMF series (std.red.potential) according to their electrode potential at 298 K.

Option 1:
$Fe^{2+} + 2e^- \rightarrow Fe_{(s)}$

Option 2:
$Au^{2+} + 3e^- \rightarrow Au_{(s)}$

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Option 3:
$K^+ + 1e^- \rightarrow K_{(s)}$

Option 4:
$Mg^{2+} + 2e^- \rightarrow Mg_{(s)}$

Correct Answer:
$Au^{2+} + 3e^- \rightarrow Au_{(s)}$

Solution:
Au is the least reactive metal among Fe, K and Mg.

Therefore, $Au^{2+} + 3e^- \rightarrow Au_{(s)}$ will have the greatest reduction potential.

Hence, the answer is the option (2).

Q. 5 Which among the following salt solutions is basic in nature ?

Option 1:
Ammonium sulphate

Option 2:
Ammonium nitrate

Option 3:
Sodium acetate

Option 4:
Ammonium Chloride

Correct Answer:
Sodium acetate

Solution:
Salts of weak acids and strong bases are basic. Sodium acetate is a salt of NaOH and $\mathrm{CH_3COOH}$.

Hence, the answer is the option (3).

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Q. 6 Match the coordination number an type of hybridisation with distribution of hybrid orbitals in space based on Valence bond theory.

Coordination number and type of hybridisation	Distribution of hybrid orbitals in space
(a) 4, sp^3	(i) trigonal bipyramidal
(b) 4, dsp^2	(ii) Octahedral
(c) 5, sp^3d	(iii) Tetrahedral
(d) 6, d^2sp^3	(iv) Square planar

Select the correct option :

Option 1:
(a) - (iii) (b) - (iv) (c) - (i) (d) - (ii)

Option 2:
(a) - (iv) (b) - (i) (c) - (ii) (d) - (iii)

Option 3:
(a) - (iii) (b) - (i) (c) - (iv) (d) - (ii)

Option 4:
(a) - (ii) (b) - (iii) (c) - (iv) (d) - (i)

Correct Answer:
(a) - (iii) (b) - (iv) (c) - (i) (d) - (ii)

Solution:
(a) sp^3 - (iii) tetrahedral

(b) dsp^2 - (iv) square planer

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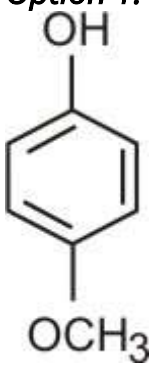
(c) sp^3d - (i) Trigonal bipyramidal

(d) d^2sp^3 - (ii) Octahedral

Hence, the answer is the option (1).

Q. 7 Which of the following substituted phenols is the strongest acid ?

Option 1:



Option 2:

Option 3:

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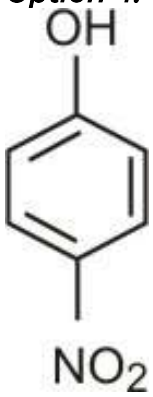
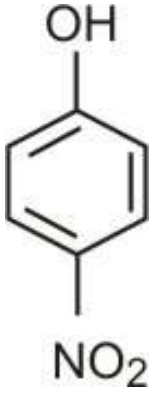
Option 4:

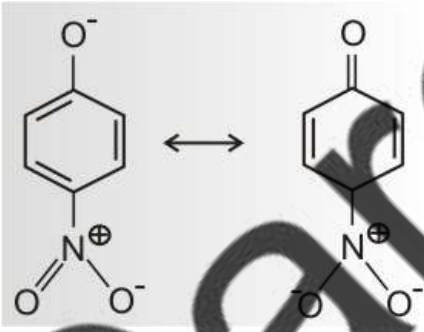


Correct Answer:



Solution:

Nitrophenol is the most acidic as its conjugate base is additionally stabilised by the electron withdrawing nature of nitro group.



Hence, the correct answer is **option 4.**

Q. 8 In collision theory of chemical reaction, Z_{AB} represents

Option 1:

the collision frequency of reactants, A and B

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Option 2:
steric factor

Option 3:
the fraction of molecules with energies equal to E_a

Option 4:
the fraction of molecules with energies greater than E_a

Correct Answer:
the collision frequency of reactants, A and B

Solution:
In the collision theory, the rate is given as the rate $= p\ Z_{AB}\ e^{-Ea/RT}$

Where Z_{AB} represents the collision frequency of reactants A and B.

p is steric factor

R is the universal gas constant.

E_a is the Activation energy required for the reaction.

Hence, the answer is the option (1).

Q. 9 The reaction of concentrated sulphuric acid with carbohydrates $(C_{12}H_{22}O_{11})$ is an example of

Option 1:
Oxidation

Option 2:
Reduction

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Option 3:
Sulphonation

Option 4:
Dehydration

Correct Answer:
Dehydration

Solution:

$$C_{12}H_{22}O_{11} \xrightarrow[Conc.]{H_2SO_4} 12C + 11H_2O$$

The above reaction is an example of dehydration.

Hence, the answer is the option (4).

Q. 10 What is the role of gypsum, $CaSO_4.2H_2O$ in setting of cement ? Identify the correct option from the following :

Option 1:
to provide water molecules for hydration process

Option 2:
to help to remove water molecules

Option 3:
to slow down the setting process

Option 4:
to fasten the settings process

Correct Answer:
to slow down the setting process

Solution:

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Gypsum is an important ingredient of cement as it slows down the setting process, without gypsum cement will instantly when comes in contact with cement.

Hence, the correct answer is **option 3.**

Q. 11 Which of the following statement is not true about glucose ?

Option 1:
It contains five hydroxyl groups.

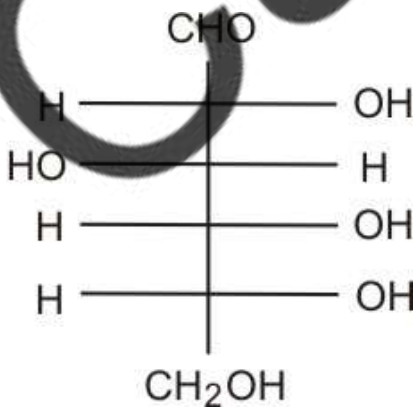
Option 2:
It is a reducing sugar.

Option 3:
It is an aldopentose.

Option 4:
It is an aldohexose.

Correct Answer:
It is an aldopentose.

Solution:
The structure of glucose is



It contains 5 hydroxyl groups, it is a reducing sugar as it is an aldohexose.

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Hence, the correct answer is **option 3.**

Q. 12 The minimum pressure required to compress $600\ dm^3$ of a gas at 1 bar to $150\ dm^3$ at 40^oC is

Option 1:
0.2 bar

Option 2:
1.0 bar

Option 3:
2.5 bar

Option 4:
4.0 bar

Correct Answer:
4.0 bar

Solution:
From Boyle's law,

$$P_1V_1 = P_2V_2$$

$$\Rightarrow 1 \times 600 = P_2 \times 150$$

$$\Rightarrow P_2 = 4\ bar$$

Hence, the correct answer is **option 4.**

Q. 13 If 8 g of a non - electrolyte solute is dissolved in 114 g of n-octane to reduce its vapour pressure to 80%, the molar mass (in g mol $^{-1}$) of the solute is

[Given that molar mass of n-octane is $114\ \text{g mol}^{-1}$]

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Option 1:
60

Option 2:
80

Option 3:
20

Option 4:
40

Correct Answer:
40

Solution:
We know that,

Relative lowering of vapour pressure = Mole fraction of solute

$$\frac{P_o - P_s}{P_o} = \frac{n_B}{n_A + n_B}$$

$$\Rightarrow 0.2 = \frac{n_B}{n_A + n_B}$$

Assuming that the solution is very dilute

i.e. $n_A + n_B \cong n_A$

We have,

$$0.2 = \frac{n_B}{n_A}$$

$$\Rightarrow n_B = 0.2 \times n_A$$

$$\Rightarrow n_B = 0.2$$

$$\Rightarrow \frac{8}{M_B} = 0.2$$

$$\Rightarrow M_B = 40$$

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Note: NTA's assumption of a very dilute solution is not correct, but no other option can be arrived at by not neglecting n_B with respect to n_A.

Hence, the answer is the option (4).

Q. 14 Which one of the following reactions does not come under hydrolysis type reaction ?

Option 1:
$Li_3N_{(s)} + 3H_2O_{(1)} \rightarrow NH_{3(g)} + 3LiOH_{(aq)}$

Option 2:
$2F_{2(g)} + 2H_2O_{(1)} \rightarrow 4HF_{(aq)} + O_{2(g)}$

Option 3:
$P_4O_{10(s)} + 6H_2O_{(1)} \rightarrow 4H_3PO_{4(aq)}$

Option 4:
$SiCl_{4(l)} + 2H_2O_{(l)} \rightarrow SiO_{2(s)} + 4HCl_{(aq)}$

Correct Answer:
$2F_{2(g)} + 2H_2O_{(1)} \rightarrow 4HF_{(aq)} + O_{2(g)}$

Solution:
$2F_{2(g)} + 2H_2O_{(1)} \rightarrow 4HF_{(aq)} + O_{2(g)}$

The above reaction is an example of a Redox reaction wherein water is oxidized by fluoride.

Hence, the answer is the option (2).

Q. 15 Which of the following statement is NOT true about acid rain ?

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Option 1:
Causes no damage to monuments like Taj Mahal.

Option 2:
It is harmful for plants

Option 3:
Its pH is less than 5.6

Option 4:
It is due to reaction of SO_2, NO_2 and CO_2 with rain water

Correct Answer:
Causes no damage to monuments like Taj Mahal.

Solution:
Acid rain causes damage to monuments like Taj Mahal as marble (calcium carbonate) is corroded due to acid rain

$CaCO_3 + H_3SO_4 \rightarrow CaSO_4 + CO_2 + H_2O$Hence, the correct answer is **option 1.**

Q. 16 Identify the wrongly matched pair.

Option 1:

Molecule	Shape or geometry of molecule
SF_6	Octahedral

Option 2:

Molecule	Shape or geometry of molecule
$BeCl_2$	Linear

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Option 3:

Molecule	Shape or geometry of molecule
NH_3	Trigonal pyramidal

Option 4:

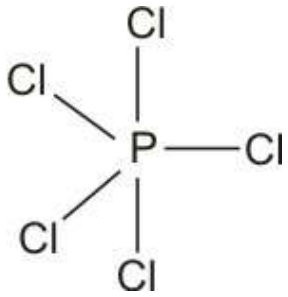
Molecule	Shape or geometry of molecule
PCl_5	Trigonal planar

Correct Answer:

Molecule	Shape or geometry of molecule
PCl_5	Trigonal planar

Solution:

The shape of PCl_5 is trigonal bipyramidal.



Hence, the correct answer is **option 4.**

Q. 17 One mole of carbon atom weighs 12 g, the number of atoms in it is equal to, (Mass of carbon - 12 is $1.9926 \times 10^{-23}\mathrm{g}$)

Option 1:

6.022×10^{22}

Option 2:

12×10^{22}

Option 3:

6.022×10^{23}

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Option 4:

1.2×10^{23}

Correct Answer:

6.022×10^{23}

Solution:

The number of atoms present in 12 g of carbon is equal to Avogadro's number (6.022×10^{23})

Hence, the answer is the option (3).

Q. 18 If for a certain reaction $\Delta_r H$ is $30 kJ mol^{-1}$ at 450 K, the value of $\Delta_r S$ (in JK^{-1} mol^{-1}) for which the same reaction will be spontaneous at the same temperature is

Option 1:

-33

Option 2:

33

Option 3:

-70

Option 4:

70

Correct Answer:

70

Solution:

For spontaneity,

$\Delta G < 0$

$\Rightarrow \Delta H - T\Delta S < 0$

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$$\Rightarrow \Delta H - T\Delta S$$

$$\Rightarrow \Delta S > \frac{\Delta H}{T}$$

$$\Rightarrow \Delta S > \frac{30 \times 10^3}{450} > 66.67\ Jk^{-1}mol^{-1}$$

Hence, the answer is the option (4).

Q. 19 A liquid compound (x) can be purified by steam distillation only if it is

Option 1:
Not steam volatile, miscible with water

Option 2:
Steam volatile, miscible with water

Option 3:
Not steam volatile, immiscible with water

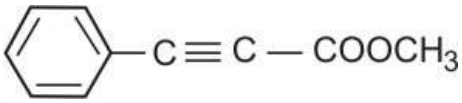
Option 4:
Steam volatile, immiscible with water

Correct Answer:
Steam volatile, immiscible with water

Solution:
Steam distillation is applicable for compounds that are steam-volatile and immiscible with water.

Hence, the answer is the option (4).

Q. 20 How many (i) sp^2 hybridised carbon atoms and (ii) π bonds are present in the following compound ?



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Option 1:
8,6

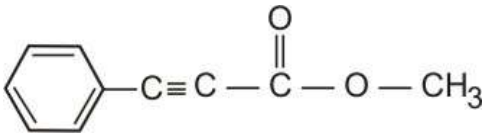
Option 2:
7,6

Option 3:
8,5

Option 4:
7,5

Correct Answer:
7,6

Solution:



Total number of sp^2 hybridised carbons = 7

Total number of $\overline{\wedge}$ bonds = 6

Hence, the correct answer is **option 2.**

Q. 21 Reaction of propanamide with ethanolic sodium hydroxide and bromine will give

Option 1:
Methylamine

Option 2:
Propylamine

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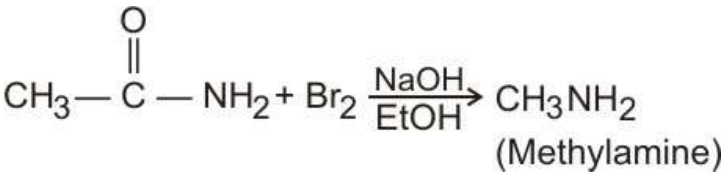
Option 3:
Aniline

Option 4:
Ethylamine

Correct Answer:
Methylamine

Solution:
This is an example of Hoffman Bromamide degradation.



Hence, the correct answer is **option 1.**

Q. 22 Identify the correct statement from the following.

Option 1:
Lithium and Magnesium show some similarities in their physical properties as they are diagonally placed in the periodic table.

Option 2:
Lithium is softer among all alkali metals.

Option 3:
Lithium chloride is deliquescent and crystallises as a hydrate, $LiCl.H_2O$.

Option 4:
The order of hydration enthalpies of alkaline earth cations
$Be^{2+} < Mg^{2+} < Ca^{2+} < Sr^{2+} < Ba^{2+}$

Correct Answer:
Lithium and Magnesium show some similarities in their physical properties as they are diagonally placed in the periodic table.

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Solution:
The softness of Alkali metals decreases down the group as the strength of the metallic bond decreases down the group.

$LiCl$ crystallizes as a hydrate $LiCl.2H_2O$ but it is deliquescent.

Hydration enthalpy decreases down the group as the size of the cations increases down group and charge to size ratio decreases down the group.

$Li\ and\ Mg$ are diagonally present to each other and lithium shows many resemblances to magnesium, which is due to the similar atomic radii of the two.

Hence, the correct answer is **option 1.**

Q. 23 Match the following aspects with the respective metal.

Aspects	Metal
(a) The metal which reveals a maximum number of oxidation states	(i) Scandium
(b) The metal although placed in 3d block is considered not as transition element	(ii) Copper
(c) The metal which does not exhibit variable oxidation states	(iii) Manganese
(d) The metal which in +1 oxidation state in aqueous solution undergoes disproportionation	(iv) Zinc

Select the correct option :

Option 1:
(a) - (iii) (b) - (iv) (c) - (i) (d) -(ii)

Option 2:
(a) - (iii) (b) - (i) (c) - (iv) (d) -(ii)

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Option 3:
(a) - (ii) (b) - (iv) (c) - (i) (d) -(iii)

Option 4:
(a) - (i) (b) - (iv) (c) - (ii) (d) -(iii)

Correct Answer:
(a) - (iii) (b) - (iv) (c) - (i) (d) -(ii)

Solution:
(a) - (iii)

Manganese shows the maximum oxidation state

(b) - (iv)

Zinc belongs to the 3d series but is not a transition element.

(c) - (i)

Sodium shows an oxidation state of +3 only

(d) - (ii)

Copper (i) is disproportionate to copper (o) and copper (ii) in aqueous solutions.

Hence, the answer is the option (1).

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Q. 24 Match the element in column I with that in Column II.

Column I	Column II
(a) Copper	(i) Non- metal
(b) Fluorine	(ii) Transition metal
(c) Silicon	(iii) Lanthanoid
(d) Cerium	(iv) Metalloid

Identify the correct match :

Option 1:
(a) - (ii) , (b) - (i) , (c) - (iv), (d) -(iii)

Option 2:
(a) - (iv) , (b) - (iii) , (c) - (i), (d) -(ii)

Option 3:
(a) - (i) , (b) - (ii) , (c) - (iii), (d) -(iv)

Option 4:
(a) - (ii) , (b) - (iv) , (c) - (i), (d) -(iii)

Correct Answer:
(a) - (ii) , (b) - (i) , (c) - (iv), (d) -(iii)

Solution:
(a) - (ii)

Copper is a transition metal

(b) - (i)

Fluorine is a non metal

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(c) - (iv)

Silicon is a metalloid

(d) - (iii)

Cerium is a lanthanoid

Hence, the answer is the option (1).

Q. 25 Among the compounds shown below which one revealed a linear structure ?

Option 1:
$HOCl$

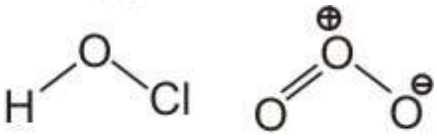
Option 2:
O_3

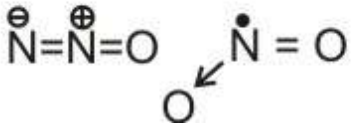
Option 3:
N_2O

Option 4:
NO_2

Correct Answer:
N_2O

Solution:
The structures are





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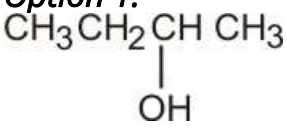
N_2O is linear among the given species.

Hence, the correct answer is **option 3.**

Q. 26 $CH_3CH_2CH=CH_2 \xrightarrow[H_2O_2OH^-]{B_2H_6} Z$

What is z ?

Option 1:



Option 2:

$CH_3CH_2CH_2CHO$

Option 3:

$CH_3CH_2CH_2CH_3$

Option 4:

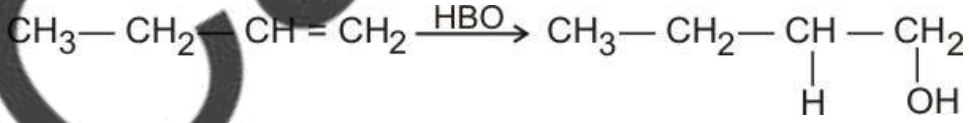
$CH_3CH_2CH_2CH_2OH$

Correct Answer:

$CH_3CH_2CH_2CH_2OH$

Solution:

Hydroboration - oxidation causes anti-markovnikov addition



Hence, the correct answer is **option 4.**

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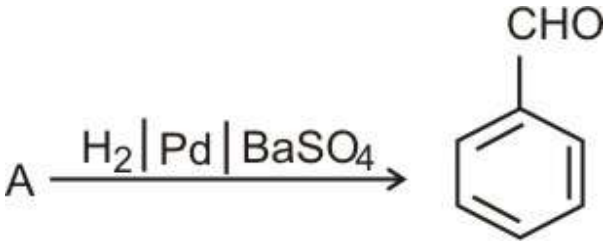
Q. 27 Identify compound (A) in the following reaction :



Option 1:
Toluene

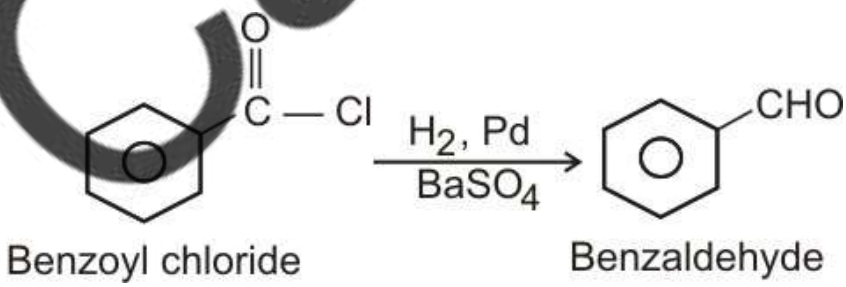
Option 2:
Acetophenone

Option 3:
Benzoic acid

Option 4:
Benzoyl chloride

Correct Answer:
Benzoyl chloride

Solution:
The given reaction is Rosenmend Reduction



Hence, the correct answer is **option 4.**

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Q. 28 Which of the following statement is correct about Bakelite?

Option 1:
It is an additional polymer

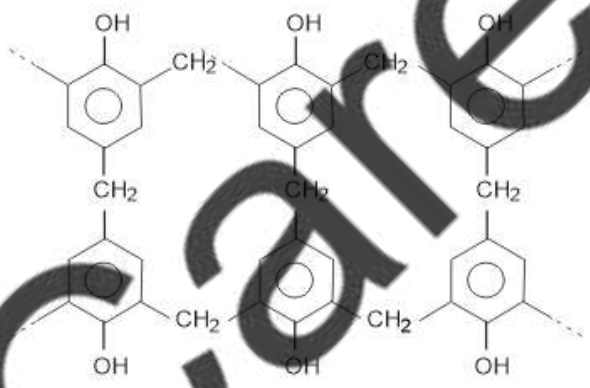
Option 2:
It is a branched chain polymer

Option 3:
It is a linear polymer

Option 4:
It is a cross linked polymer

Correct Answer:
It is a cross linked polymer

Solution:
Bakelite is a cross linked polymer of Phenol and formeldehyde



Hence the correct option is option 4.

Q. 29 Which of the following compound is most reactive in electrophilic aromatic substitution?

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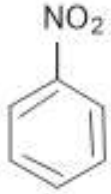
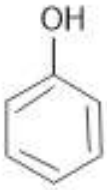
Option 1:

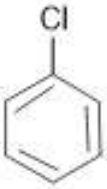


Option 2:

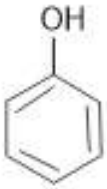
Option 3:



Option 4:



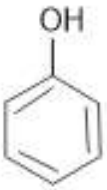
Correct Answer:



Solution:

Electron donating group activate the benzene ring towards electrophilic aromatic substitution

Thus, the most reactive compound is



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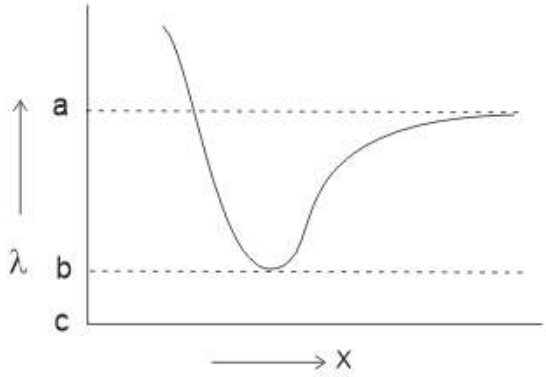
Hence, the correct answer is option 3

Q. 30 The potential energy (y) curve for H_2 formation as a function of internuclear distance (x) of the H atoms is shown below.



The bond energy of H_2 is

Option 1:
$\frac{(c-a)}{2}$

Option 2:
$\frac{(b-a)}{2}$

Option 3:
$(c-a)$

Option 4:
$(b-a)$

Correct Answer:
$(b-a)$

Solution:

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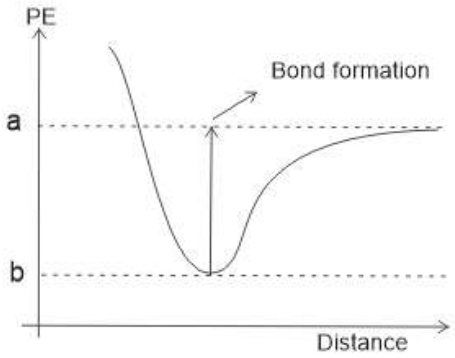

The minima in the above curve represent bond formation.

Thus, the bond energy is (b-a), which is the amount of energy released when two-mole hydrogen atoms are forming one mole of hydrogen molecule.

Hence, the correct answer is option 4.

Q. 31 Which of the following compounds shows both, Frenkel as well as Schottky defects?

Option 1:
AgI

Option 2:
$NaCl$

Option 3:
ZnS

Option 4:
$AgBr$

Correct Answer:
$AgBr$

Solution:
AgBr shows both Frenkel as well as Schottky defect.

Hence, the correct answer is option 4.

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Q. 32 Identify the incorrect statement from the following

Option 1:
Lanthanoids reveal only +3 oxidation state.

Option 2:
The lanthanoids ions other than the f^0 type and f^{14} type all paramagnetic.

Option 3:
The overall decrease in atomic and ionic radii from lanthanum to lutetium is called lanthanoid contraction.

Option 4:
Zirconium and Hafnium have identical radii of 160 pm and 159 pm, respectively as a consequence of lanthanoid contraction.

Correct Answer:
Lanthanoids reveal only +3 oxidation state.

Solution:
Lanthanoids can show oxidation states of +2, +3 as well as +4.

Hence, the answer is the option (1).

Q. 33 At standard conditions, if the change in the enthalpy for the following reaction is $-109kJmol^{-1}$.

$$H_{2(g)} + Br_{2(g)} \rightarrow 2HBr_{(g)}$$

Given that bond energy of $H_2 and Br_2$ is $435kJmol^{-1}\ and\ 192kJmol^{-1}$ respectively, What is the bond energy (in kJ mol^{-1}) of HBr?

Option 1:
736

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Option 2:

518

Option 3:

259

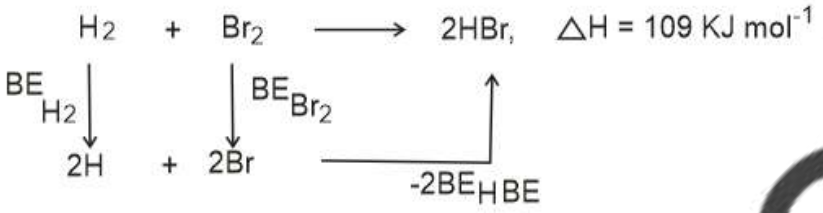
Option 4:

368

Correct Answer:

368

Solution:



$$H_2 \rightarrow 2H \quad B.E_{H_2}$$
$$Br_2 \rightarrow 2Br \quad B.E_{Br_2}$$
$$2H + 2Br \rightarrow 2HBr \quad -2B.E_{HBr}$$
$$H_2 + Br_2 \rightarrow 2HBr$$

From the Bond energy concept, we have

$$BE_{H_2} + BE_{Br_2} - 2BE_{HBr} = \Delta H_{HBr}$$
$$\rightarrow 435 + 192 - 2x = -109$$
$$\rightarrow 2x = 736$$
$$\rightarrow x = 368\ kJ\ mol^{-1}$$

Hence the correct answer is option 4

Q. 34 Which of the following oxide is amphoteric in nature?

Option 1:

SiO_2

Option 2:

GeO_2

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Option 3:
CO_2

Option 4:
SnO_2

Correct Answer:
SnO_2

Solution:
SnO_2 is amphoteric.

SiO_2, GeO_2, and CO_2 are acidic.

Going down the group acidic nature of oxides decreases as the electronegativity difference between oxygen and element increases.

Hence, the answer is the option (4).

Q. 35 The half-life for a zero-order reaction having $0.02M$ initial concentration of reactant is $100s$. The rate constant $(in\ molL^{-1}s^{-1})$ for the reaction is

Option 1:
2.0×10^{-4}

Option 2:
2.0×10^{-3}

Option 3:
1.0×10^{-2}

Option 4:
1.0×10^{-4}

Correct Answer:
1.0×10^{-4}

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Solution:
For a zero order reaction

$$t_{\frac{1}{2}} = \frac{[A_o]}{2K}$$

$$\Rightarrow 100 = \frac{0.02}{2K}$$

$$\Rightarrow K = 10^{-4}\frac{mol}{lit\ \ sec}$$

Hence the correct answer is option 4.

Q. 36 Which of the following is not true about chloramphenicol?

Option 1:
It is a broad spectrum antibiotic.

Option 2:
It is not bactericidal.

Option 3:
It is bacteriostatic.

Option 4:
It inhibits the growth of the only gram positive bacteria.

Correct Answer:
It inhibits the growth of the only gram positive bacteria.

Solution:
Chloramphenicol is a broad-spectrum bacteriostatic antibiotic. It inhibits the growth of gram-positive as well as gram-negative bacteria.

Hence the correct answer is option 4.

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Q. 37 Match the elements in Column I with methods of purification in Column II

Column I	Column II
(a) Boron	(i) Van Arkel method
(b) Tin	(ii) Mond's Process
(c) Zirconium	(iii) Liquation
(d) Nickel	(iv) Zone refining

Option 1:
(a)-(iv) (b)-(iii) (c)-(ii) (d)-(i)

Option 2:
(a)-(ii) (b)-(i) (c)-(iv) (d)-(iii)

Option 3:
(a)-(iii) (b)-(iv) (c)-(i) (d)-(ii)

Option 4:
(a)-(iv) (b)-(iii) (c)-(i) (d)-(ii)

Correct Answer:
(a)-(iv) (b)-(iii) (c)-(i) (d)-(ii)

Solution:
(a) - (iv)

Boron - Zone refining

(b) - (iii)

Tin - Liquation

(c) - (i)

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Zirconium - Van Arkel method

(d) - (ii)

Nickel - Mond's process

Hence the correct answer is option 4.

Q. 38 In a typical fuel cell, the reactants (R) and product (P) are

Option 1:
$R = H_{2(g)}, O_{2(g)}; P = H_2O_{(l)}$

Option 2:
$R = H_{2(g)}, O_{2(g)}; Cl_{2(g)}; P = HClO_{4(aq)}$

Option 3:
$R = H_{2(g)}, N_{2(g)}; P = NH_3(aq)$

Option 4:
$R = H_{2(g)}, O_{2(g)}; P = H_2O_2(l)$

Correct Answer:
$R = H_{2(g)}, O_{2(g)}; P = H_2O_{(l)}$

Solution:
Fuel cell reaction

$$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O(l)$$

Hence, the answer is the option (1).

Q. 39 Isotonic solution have same

Option 1:
freezing temperature

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Option 2:
osmotic pressure

Option 3:
boiling temperature

Option 4:
vapour pressure

Correct Answer:
osmotic pressure

Solution:
Isotonic solutions have the same Osmatic Pressure

Hence, the answer is the option (2).

Q. 40 The solubility product for a salt of the type AB is 4×10^{-8}. What is the molarity of its standard solution?

Option 1:
$16 \times 10^{-16}\ \mathrm{mol/L}$

Option 2:
$2 \times 10^{-16}\ \mathrm{mol/L}$

Option 3:
$4 \times 10^{-4}\ \mathrm{mol/L}$

Option 4:
$2 \times 10^{-4}\ \mathrm{mol/L}$

Correct Answer:
$2 \times 10^{-4}\ \mathrm{mol/L}$

Solution:

$$\underset{}{\mathrm{AB}} \rightleftharpoons \underset{\mathrm{s}}{\mathrm{A^{x+}}} + \underset{\mathrm{s}}{\mathrm{B^{x-}}}, \mathrm{K_{sp}} = 4 \times 10^{-8}$$

We have,

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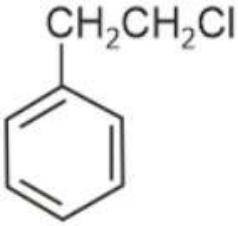
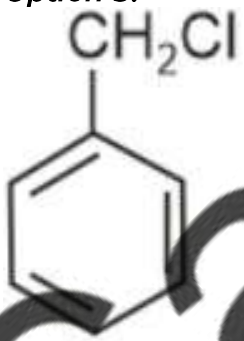
$S^2 = 4 \times 10^{-8}$
$S = 2 \times 10^{-4} M$

Hence, the answer is the option (4).

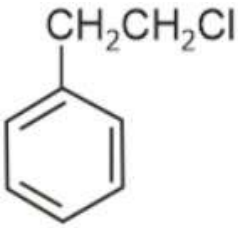
Q. 41 Which of the following will not undergo $S_N 1$ reaction with $\bar{O}H$?

Option 1:
$(CH_3)_3 CCl$

Option 2:


Option 3:


Option 4:
$CH_2 = CH - CH_2Cl$

Correct Answer:


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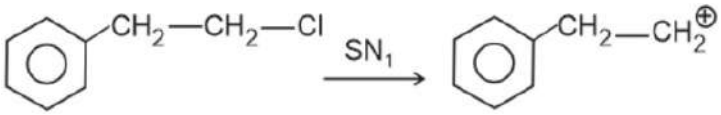
Solution:



Primary $C^{\oplus}$ is not formed very easily and hence, it does not give S_N, reaction .

Hence the right answer is option 2

Q. 42 Which of the following is a free radical substitution reaction?

Option 1:
Acetylenewith HBr

Option 2:
Methane with Br_2/hv

Option 3:
Propene with HBr/$(C_6H_5COO)_2$

Option 4:
Benzene with Br_2/$AlCl_3$

Correct Answer:
Methane with Br_2/hv

Solution:

$$CH_4 + Br_2 \underset{hv}{\rightarrow} CH_3Br + HBr$$

is an example of a free radical substitution reaction

Hence, the answer is the option (2).

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Q. 43 Match the compounds of Xe in column I with the molecular structure in column II

	Column I		Column II
(a)	XeF_2	(i)	Square Planar
(b)	XeF_4	(ii)	Linear
(c)	XeO_3	(iii)	Square pyramidal
(d)	$XeOF_4$	(iv)	Pyramidal

Option 1:
(a) - (ii) (b) - (iv) (c) - (iii) (d) - (i)

Option 2:
(a) - (ii) (b) - (iii) (c) - (i) (d) - (iv)

Option 3:
(a) - (ii) (b) - (i) (c) - (iv) (d) - (iii)

Option 4:
(a) - (ii) (b) - (i) (c) - (iii) (d) - (iv)

Correct Answer:
(a) - (ii) (b) - (i) (c) - (iv) (d) - (iii)

Solution:
The hybridization of the com

Molecule	Hybridization	Bond pairs	lone pairs	Shape
XeF_2	sp^3d	2	3	Linear
XeF_4	sp^3d^2	4	2	Square Planar

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XeO_3	sp^3	1	3	Pyramidal
$XeOF_4$	sp^3d^2	1	5	Square pyramidal

Hence, the answer is the option (3).

Q. 44 The number of angular nodes and radial nodes in 3s orbital are

Option 1:
1 and 0, respectively

Option 2:
3 and 0, respectively

Option 3:
0 and 1, respectively

Option 4:
0 and 2, respectively

Correct Answer:
0 and 2, respectively

Solution:
For 3s orbit

n = 3
l = 0

Angular nodes = l = 0

Radial nodes = (n - l - 1) = 2

Hence, the answer is the option (4).

Q. 45 The oxidation number of the underlined atom in the following species

Identify the incorrect option.

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Option 1:
$\underline{Cl}O_3^- \ is \ +5$

Option 2:
$K_2 \ \underline{Cr}_2O_7 \ \ is +6$

Option 3:
$H \ \underline{Au} \ Cl_4 \ \ is +3$

Option 4:
$Cu_2 \ \ \underline{O} \ is -1$

Correct Answer:
$Cu_2 \ \ \underline{O} \ is -1$

Solution:
In Cu_2O, Cu is in $+1$ oxidation state and oxygen is in -2 oxidation state.

Hence, the answer is the option ().

Biology

Q. 1 After about how many years of formation of earth, life appeared on this planet ?

Option 1:
50 million years

Option 2:
500 millions years

Option 3:

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Option 4:
500 billion years

Correct Answer:
500 millions years

Solution:
- Earth originated 4.5 billion years ago
- Life originated 4 bilion years ago
- Thus Life originated 500 million years after Origin of Earth

---

Q. 2 Which of the following statement about cork cambium is incorrect ?

Option 1:
It forms a part of periderm

Option 2:
It is responsible for the formation of lenticels

Option 3:
It is couple of layers thick

Option 4:
It forms secondary cortex on its outerside

Correct Answer:
It forms secondary cortex on its outerside

Solution:
Cork cambium or phellogen forms during secondary growth in the extra-stellar region of plants. It forms the cork or phellem outer side and the secondary cortex or phelloderm inner side.

Hence, option 4 is incorrect.

Q. 3 Select the incorrect statement

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Option 1:
Elements most easily mobilized in plants from one region to another are : phosphorus, sulphur, nitrogen and potassium

Option 2:
Transport of molecules in phloem can be bidirectional

Option 3:
Movement of minerals in xylem is unidirectional

Option 4:
Unloading of sucrose at sink does not involve the utilization of ATP

Correct Answer:
Unloading of sucrose at sink does not involve the utilization of ATP

Solution:
The solutes present in the phloem get unloaded with the help of companion cells to transport them into sinks. This process happens actively with the utilization of ATP.

Q. 4 Which of the following statements is incorrect regarding the phosphorus cycle?

Option 1:
It is sedimentary cycle

Option 2:
Phosphates are the major form of phosphorus reservoir

Option 3:
Phosphorus solubilising bacteria facilitate the release of phosphorus from organic remains

Option 4:
There is appreciable respiratory release of phosphorus into atmosphere

Correct Answer:
There is appreciable respiratory release of phosphorus into atmosphere

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Solution:
Phosphorus cycle occurs on land

Q. 5 Match the items in Column I with those in Column II:

Column I	Column II
(a) Herbivores -Plants	(i) Commensalism
(b) Mycorrhiza -Plants	(ii) Mutualism
(c) Sheep - Cattle	(iii) Predation
(d) Orchid - Tree	(iv) Competition

Select the correct option from the following:

Option 1:
a-iii, b-ii, c-iv, d-i

Option 2:
a-ii, b-i, c-iii, d-ii

Option 3:
a-i, b-iii, c-iv, d-ii

Option 4:
a-iv, b-ii, c-i, d-iii

Correct Answer:
a-iii, b-ii, c-iv, d-i

Solution:
Predation: It is a type of interaction in which the members of one species capture, kill and eat up members of other species. The sapecies that capture is called **predator** and the other that is captured is called **prey**.

Herbivory is a form of predation in which the prey organism is a plant.

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Mutualism: It is an obligate association of two organisms in which each derives benefit from the other. The two organisms often live together and can't live separately.

Mycorrhiza, which is derived from the Greek words **myco** meaning **fungus** and **rhizo** meaning **roots**, refers to fungal partner showing mutal relationship with roots of vascular plants. Fungus allows plant to draw more nutrients and water from the soil and increases the abiotic and biotic stress resistant of the host and in return fungus colonizes in the roots of the plant.

Competition: It is the struggle between two or more organisms for obtaining various requirements for their survival. It is both **intraspecific** (between organisms of same species) and **interspecific** (between organisms of different species).

Sheep (cattle) shows **interspecific competition** as this competition arises when organisms of different species belong to same trophic level or having similar feeding habit e.g. in **grassland**, severe competition occurs between herbivores like rabbits, deer, sheep etc., as all feed upon grass.

Commensalism: It is an interaction between two orgaanisms of different species, benefitting only one species, the other species is neither benefitted nor harmed. The species which is benefitted is termed **commensal** and the other species is called **host**.

Orchids show a healthy relationship with tress in which they get nutrition from them without harming the trees and there is no benefit to trees from orchids.Such a type of relationship is called **Commensalism**.

Hence, the correct option is (a)

Q. 6 During non cyclic photophosphorylation, when electrons are lost from the reaction center at PS II, what is the source which replaces these electrons?

Option 1:
water

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Option 2:
Carbon dioxide

Option 3:
light

Option 4:
Oxygen

Correct Answer:
water

Solution:
During acyclic photophosphorylation, the reaction center of PS II absorbs specific wavelengths of red light and releases energetic electrons.These electrons travel along the electron transport chain and eventually reach PS I. The voids created in PS II due to loss of electrons are filled by water separation (photolysis of water) releasing electrons, protons and oxygen. Hence, option 1 is correct.

Q. 7 Which of the following statements is incorrect?

Option 1:
Energy content gradually increases from the first to fourth trophic level.

Option 2:
Number of the individuals decreases from first trophic level to fourth trophic level

Option 3:
Energy content gradually decreases from first to fourth trophic level,

Option 4:
Biomass decrease from first to fourth trophic level

Correct Answer:
Energy content gradually increases from the first to fourth trophic level.

Solution:
Only 10% of energy passes to the next trophic level. The remaining energy is used for various processes

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Q. 8 Male and female gametophytes do not have an independent free-living existence in:

Option 1:
Algae

Option 2:
Angiosperms

Option 3:
Bryophytes

Option 4:
Pteridophytes

Correct Answer:
Angiosperms

Solution:
Male and female gametophytes do not have independent existence in Angiosperms

Q. 9 Identify the statement which is incorrect.

Option 1:
Glycine is an example of lipids.

Option 2:
Lecithin contains phosphorous atom in its structure

Option 3:
Tyrosine possess aromatic ring in its structure.

Option 4:
Sulphur is an integral part of cystein

Correct Answer:
Glycine is an example of lipids.

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Solution:

- Glycine is the simplest Amino acid as it contains both COOH and NH2 groups
- It does not possess chiral carbon as the structure shows two H groups

---

Q. 10 In the following in each set a conservation approch and an example of method of conservation are given

(a) In-situ conservation - Biosphere Reserve

(b) Ex-situ conservation - Sacred groves

(c) In situ conservation - Seed bank

(d) Ex-situ conservation - Cryopreservation

Select the option with correct match of approch and method

Option 1:
(a) and (d)

Option 2:
(b) and (d)

Option 3:
(a) and (b)

Option 4:
(a) and (c)

Correct Answer:
(a) and (d)

Solution:
In-situ conversation is carried out in natural ecosystem while ex-situ is in artificial ecosystem.

Examples of in-situ are national parks(biosphere reserves) and wildlife sanctuaries.

Examples of ex-situ are gene banks, seed banks abd zoological gardens

Hence, the correct option is (a).

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Q. 11 In mitotic cycle, the correct sequence of phases is

Option 1:
G_1, S, G_2, M

Option 2:
M, G_1, G_2, S

Option 3:
G_1, G_2, S, M

Option 4:
S, G_1, G_2 , M

Correct Answer:
G_1, S, G_2, M

Solution:
- The cell cycle consists of Interphase and Mitotic phase.
- Interphase consists of G1, S and G2 phase
- Mitotic phase consists of karyokinesis and cytokinesis

Q. 12 Which of the following elements helps in maintaining the structure of ribosomes?

Option 1:
Zinc

Option 2:
Copper

Option 3:
Molybdeum

Option 4:
Magnesium

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Correct Answer:
Magnesium

Solution:
The two ribosomal subunits remain united with each other due to a specific concentration of the **Mg^{++} ions**. When the concentration of Mg^{++} ions reduces below a critical level both ribosomal subunits get separated. But, by raising the concentration of Mg^{++} ions in the matrix, the two ribosomes become associated with each other and is known as the **dimer.**

Q. 13 Phycoerythrin is the major pigment in:

Option 1:
Blue green algae

Option 2:
Green algae

Option 3:
Brown algae

Option 4:
Red algae

Correct Answer:
Red algae

Solution:
In **Red Algae (Rhodophyceae)** photosynthetic pigments include **cholorphyll a,d, carotenoids** and **phycobilins**. Phycobilins are water soluble pigments namely, **red coloured phycoerythrin**, blue coloured **phycocyanin** and **allophycocyanin**.

The red colour of red algae is due to abundant formation of phycoerythrin. Phycoerythrin is able to absorb blue green wavelengths of light. Being shorter, these wavelengths are able to reach the maximum depth in water. Therfore, red algae reach the maximum depth in sea where no other type of photosynthetic organisms grow. However, organisms living in shallower water do not appear reddish due to lesser synthesis of phycoerythrin.

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Q. 14 Chromosomal theory of inheritance was proposed by:

Option 1:
Watson and Crick

Option 2:
Bateson and Punnet

Option 3:
Sutton and Boveri

Option 4:
T.H. Morgan

Correct Answer:
Sutton and Boveri

Solution:
In 1902 and 1903, Sutton and Boveri published independent papers proposing what we now call the chromosome theory of inheritance.

Q. 15 A species which was introduced for ornamentation but has become a trouble some weed in India:

Option 1:
Eichhornia crassipes

Option 2:
Prosopis juliflora

Option 3:
Trapa spinosa

Option 4:
Parthenium hysterophorus

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Correct Answer:
Eichhornia crassipes

Solution:
Eichhornia crassipes was introduced in India by Queen Victoria for ornamentation but has become a troublesome weed in India.

Itis known as water hyacinth (terror of bengal)

Q. 16 Attachment of spindle fibers to kinetochores of chromosomes becomes evident in:

Option 1:
Teleophase

Option 2:
Prophase

Option 3:
Metaphase

Option 4:
Anaphase

Correct Answer:
Metaphase

Solution:
Spindle fibres attached tothe kinetochore are clearly evident in Metaphase

---

Q. 17 First discovered resistration endonuclease that always cuts DNA molecules at a particular point by recognising a specific sequence of six base pairs is:

Option 1:
Adenosine deaminase

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Option 2:
Thermostable DNA polymerase

Option 3:
Hind II

Option 4:
EcoR I

Correct Answer:
Hind II

Solution:
- Hind II is an example of restriction endonuclease that cuts the DNA molecules at particular sequence
- The Restriction endonuclease follow certain method to name the enzymes
- First letter denotes the Genus, Second and Third letter denotes the species and Fourth letter denotes the strain
- Hind II is isolated from Haemophilus influenzae thus first letter is 'H', second and third letters are 'IN' and fourth letter is 'D' which means it is D strain. The roman number denotes the order in which it is isolated

---

Q. 18 In *Glycine max*, the product of biological nitrogen fixation is transported from root nodules to other parts as:

Option 1:
Glutamate

Option 2:
Nitrates

Option 3:
Ureides

Option 4:
Ammonia

Correct Answer:
Ureides

Solution:
In *Glycine max* (soya bean) , the nitrogen is translocated as ureides

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Q. 19 The term 'Nuclein' for the genetic materials was used by:

Option 1:
Mendel

Option 2:
Franklin

Option 3:
Meischer

Option 4:
Chargaff

Correct Answer:
Meischer

Solution:
The term nuclein was given by Friedrich Meishner

Q. 20 Match the following concerning the activity/function and the phytohormone involved.

(a) Fruit ripener	(i) Abscisic acid
(b) Herbicide	(ii) GA_3
(c) Bolting agent	(iii) 2, 4-D
(d) Stress hormone	(iv) Ethephon

Select the correct option from the following:

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Option 1:
a-iii, b-iv, c-ii, d-i

Option 2:
a-iv, b-iii, c-ii, d-i

Option 3:
a-iv, b-ii, c-i, d-iii

Option 4:
a-ii, b-iii, c-iv, d-iv

Correct Answer:
a-iv, b-iii, c-ii, d-i

Solution:
Ethephon is used as fruit ripener as it produces Ethylene gas

---

Q. 21 According to Alxender Von Humbolt:

Option 1:
Species richess increase with increasing area, but only up to limit

Option 2:
There is no relationship between the richness and area explored

Option 3:
Species richness goes on increasing with increasing area of exploration

Option 4:
Species richness decreases with increasing area of exploration

Correct Answer:
Species richess increase with increasing area, but only up to limit

Solution:
Species richness increases with increasing area upto a certain limit

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Q. 22 The impact of immigration on population density is:

Option 1:
Both positive and negative

Option 2:
Neturalized by natality

Option 3:
Positive

Option 4:
Negative

Correct Answer:
Positive

Solution:
Immigration is movement of same species intothe habitate from somewhere else. This increases the density and hence it has a positive impact.

Q. 23 Which of the following is incorrect for wind-pollinated plants?

Option 1:
many ovules in each ovary

Option 2:
Flowers are small and not brightly coloured

Option 3:
Pollen grains are light and non-sticky

Option 4:
Well exposed stamens and stigma

Correct Answer:
many ovules in each ovary

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Solution:
Wind pollination is Anemophily

Q. 24 Embryological support for evolution was propsed by:

Option 1:
Karl Ernst von Baer

Option 2:
Charles darwin

Option 3:
Alfred Wallace

Option 4:
Ernst Heckel

Correct Answer:
Ernst Heckel

Solution:
Embyological evidence of evolution was by Ernst haeckel

Q. 25 Which of the following is incorrect about Cynobacteria?

Option 1:
They lack heterocysts

Option 2:
They often form blooms in polluted water bodies

Option 3:
They chlorophyll A similar to grown plants

Option 4:
They are photoautotrophs

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Correct Answer:
They lack heterocysts

Solution:
They have heterocysts to fix nitrogen in plants

Q. 26 Match the following:

(a) Aquaporin	(i) Amide
(b) Asparagine	(ii) Polysaccharide
(c) Abscisic acid	(iii) Polypeptide
(d) Chitin	(iv) Carotenoids

Select the correct option

Option 1:
(a)-(ii) (b)-(iii) (c)-iv) (d)-(iv)

Option 2:
(a)-(ii) (b)-(i) (c)-iv) (d)-(iii)

Option 3:
(a)-(iii) (b)-(i) (c)-ii) (d)-(iv)

Option 4:
(a)-(iii) (b)-(i) (c)-iv) (d)-(ii)

Correct Answer:
(a)-(iii) (b)-(i) (c)-iv) (d)-(ii)

Solution:

(a) Aquaporin	polypeptide
(b) Asparagine	amide
(c) Abscisic acid	carotenoids
(d) Chitin	polysaccaride

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Q. 27 The biosynthesis of ribosomal RNA occurs in:

Option 1:
Golgi apparatus

Option 2:
Microbodies

Option 3:
Nucleolus

Option 4:
Ribosomes

Correct Answer:
Nucleolus

Solution:
Nucleolus is considered a protein factory

Q. 28 Who coined the term 'Kinetin' ?

Option 1:
Darwin

Option 2:
Went

Option 3:
Kurosawa

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Option 4:
Skoog and Miller

Correct Answer:
Skoog and Miller

Solution:
skoog and miller coined the term Kinetin

Q. 29 The number of contrasting character studied by Mendel for his experiments was:

Option 1:
4

Option 2:
2

Option 3:
7

Option 4:
14

Correct Answer:
7

Solution:
7 pairs of contracting characters were studied by mendel

Q. 30 Correct position of floral parts over thalamus in mustard plant is

Option 1:
Margin of the thalamus grows upward, enclosing the ovary completely, and other parts arise below the ovary.

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Option 2:
Gynoecium is present in the centre and other parts cover it partially.

Option 3:
Gynoecium is situated in the centre, and the other parts of flower are located at the rim of the thalamus, at the same level.

Option 4:
Gynoecium occupies the highest position, while the other parts are situated under it.

Correct Answer:
Gynoecium occupies the highest position, while the other parts are situated under it.

Solution:
Mustard is hypogynous flower

Q. 31 In a mixture, DNA fragments are seperated by:

Option 1:
Restriction digestion

Option 2:
Electrophoresis

Option 3:
Polymerase chain reaction

Option 4:
Bioprocess engineering

Correct Answer:
Electrophoresis

Solution:
After the cutting of DNA by restriction enzymes, fragments of DNA are formed. These fragments can be separated by a technique called **gel electrophoresis.**

Electrophoresis is a technique of separation of charged molecules under the influence of electric field through a medium/ matrix.The most commonly used matrix is **Agarose**.

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Q. 32 In Recombinant DNA technology antibiotics are used:

Option 1:
to detect alien DNA

Option 2:
to impact disease -resistance to the host plant

Option 3:
as selectable markers

Option 4:
to keep medium bacteria-free

Correct Answer:
as selectable markers

Solution:
Antibiotics act as selectable markers

Q. 33 Which of the following statements is incorrect?

Option 1:
In C_4 plants, the site of RuBisCo activity is in mesophyll cell

Option 2:
The substrate molecule for RuBisCo activity is a 5-carbon compound

Option 3:
RuBisCo action requires ATP and NADPH

Option 4:
RuBisCo is a bifunctional enzymes

Correct Answer:
In C_4 plants, the site of RuBisCo activity is in mesophyll cell

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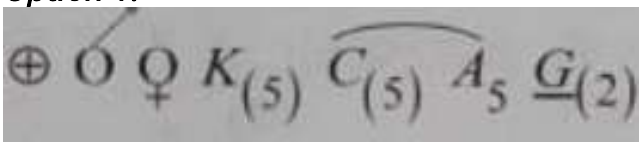
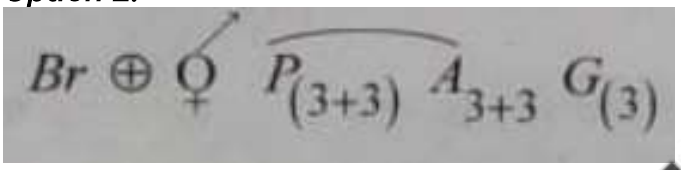
Solution:
C4 plants have specialized leaf anatomy with two different types of photosynthetic cells: mesophyll cells (on the exterior of the leaf, near stomata) and bundle sheath cells (in the interior of the leaf, far away from stomata). Rubisco is located in **bundle sheath cells**, but not in mesophyll cells.

Q. 34 Which of the followig is correct floral formula of Liliaceae?

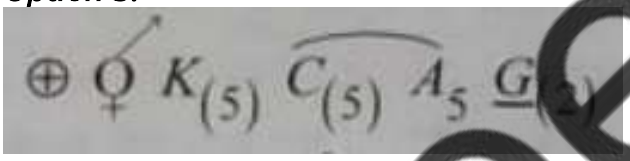
Option 1:



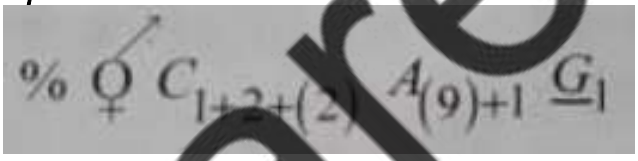
Option 2:



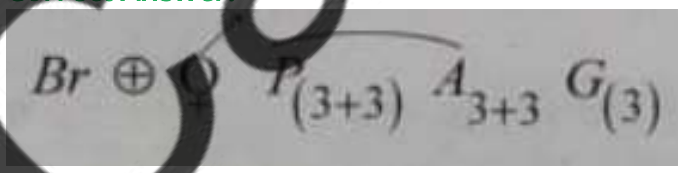
Option 3:



Option 4:



Correct Answer:



Solution:
The plants belong to family 'Liliaceae' have following floral characters:

Bracteate,

Flower: actinomorphic and Bisexual,

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Perianth: tepal six (3+3), often united into tube, valvate aestivation,

Androecium: stamen six, 3+3, epitepalous,

Gynoecium: tricarpellary, syncarpous, ovary superior, trilocular with many ovules; axile placentation,

Fruit: capsule, rarely berry,

Seed: endospermous.

Q. 35 Inhibitory substance in dormant seeds cannot be removed by subjecting seeds to:

Option 1:
Nitrate

Option 2:
Ascorbic acid

Option 3:
Chilling conditions

Option 4:
Gibberellic acid

Correct Answer:
Ascorbic acid

Solution:
Ascorbic acid promotes seed dormancy

Q. 36 Pyruvate dehydrogenase activity during aerobic respiration requires:

Option 1:
Iron

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Option 2:
Cobalt

Option 3:
Magnesium

Option 4:
Calcium

Correct Answer:
Magnesium

Solution:
- During respiration, 1 molecule of Glucose forms 2 molecules of Pyruvic acid
- In Aerobic respiration, Pyruvic acid further forms Acetyl Co A and this enters Krebs Cycle. This is the link reaction.
- During link reaction Pyruvic acid is converted into Acetyl CoA with the help of Pyruvate dehydrogenase complex. For this conversion Pyruvate dehydrogenase requires magnesium, CoA, NAD^+, TPP and lipoic acid.

Q. 37 Large, empty colourless cells of the adaxial epidermis along the veins of grass leaves are:

Option 1:
Gurad cells

Option 2:
Bundle sheath cells

Option 3:
Bulliform cells

Option 4:
Lenticels

Correct Answer:
Bulliform cells

Solution:
These are bulliform cells

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Q. 38 Identify the correct features of Mango and Coconut fruits:

(i) In both, fruit is a drupe

(ii) Endocrap is edible in both

(iii) Mesocrap in coconut is fibrous and in mango it is fleshy

(iv) In both, fruit develops from monocarpellary ovary

Select the correct option from below:

Option 1:
i,ii and iii only

Option 2:
i and iv only

Option 3:
i and ii only

Option 4:
i, iii and iv only

Correct Answer:
i, iii and iv only

Solution:
Edible part in **Mango** is **Fleshy mesocarp.**

Edible part in **Coconut** is **Endosperm.**

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Q. 39 Match the following techiques or instruments with their usage:

(a) Bioreactor	(i) Sepration of DNA fragements
(b) Electrophoresis	(ii) Production of large quantities of products
(c) PCR	(iii) Detection of pathogen based on antigen - antibidy reaction
(d) ELISA	(iv) Amplification of nucleic acids

Select the correct option from the following:

Option 1:
a-ii, b-i, c-iv, d-iii

Option 2:
a-iv, b-iii, c-ii, d-i

Option 3:
a-ii, b-i, c-iii, d-iv

Option 4:
a-iii, b-ii, c-iv, d-i

Correct Answer:
a-ii, b-i, c-iv, d-iii

Solution:

(a) Bioreactor	Production of large quantities of products
(b) Electrophoresis	seperation of DNA
(c) PCR	Amplication
(d) ELISA	Detection of pathogen based on antigen - antibidy reaction

Hence, the correct option is (a).

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Q. 40 Which of the following statements is incorrect about gymnosperms?

Option 1:
Male and female gametophytes are free living

Option 2:
Most of them have narrow leaves with thick cuticle.

Option 3:
Their seeds are not covered

Option 4:
They are heterosporous

Correct Answer:
Male and female gametophytes are free living

Solution:
Unlike bryophytes and pteridophytes, in gymnosperms the male and female gametophytes do not have an independent free living existence. They are present on seperate male and female **cones/strobili.**

Q. 41 In a polynucleotide chain of DNA, a nitrogenous base linked to the -OH of:

Option 1:
3'C pentose sugar

Option 2:
5'C pentose sugar

Option 3:
1'C pentose sugar

Option 4:
2'C pentose sugar

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Correct Answer:
1'C pentose sugar

Solution:

- Nucleotide is the monomer that forms DNA molecule
- It consists of Phosphorus + Pentose sugar + Nitrogen Base
- Nitrogen bases can be Purines (Double rings) or Pyrimidines (Single ring)
- These nitrogen bases are attached to the sugar at 1C

Q. 42 Air(Prevention and Control of Pollution) Act was amended in 1987 to include which pollutants

Option 1:
Allergy causing pollen

Option 2:
Noise

Option 3:
Particlalutes of size 2.5 micrometer or below

Option 4:
Vehicular exhaust

Correct Answer:
Noise

Solution:
Air(Prevention and Control of Pollution) Act was amended in 1987 to include noise as pollutants

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Q. 43 Inclusion bodies of blue-green, purple and green photosynthetic bacteria are:

Option 1:
Gas Vaculoes

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Option 2:
Centrioles

Option 3:
Microtubules

Option 4:
Contraticle vacuoles

Correct Answer:
Gas Vaculoes

Solution:
- Inclusion bodies are reserved materials
- These are the gas vacuoles found in prokaryotic cells

Q. 44 Vegetative propagule in *Agvae* is termed as:

Option 1:
Bulbil

Option 2:
Offset

Option 3:
Eye

Option 4:
Rhizome

Correct Answer:
Bulbil

Solution:
Bulbils are swollen buds that seperate from parent plants when it gets favourable conditions

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Q. 45 In some plants thalamus contributes to fruit formation. Such fruits are termed as:

Option 1:
Aggregate fruits

Option 2:
True fruits

Option 3:
Pathenocarpic fruit

Option 4:
False fruit

Correct Answer:
False fruit

Solution:
Apple, Pear etc are examples of false fruits

Q. 46 Spooling is-

Option 1:
Cutting of seprated DNA bands from the agarose gel

Option 2:
Transfer of sperated DNA fragments to sythetic membranes

Option 3:
Collection of isolated DNA

Option 4:
Amplification of DNA

Correct Answer:
Collection of isolated DNA

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Solution:

- DNA spooling is a method to obtain DNA in the form of spool
- This is the last step of DNA extraction
- A thin glass rod is inserted in the glass that contains purified DNA and the rod is slightly rotated so that DNA adheres to the glass rod and this is spooling

---

Q. 47 The laws and rules to prevent unauthorised exploitation of bio-resoyrces are termed as-

Option 1:
Bioethics

Option 2:
Bioengineering

Option 3:
Biopiracy

Option 4:
Biopatenting

Correct Answer:
Bioethics

Solution:
The laws and rules to prevent unauthorised exploitation of bio-resoyrces are termed as-BIOETHICS

Q. 48 Select the correct statements:

Option 1:
Angiotensin II is a powerful vasodilator

Option 2:
Counter current pattern blood flow is not observed in vasa recta

Option 3:
Reduction in Glomerular filtration rate activates JG cells to release renin.

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Option 4:
Atrial natriuretic Factor increases the blood pressure

Correct Answer:
Reduction in Glomerular filtration rate activates JG cells to release renin.

Solution:
- Reduction in GFR causes stimulation of RAAS
- RAAS is Renin Angiotensin Aldosterone System
- Low GFR activates JG cells to secrete Renin
- Renin then forms Angiotensin I and II
- Angiotensin II is a potent vasoconstrictor
- This causes increase in blood pressure
- Angiotensin II also causes release of Aldosterone that controls water and electrolyte balance
- Thus GFR is increased
- ANF from heart controls the incresed GFR

Q. 49 In cockrach, identify the parts of foregut in correct sequence:

Option 1:
Mouth → Crop → Pharynx → Oesophagus → Gizzard

Option 2:
Mouth → Gizzard → Crop → Pharynx →Oesophagus

Option 3:
Mouth → Pharynx → Oesophagus → Crop → Gizzard

Option 4:
Mouth → Oesophagus → Pharynx → Crop → Gizzard

Correct Answer:
Mouth → Pharynx → Oesophagus → Crop → Gizzard

Solution:
- The alimentary canal in Cockroach is divided into Foregut, Midgut and Hindgut
- Foregut includes Mouth, Pharynx, Esophagus, Crop and Gizzard

---

Q. 50 Select the correct statement from the following:

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Option 1:
The polymerase enzyme joins the gene of intrest and the vector DNA

Option 2:
Restriction enzyme digestions are performed by incubating purified DNA molecules with restriction enzymes of optimum conditions

Option 3:
PCR is used for isolation and sepration of gene of interest

Option 4:
Gel electrophoresis is used for amplification of DNA segment

Correct Answer:
Restriction enzyme digestions are performed by incubating purified DNA molecules with restriction enzymes of optimum conditions

Solution:
The vector and the target DNA fragment can be sepeartely digested with the same restriction enzyme. The digested vector and the target DNA fragment are then incubated together in the presence of DNA **ligase enzyme.** Incubation results in bonding two types of DNA by **phosphodiester bonds** between them. Thus forming a **recombinant DNA molecule.**

Q. 51 Which is the basis of genetic mapping of human genome as well as DNA finger printing?

Option 1:
Single nucleotide ppolymorphism

Option 2:
Polymorphism and hnRNA sequence

Option 3:
Polymorphism in RNA sequence

Option 4:
Polymorphism in DNA sequence

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Correct Answer:
Polymorphism in DNA sequence

Solution:
Polymorphism in DNA sequence is the basis

Q. 52 A Hominid fossil discovered in Java in 1891, now extinct, having cranial capacity of about 900 cc was:

Option 1:
Neanderthal man

Option 2:
Homo sapiens

Option 3:
Australopithecus

Option 4:
Homo erectus

Correct Answer:
Homo erectus

Solution:
- Homo erectus fossilnwere discovered in Java near Indonesia in 1891. The cranial capacity was 900 c.c
- This was named by Eugene Dubois
- Mayer named it as Homo erectus erectus
- THey were the first to use fire for hunting and cooking

Hence, the correct option is (d).

Q. 53 The phenomenon of evolution of different species in a given geographical area starting from a point and spreading to other habits is called -

Option 1:
Co-evolution

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Option 2:
natural selection

Option 3:
Adaptive radiation

Option 4:
Saltation

Correct Answer:
Adaptive radiation

Solution:
- When one organism evolves with respect to evolution in another organism, it is co-evolution
- Saltation is single step large evolution
- Natural selection is the process through which population of living organism that adapt are selected by nature
- Hence Adaptive radiation is the correct answwer

Q. 54 According to Central Pollution Control Board (CPCB) what size (in diameter) of particulae is responsible for causing greater harm to human health?

Option 1:
2.5 micrometers

Option 2:
4.0 micrometers

Option 3:
3.0 micrometers

Option 4:
3.5 micrometers

Correct Answer:
2.5 micrometers

Solution:
- The effect of particulate air pollutant depends on the size of the particles

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- According to CPCB, Central Pollution Control Board, particluate matter of 2.5 or less causes more harm
- THese particles are inhaled deep and thus affects the respiratory tract and lungs

---

Q. 55 RNA interference is used for which of the following purposes in the field of biotechnology?

Option 1:
to develop a pest resistant plant against infestation by nematode

Option 2:
to enhance the mineral usage by the plant

Option 3:
to reduce post harvest losses

Option 4:
to develop a plant tolerant to abiotic stresses

Correct Answer:
to develop a pest resistant plant against infestation by nematode

Solution:
RNA interference is used to stop protein synthesis by binding to messenger RNA . It is used to develop resistant plants against nematodes. So the correct option is a.

Q. 56 Which of the following STDs are not curable?

Option 1:
Chlamydiasis, Syhilis, Genital warts

Option 2:
HIV, Gonorrhoea, Trichromoniasis

Option 3:
Gonorrhoea, Trichromoniasis, Hepatitis B

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Option 4:
Genetial herpes, Hepatitis B, HIV infection

Correct Answer:
Genetial herpes, Hepatitis B, HIV infection

Solution:
- Sexualy transmitted diseases are Veneeral diseases
- Hepatitis, HIV and Herpes are incurable diseases
- Chlamydiasis, Gonorrhea are curable diseases

Q. 57 *E.Coli*. has only 4.6×10^6 base pairs and completes the process of replication within 18 minutes; then the average rate of polymerization is approximately

Option 1:
3000 base pairs /second

Option 2:
4000 base pairs /second

Option 3:
1000 base pairs /second

Option 4:
2000 base pairs /second

Correct Answer:
4000 base pairs /second

Solution:
E. coli has only 4.6 ×10^6 bp and completes the process of replication within 18 minutes.
Rate of polymerization = nucleotides / time
= 4.6 ×106 / 18 * 60
= 4259 bp per second

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Q. 58 Match the following colums and select the correct options:

Column I	Column II
(a) *Aptenodytes*	(i) Flying fox
(b) *Pteropus*	(ii) Angel fish
(c) *Pterophyllum*	(iii) Lamprey
(d) *Petromyzon*	(iv) Penguin

Option 1:
a-iii, b-iv, c-i, d-ii

Option 2:
a-iv, b-i, c-ii, d-iii

Option 3:
a-ii, b-i, c-iv, d-iii

Option 4:
a-iii, b-iv, c-ii, d-i

Correct Answer:
a-iv, b-i, c-ii, d-iii

Solution:

Column I	Column II
(a) *Aptenodytes*	Penguin
(b) *Pteropus*	Flying fox
(c) *Pterophyllum*	Angel fish
(d) *Petromyzon*	Lamprey

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Q. 59 Match the following columns and select the correct option:

Column I	**Column II**
(a) Pituitary hormone	(i) Steriod
(b) Epinephrine	(ii) Neuropeptides
(c) Endorphins	(iii) Peptide, proteins
(d) Cortisol	(iv) Biogenic amines

Option 1:
a-iii, b-iv, c-ii, d-i

Option 2:
a-iv, b-iii, c-i, d-ii

Option 3:
a-iii, b-iv, c-i, d-ii

Option 4:
a-iv, b-i, c-ii, d-iii

Correct Answer:
a-iii, b-iv, c-ii, d-i

Solution:
Hormones can be classified based on their biochemical nature as follows:

1. Steroids – These types of hormones are composed of cholesterol. Cortisol is a steroid.

2. Neuropeptides – Neuropeptides comprise small chains of amino acids which are released by the brain. Endorphins are an example of this type of hormone.

3. Peptides, proteins – They consist of two or more Amino acids. Pituitary hormones come under this category.

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4. Biogenic amines – Biogenic hormones consists of amine groups. Epinephrine is one such example.

Column I	**Column II**
(a) Pituitary hormone	Peptide, Proteins
(b) Epinephrine	Biogenic amines
(c) Endorphins	Neuropeptides
(d) Cortisol	Steroids

Hence, the correct answer is option 1 - a-iii, b-iv, c-ii, d-i

Q. 60 Match the following columns with reference to cockroach and select correct option:

Column I	Column II
(a) Grinding of the food particles	(i) Hepatic caecal
(b) Secrete gastric juice	(ii) 10^{th} segment
(c) 10 pairs	(iii) Proventriculus
(d) Anal cerci	(iv) Spiracles
	(v) Alary muscles

Option 1:
a-iv, b-iii, c-v, d-ii

Option 2:
a-i, b-iv, c- iii, d-ii

Option 3:
a-ii, b-iii, c-i, d-iv

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Option 4:
a-iii, b-i, c-iv, d-ii

Correct Answer:
a-iii, b-i, c-iv, d-ii

Solution:

A	B
a. Grinding of food particles	Proventriculus
b. Secrete Gastric juice	Hepatic caece
c. Anal cerci	10th segment
d. 10 pairs	Spiracles

Q. 61 Match the following columns and select the correct option:

Column I	Column II
(a) Ovary	(i) Human chrionic Gonadotropin
(b) Placenta	(ii) Estrogen & Progestron
(c) Corpus luteum	(iii) Androgens
(d) Leydig cells	(iv) Progestrone only

Option 1:
a-i, b-ii, c-iii, d-iv

Option 2:
a-i, b-iii, c-ii, d-iv

Option 3:
a-ii, b-i, c-iv, d-iii

Option 4:
a-iv, b-iii, c-ii, d-i

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Correct Answer:
a-ii, b-i, c-iv, d-iii

Solution:
The correctanswer is optionc that is ,

a-ii, b-i, c-iv, d-iii

Column I	Column II
(a) Ovary	Estrogen and Progesterone
(b) Placenta	Human chorionic gonadotrophin
(c) Corpus luteum	Progesterone only
(d) Leydig cells	Androgens

Q. 62 For the commerical and industrial production of Citric Acid, which of the following microbes is used?

Option 1:
Lactobacillus sp.

Option 2:
Saccharomyces cerevisiae

Option 3:
Clostridium butylicum

Option 4:
Aspergillus niger

Correct Answer:
Aspergillus niger

Solution:
Aspergillus niger is a fungus that is used for production of citric acid

Q. 63 In human beings, at the end 12 weeks (first trimster) of pregnancy, the following is observed:

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Option 1:
Most of the major organ systems are formed

Option 2:
The head is coverded with fine hair

Option 3:
Movement of the foetus

Option 4:
Eyelids and eyelashes are formed

Correct Answer:
Most of the major organ systems are formed

Solution:
Most of the body organs are formed in 12 weeks

Q. 64 Inbreeding depression is -

Option 1:
Decreased productivity due to mating of superior male and inferior female.

Option 2:
Decrease in body mass of progeny due to continued close inbreeding

Option 3:
Reduced fertility and productivity due to continued close inbreeding

Option 4:
Reduced motility and immunity due to close inbreeding

Correct Answer:
Reduced fertility and productivity due to continued close inbreeding

Solution:
- Inbreeding is mating of closely related organisms

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- Due to continued inbreeding for generations, Inbreeding depression is seen
- This results in reduced fertility and survival rate of progeny
- This is corrected by Outcrossing techniques

Q. 65 Match the following column and select the correct option.

Column I	Column II
(i) Typhoid	(a) Haemophilus influenzae
(ii) Malaria	(b) wuchereria bancrofti
(iii) Pneumonia	(c) Plasmodium vivax
(iv) Filariasis	(d) Salmonella typhi

Option 1:
(i)-(c), (ii)-(d), (iii)-b), (iv)-(a)

Option 2:
(i)-(a), (ii)-(c), (iii)-(b), (iv)-(d)

Option 3:
(i)-(a), (ii)-(b), (iii)-(d), (iv)-(c)

Option 4:
(i)-(d), (ii)-(c), (iii)-(a), (iv)-(b)

Correct Answer:
(i)-(d), (ii)-(c), (iii)-(a), (iv)-(b)

Solution:

Column I	Column II
(i) Typhoid	Salmonella typhi
(ii) Malaria	Plasmodium
(iii) Pneumonia	Haemophilus

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(iv) Filariasis wuchereria bancrofti

Q. 66 The increase in osmolarity from outer to inner medullary interstitium is maintained due to -:

(i) Close proximity between Henle's loop and vasa recta

(ii) Counter current mechanism

(iii) Selective secretion of HCO_3^- and hydrogen ions in PCT

(iv) Higher blood pressure in glomerular capillaries

Option 1:
(iii) and (iv)

Option 2:
(i) (ii) and (iii)

Option 3:
(i) and (ii)

Option 4:
only (ii)

Correct Answer:
(i) and (ii)

Solution:

- Both (i) and (ii) are correct
- Counter current mechanism is important to maintain concentration gradient
- This mechanism is because of Henle's loop and Vasa recta

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- The ascending limp of Loop of henle is parallel to descending vasa recta and vice versa
- Selective reabsorption is by PCT
- High blood pressure in glomerular capillaries caused glomerular filtration and not Counter current

---

Q. 67 The proteolytic enzyme rennin is found in:

Option 1:
Bile juice

Option 2:
Gastric juice

Option 3:
Pancreatic juice

Option 4:
Intestinal juice

Correct Answer:
Gastric juice

Solution:
Rennin is present in gastric juice and it brings about digestion of milk protein (caesin) in children

---

Q. 68 The rate of decomposition is faster in the ecosystem due the following factors EXCEPT:

Option 1:
Warm and mosit environment

Option 2:
Presence of aerobic soil microbes

Option 3:
Detritus richer in ligin and chitin

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Option 4:
Detritus rich in sugar

Correct Answer:
Detritus richer in ligin and chitin

Solution:
The rate of decomposition is slow if it contains lignin, chitin, tannins (Phenolics) and cellulose.

Q. 69 Match the following columns and select the correct option:

Column I	Column II
(a) Pneumotaxic centre	(i) Alveoli
(b) O_2 Dissociation curve	(ii) Pons region of brain
(c) Carbonic Anhydrase	(iii) Haemoglobin
(d) Primary site of exchange of gases	(iv) RBC

Option 1:
a-ii, b-iii, c-iv, d-i

Option 2:
a-iii, b-ii, c-iv, d-i

Option 3:
a-iv, b-i, c-iii, d-ii

Option 4:
a-i, b-iii, c-ii, d-iv

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Correct Answer:
a-ii, b-iii, c-iv, d-i

Solution:
- Pneumotaxic centre is in the pons
- Oxygen dissociation curve is for Hb saturation
- Carbonic Anhydrase is an enzyme present in RBC and Plasma
- Alveoli is the site of exchange of gases

Q. 70 Match the following events that occur in their respective phases cell cycle and select the correct option:

(a) G_1 Phase	(i) Cell grows and organelle duplication
(b) S phase	(ii) DNA replication and chromosome duplication
(c) G_2 phase	(iii) Cytoplasmic growth
(d) Metaphase in M-phase	(iv) Alignment of chromosomes

Option 1:
a-iii, b-iv, c-i, d-ii

Option 2:
a-iv, b-i, c-ii, d-iii

Option 3:
a-i, b-ii, c-iii, d-iv

Option 4:
a-ii, b-iii, c-iv, d-i

Correct Answer:
a-i, b-ii, c-iii, d-iv

Solution:
G1, S and G2 are stages of Interphase while M phase is the mitotic phase

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(a) G_1 Phase	(i) Cell grows and organelle duplication
(b) S phase	(ii) DNA replication and chromosome duplication
(c) G_2 phase	(iii) Cytoplasmic growth
(d) Metaphase in M-phase	(iv) Alignment of chromosomes

Q. 71 Match the following Columns and select the correct options:

Column I	Column II
(a) Dragonflies	(i) Biocontrol agents of several plant pathogens
(b) Bacillus thuringenesis	(ii) Get rid of Aphids and mosquitoes
(c) Glomus	(iii) Narrow spectrum incesticidal application
(d)Baculoviruses	(iv) Biocontrol agents of lepidopteran plants pest
	(v) Absorbs phosphorous from soil

Option 1:
a-ii, b-i, c-iii, d-iv

Option 2:
a-ii, b-iii, c-iv, d-v

Option 3:
a-ii, b-iv, c-v, d-iii

Option 4:
a-iii, b-v, c-iv, d-i

Correct Answer:
a-ii, b-iv, c-v, d-iii

Solution:
Dragonflies are used to get rid of mosquitoes.

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Bacillus thuringiensis is a microbial biocontrol agent used against pests.

Baculovirus is used as a biological control agent for insecticidal applications.

Glomus(fungus) forms a symbiotic association with plants. It helps plants by absorbing phosphorus from the soil. The correct option is c.

Q. 72 Match the group of organisms with their respective distinctive charachteristics and select the correct option:

Column I	Column II
(a) Plathyhelminthes	(i) Cylindrical body with no segmentation
(b) Echinoderms	(ii) Warm blooded animals with direct development
(c) Hemichrodates	(iii) Bilateral symmetry with incomplete digestive system
(d) Aves	(iv) Radial symmetry with indirect development

Option 1:
a-(i), b-(ii), c-(iii), d-(iv)

Option 2:
a-(ii), b-(iii), c-(iv), d-(i)

Option 3:
a-(iii), b-(iv), c-(i), d-(ii)

Option 4:
a-(ii), b-(iv), c-(i), d-(iii)

Correct Answer:
a-(iii), b-(iv), c-(i), d-(ii)

Solution:

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a-(iii), b-(iv), c-(i), d-(ii)

Q. 73 Select the correct option of haploid cells from the following groups:

Option 1:
Secondary spermatocyte, First polar body, Ovum

Option 2:
Spermatogonia , Primary spermatocyte, spermatid

Option 3:
Primary spermatocyte, Secondary spermatocyte, Second polar body

Option 4:
Primary oocyte, Secondary oocyte, Spermatid

Correct Answer:
Secondary spermatocyte, First polar body, Ovum

Solution:

- Primary spermatocyte undergoes Meiosis 1 and forms secondary spermatocyte and thus secondary spermatocyte is haploid
- Primary oocyte forms Secondary oocyte and first polar body by Meiosis 1 and hence are haploid too

---

Q. 74 The Total Lung Capacity (TLC) is the total volume of air accomodated in the lungs at the end of a forced inspiration. This includes:

Option 1:
RV; ERV; IC and EC

Option 2:
RV; ERV; VC (Vital Capacity) and FRC(Functional Residual Capacity)

Option 3:
RV ; ERV; TV; IRV

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Option 4:
RV; IC; EC and ERV

Correct Answer:
RV ; ERV; TV; IRV

Solution:
Total lung capacity includes all the lung volumes : Tidal volume, Residual volume, IRV and ERV

Q. 75 Match the following columns and select the correct option:

Column I	Column II
(a) Smooth Endoplasmic Recticulum	(i) Protein synthesis
(b) Rough Endoplasmic Recticulum	(ii) Lipid synthesis
(c) Golgi complex	(iii) Glycosylation
(d) Centriole	(iv) Spindle formation

Option 1:
a-iii, b-i, c-ii, d-iv

Option 2:
a-iv, b-ii, c-i, d-iv

Option 3:
a-i, b-ii, c-iii, d-iv

Option 4:
a-ii, b-i, c-iii, d-iv

Correct Answer:
a-ii, b-i, c-iii, d-iv

Solution:

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Smooth endoplasmic reticulum- Lipid synthesis

Rough endoplasmic reticulum - Protein synthesis

Golgi apparatus - Glycosylation

Centriole - spindle formation

Q. 76 Select the incorrectly matched pair form following:

Option 1:
Neurons - Nerve cells

Option 2:
Fibroblast - Areolar tissue

Option 3:
Osteocytes - Bone cells

Option 4:
Chondrocytes - Smooth muscle cells

Correct Answer:
Chondrocytes - Smooth muscle cells

Solution:

- Neurons are structural and functional units of Nervous system. These are modified Nerve cells
- Areolar tissue is a Loose connective tissue and it consists of Matrix,Cells and Fibres
- The cells are Fibroblasts,Monocytes, Macrophages etc
- Fibroblasts are cells that form fibres
- Osteocytes are mature cells of bones
- Chondrocytes are cartilage forming cells

Q. 77 Which of the following is associated with decrease in cardiac output?

Option 1:
Parasympathetic neutral signals

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Option 2:
Pneumotaxic centre

Option 3:
Adrenal medullary hormones

Option 4:
Sympathetic nerves

Correct Answer:
Parasympathetic neutral signals

Solution:
- Parasympathetic neural signals decrease the rate of heart beat, speed of conduction, stroke volume and thus cardiac output
- Sympathetic system is antogonistic to parasympathetic and thus increase the heart rate and thus cardiac output
- Adrenal medullary hormones also increase the cardiac output
- Pnemotaxic centre is related to respiration

Q. 78 Intrinsic factor that helps in the absorption of vitamin B_{12} is secreted by:

Option 1:
Hepatic cells

Option 2:
Oxyntic cells

Option 3:
Cheif cells

Option 4:
Goblet cells

Correct Answer:
Oxyntic cells

Solution:
- Gastric glands consist os many cells that secrete various substances
- These cells are

1. Chief cells that secrete inactive pepsinogen

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2. Oxyntic cells or Parietal cells that secrete HCl and Intrinsic Factor
3. Mucous cells that secrete Mucous

- Intrinsic factr is responsible for absorption of Vit B12

Q. 79 Match the following Columns and select the correct option:

Column I	Column II
(a) Gout	(i) Decreased levels of estrogen
(b) Osteoporosis	(ii) Low Ca^{++} ions in the blood
(c) Tetany	(iii) Accumulation of uric acid crystals
(d) Muscular dystrophy	(iv) Auto immune disorder
	(v) Genetic disorder

Option 1:
a-iii, b-i, c-ii, d-v

Option 2:
a-iv, b-v, c-i, d-ii

Option 3:
a-i, b-ii, c-iii, d-iv

Option 4:
a-ii, b-i, c-iii, d-iv

Correct Answer:
a-iii, b-i, c-ii, d-v

Solution:
- GOUT : Increase uric acid crystals in the blood
- OSTEOPOROSIS : Bones become brittle due to low levels of estrogen

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- TETANY : It is because of low calcium levels
- MUSCULAR DYSTROPHY : Generalised weakness of muscles and it is a genetic disorder

Q. 80 The best example for pleiotrophy is:

Option 1:
Phenylketnoneuria

Option 2:
Colour blindness

Option 3:
ABO blood group

Option 4:
Skin colour

Correct Answer:
Phenylketnoneuria

Solution:
- Pleiotrophy is genetic effect of single gene on multiple phenotypic traits
- Phenylketonuria, sickle cell anemia, cystic fibrosis are examples of Pleiotrophy

Q. 81 Match the following columns and select the correct option:

Column I	Column II
(a) Rods and Cones	(i) Absence of photoreceptor
(b) Blind spot	(ii) Cones are densly packed
(c) Fovea	(iii) Photoreceptor cells
(d) Iris	(iv) Visible coloured portion of the eye

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Option 1:
a-ii, b-iii, c-i, d-iv

Option 2:
a-iii, b-iv, c-ii, d-i

Option 3:
a-ii, b-iv, c-iii, d-i

Option 4:
a-iii, b-i, c-ii, d-iv

Correct Answer:
a-iii, b-i, c-ii, d-iv

Solution:

- Retina is the innermost layer of the eye
- It consists of three layers from inner to outer
- These layers are Ganglion layer, Bipolar layer and Photoreceptors
- The photoreceptors are Rods and Cones
- Retina shows an area called blind spot from where optic nerve arises. This area does not show presence of any Photoreceptors
- Lateral to the blind spot, there is an area called as Macula densa that shows presence of only cons
- The centre of Macula is Fovea centralis which shows maximum number of cones
- Iris is the middle layer of eye and it is the coloured portion of the eye

---

Q. 82 Which of the following conditions cause erythroblastosis foetalis?

Option 1:
Mother - Rh^{-ve} and foetus - Rh^{+ve}

Option 2:
Both mother and foetus Rh^{-ve}

Option 3:
Both mother and foetus Rh^{+ve}

Option 4:
Mother - Rh^{+ve} and foetus Rh^{-ve}

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Correct Answer:
Mother - Rh^{-ve} and foetus - Rh^{+ve}

Solution:

- Rh Antigen is a self Antigen seen on the surface of RBC
- Based on absence and presence of Rh Antigen, the blood groups are Rh negative and Rh positive respectively
- **Erythroblastosis fetalis** results from Rh incompatibility, which may develop when a woman with Rh-negative blood is pregnant and conceives a fetus with Rh-positive blood, resulting in hemolysis of RBC's

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Q. 83 During Meiosis 1, in which stage synapsis takes place?

Option 1:
Zygotene

Option 2:
Diplotene

Option 3:
Leptotene

Option 4:
Pachytene

Correct Answer:
Zygotene

Solution:
During meiosis, synapsis occur during Zygotene stage

Q. 84 The size of Pleuropneumonia - like Organism (PPLO) is

Option 1:
1-2μm

Option 2:
10 -20 μm

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Option 3:
0.1 μm

Option 4:
0.02 μm

Correct Answer:
0.1 μm

Solution:
PPLO is pleuropneumonia-like organisms. It is also known as mycoplasma (jokers of microbiology) and has a diameter of 0.1 micrometers.

Q. 85 The yellowish fluid "colostrum" secreted by mammary glands of mother during the initials days of lactation has abundant antiboides (IgA) to protect the infant. This type of immunity is called as:

Option 1:
Active immunity

Option 2:
Acquired immunity

Option 3:
Autoimmunity

Option 4:
Passive immunity

Correct Answer:
Passive immunity

Solution:

- In Passive immunity, the Antibodies are directly inserted in the body
- Colostrum is the first milk to infants after birth.It consists of IgA
- Thus this is Passive immunity
- This is a natural phenomenon and hence it is Acquired Natural Passive immunity
- Hence, the correct option is (d).

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Q. 86 Which of the following options does correctly represents the characteristics feature of phylum Annelida?

Option 1:
Triploblastic, segemented body and bilaterally symmetrical

Option 2:
Triploblastic, flattend body and acoelomate condition

Option 3:
Diploblastic, mostly marine and radially symmetrical

Option 4:
Triploblastic, unsegmentaed body and bilaterally symmetrical

Correct Answer:
Triploblastic, segemented body and bilaterally symmetrical

Solution:

- Phylum Annelids are Segmental worms
- These show organ systems due to presence of three germ layers and hence are Triploblastic organisms
- These are bilaterally symmetrical
- Examples are Earthworms, Nereis etc

Q. 87 Cyclosporin A, used as immunosuppression agent, is produced from:

Option 1:
Saccharomyces cerevisiae

Option 2:
Penicillium notatum

Option 3:
Trichoderma polysporum

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Option 4:
Monascus purpureus

Correct Answer:
Trichoderma polysporum

Solution:
- Cyclosporin A is produced by submerged fermentation of aerobic fungi called *Trichoderma polysporum*

Q. 88 Hormones stored and relased from neurohypophysis are:

Option 1:
Oxytoxin and Vasopressin

Option 2:
Follicle stimulating hormone and Leutinizing hormone

Option 3:
Prolactin and Vasopressin

Option 4:
Thyroid stimulating hormone and Oxytocin

Correct Answer:
Oxytoxin and Vasopressin

Solution:
- Pituitary gland is a master endocriine gland
- It consists of Anterior and Posterior parts
- Anterior part (Adenohypophysis) releases hormones under the influence of Hypothalamus
- Posterior part (Neurohypophysis) does not produce any hormone.It stores and releases hormones of Hypothalamus
- The two main hormones are Oxytocin and Vasopressin
- Oxytocin is the birth hormone and Vasopressin is ADH (Anti diuretic hormone) that controls the concentration of urine

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Q. 89 Progestrone is alone or in combination with estrogen can be used as a contraceptive in the form of -

Option 1:
Injections only

Option 2:
Pills, injections and implants

Option 3:
Pills only

Option 4:
Implants only

Correct Answer:
Pills, injections and implants

Solution:
- Pills, Injections and IMplants contain either Progesterone or combination of both Estrogen and Progesterone

Q. 90 All veretebrates are chordates but all chordates are not vertebrates, why?

Option 1:
Ventral hollow nerve cord remains throughout life in some chordates

Option 2:
All chordates posses vetebral column

Option 3:
All chordates posses notochord throughout their life.

Option 4:
Notochord is replaced by vertebral column in adult of some chordates

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Correct Answer:

Notochord is replaced by vertebral column in adult of some chordates

Solution:

- Animals can be non chordates or chordates depending on absence and presence of notochord
- Chordates can be invertebrates or vertebrates
- In some animals notochord is replaced by vertebral column and these are known as Vertebrates while in some animals notochord remains as it is as in invertebrates
- Thus all Chordates are not verterates but all Vertebrrates are Chordates

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